CHRONIC DISEASES AND NONCOMMUNICABLE DISEASES IN TIMOR-LESTE

Global and Local Trends, Epidemiology, Disease Burden, and Health System Challenges



Authors

1. Dr. Ns. Priyanto, M.Kep., M.B
2. Adelaide Edina Almeida da Costa, L.Enf.
3. Irenia dos Anjos, S.Kep., Ns
4. Lindalva Zelita Ornai, Lic. Enf.

LUSOFONIA PRESS

COPYRIGHT PAGE

CHRONIC DISEASES AND NONCOMMUNICABLE DISEASES IN TIMOR-LESTE

Global and Local Trends, Epidemiology, Disease Burden, and Health System Challenges

Authors

Copyright © 2026

All rights reserved.

No part of this publication may be reproduced, distributed, stored in a retrieval system, or transmitted in any form or by any means, including electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the copyright holder, except for brief quotations used in scholarly works, academic criticism, reviews, and other uses permitted under applicable copyright law.

Authors

1. Priyanto (priyanto@unw.ac.id)
2. Adelaide Edina Almeida da Costa(costaade27@gmail.com)
3. Irenia dos Anjos (dosanjosirenia@gmail.com)
4. Lindalva Zelita Ornai (lindalvazelitaornai@gmail.com)

Publisher :Lusofonia Press
Place of Publication:Brazil
Year of Publication :2026
Language :English
ISBN : 978-65-977074-1-6
DOI :10.68609/chronic-diseases-ncd-timor-leste-2026
Format :Print Edition
Dimensions :14.5 × 20.5 cm
Number of Pages ::191 pages
Text Editing :Lusofonia Press
Language Editing :Lusofonia Press
Typesetting :Lusofonia Press
Graphic Design :Lusofonia Press
Cover Design :Lusofonia Press

Published by:
Lusofonia Press
Brazil
2026

DEDICATION

This book is dedicated to the people of Timor-Leste who live with chronic illness, whose daily struggles, courage, and resilience remind us that health is not merely the absence of illness, but an essential foundation of human dignity, well-being, and hope.

It is dedicated to patients and their families, who bear the physical, emotional, social, and economic burdens of long-term illness; to those whose lives have been affected by preventable complications and premature death; and to all who continue to face health challenges with courage and determination, often under circumstances shaped by limited resources and unequal access to healthcare.

We also dedicate this work to the healthcare professionals, nurses, physicians, community health workers, researchers, educators, and policymakers who devote their knowledge, compassion, and commitment to preventing illness, promoting healthy living, strengthening health systems, and improving the quality of life of individuals and communities throughout Timor-Leste.

Special recognition is extended to the communities of sucos and aldeias, including those in remote and underserved areas, whose experiences and needs must remain central to every effort toward equitable healthcare. Their voices remind us that effective public health begins not only with scientific evidence and clinical intervention, but also with listening, understanding, solidarity, and respect for human dignity.

Finally, this book is dedicated to the present and future generations of Timor-Leste. May the knowledge presented in these pages contribute to a stronger culture of prevention, earlier diagnosis, equitable access to treatment, and more resilient healthcare systems. May it inspire collective responsibility and sustained action so that preventable illness and premature mortality no longer impose an avoidable burden on human potential, family well-being, and national development.

Because every life matters, every community deserves equitable healthcare, and every generation deserves the opportunity to live a healthier and more dignified life.

FOREWORD

Chronic diseases represent one of the major challenges facing public health systems worldwide, including Timor-Leste. Their growing impact on individuals, families, communities, and national development highlights the urgent need for effective prevention, early diagnosis, equitable treatment, and sustainable healthcare policies.

This book, *Chronic Diseases and Noncommunicable Diseases in Timor-Leste: Global and Local Trends, Epidemiology, Disease Burden, and Health System Challenges*, provides an important contribution to understanding these challenges through the integration of global perspectives and local realities. It emphasizes the importance of evidence-based practice, health system strengthening, community participation, and equitable access to healthcare, particularly for vulnerable populations and communities in remote areas.

Beyond its scientific and academic value, this work affirms that healthcare is fundamentally a matter of human dignity, social justice, and collective responsibility. It is our hope that this book will support healthcare professionals, researchers, educators, students, and policymakers in developing more responsive and resilient health systems for the people of Timor-Leste.

May this work contribute to a healthier future in which prevention is prioritized, care is accessible to all, and no individual or community is left behind.

PREFACE

Chronic diseases and noncommunicable diseases (NCDs) represent one of the defining public health challenges of the twenty-first century. Their consequences extend far beyond clinical diagnosis and treatment, affecting life expectancy, functional capacity, family well-being, economic productivity, and the sustainability of health systems. Although these conditions are frequently associated with ageing populations, their causes and consequences are shaped by a complex interaction of behavioural, metabolic, environmental, social, and structural determinants (World Health Organization [WHO], 2024).

For Timor-Leste, understanding this challenge requires attention to the country's distinctive demographic, geographic, socioeconomic, and health system context. A young population, differences in access to health services, and the practical demands of delivering continuous care across urban, rural, and remote communities create important considerations for the prevention and management of chronic diseases. These challenges should not, however, be interpreted solely as evidence of institutional limitations. They also provide opportunities to strengthen primary health care, community participation, health education, disease surveillance, and evidence-informed policy.

This book brings together four closely connected areas of inquiry: global and local trends in chronic diseases and NCDs; epidemiological patterns and risk factors; disease burden measured through mortality, disability, and disability-adjusted life years (DALYs); and the implications of these findings for health system

development in Timor-Leste. Particular attention is given to the relationship between evidence and action, recognizing that epidemiological knowledge becomes most valuable when it informs equitable and sustainable health interventions.

National evidence is essential to this undertaking. The Timor-Leste STEPS survey conducted in 2023 provides population-based information on behavioural and biological risk factors among adults aged 18–69 years. Such evidence offers an important foundation for understanding local health needs, identifying research gaps, and developing more responsive public health strategies (Ministry of Health, Timor-Leste, 2023; WHO, 2023).

The book is intended for students, researchers, health professionals, policymakers, educators, and readers interested in the future of population health in Timor-Leste. It does not seek to present epidemiological statistics as isolated numbers. Rather, it seeks to interpret disease patterns in relation to human lives, social circumstances, institutional responsibilities, and the broader goals of health equity and sustainable development.

Ultimately, the prevention and control of chronic diseases require more than medical technologies or individual behaviour change. They demand sustained political commitment, reliable information, effective health services, intersectoral collaboration, and recognition of every person's right to live with dignity and the best health attainable. It is this perspective that guides the present work.

Acknowledgements

The development of this book has been guided by the conviction that reliable health evidence should contribute to the well-being of communities and the advancement of equitable health systems. The author acknowledges the contributions of researchers, public health professionals, epidemiologists, clinicians, health educators, and institutions whose work has expanded understanding of chronic diseases and noncommunicable diseases.

Particular recognition is due to the Ministry of Health of Timor-Leste and the World Health Organization for supporting the collection, documentation, and dissemination of information relevant to the prevention and control of NCDs. National surveillance initiatives, including the STEPS survey, provide an important foundation for evidence-informed discussion of health priorities in Timor-Leste.

The author also recognizes the international scientific community whose research has advanced the measurement of mortality, disability, disease burden, and health system performance. These contributions make it possible to examine national health challenges within a wider comparative and methodological framework.

Finally, this book is dedicated to the health professionals, community workers, families, and individuals who live with chronic diseases and whose experiences remind us that health statistics represent human lives, aspirations, and needs. Their realities reinforce the importance of accessible, compassionate, continuous, and culturally responsive health care.Any errors of interpretation or omission remain the responsibility of the author.

List of abbreviations

Abbreviation	Meaning
BMI	Body Mass Index
DALYs	Disability-Adjusted Life Years
CVDs	Cardiovascular Diseases
GBD	Global Burden of Disease
HALE	Healthy Life Expectancy
LMICs	Low- and Middle-Income Countries
NCDs	Noncommunicable Diseases
PHC	Primary Health Care
SDGs	Sustainable Development Goals
STEPS	STEPwise Approach to NCD Risk Factor Surveillance
UHC	Universal Health Coverage
WHO	World Health Organization
YLDs	Years Lived with Disability
YLLs	Years of Life Lost

TABLE OF CONTENTS

COPYRIGHT PAGE	i
DEDICATION	iv
FOREWORD	vi
PREFACE	vii
Acknowledgements	ix
List of abbreviations	x
CHAPTER 1	1
GLOBAL AND LOCAL LANDSCAPES	1
INTRODUCTION	1
Concepts, Definitions, and the Global Burden	2
Global Trends, Epidemiological Transition	7
The Timor-Leste Context	16
CHAPTER 2. EPIDEMIOLOGY, RISK FACTORS	26
Cardiovascular Diseases, Diabetes, Cancer	26
Epidemiological Measures, Risk Distribution	37
NCD Risk Factors and Health Inequalities	45

CHAPTER 3. DISEASE BURDEN, DALYs, AND THE SOCIOECONOMIC CONSEQUENCES 59

Measuring Disease Burden: Mortality, Disability 59

The Global and Timor-Leste Burden of NCDs 70

Economic, Social, and Human Development 85

CHAPTER 4 99

HEALTH SYSTEM CHALLENGES 99

Health System Capacity 100

Prevention, Early Detection, and Integrated Community 113

Policy Priorities, Strategic Recommendations, 131

APPENDIX A 154

REFERENCES 175

CHAPTER 1

GLOBAL AND LOCAL LANDSCAPES OF CHRONIC DISEASES AND NONCOMMUNICABLE DISEASES

INTRODUCTION

Chronic diseases and noncommunicable diseases (NCDs) have become central concerns in contemporary public health because of their extensive effects on mortality, disability, quality of life, social development, and health system sustainability. Their importance lies not only in the number of people affected but also in the prolonged care they often require, the possibility of preventable complications, and the inequalities that shape people's opportunities to obtain timely diagnosis and effective treatment. The World Health Organization (WHO, 2024) identifies cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes as the four major groups of NCDs responsible for a substantial proportion of premature mortality worldwide.

Understanding these conditions requires an approach that connects clinical knowledge with epidemiology, social determinants, health policy, and the organization of health services. A person living with hypertension or diabetes may require years of monitoring, medication, lifestyle support, and periodic assessment for complications. When these services are inaccessible, interrupted, or unaffordable, a condition that could have been managed effectively may progress toward stroke, kidney disease, cardiovascular complications, disability, or premature death. Consequently, chronic disease prevention and management must be understood as continuing responsibilities rather than isolated clinical interventions (Atun et al., 2013; WHO, 2020a).

For Timor-Leste, this international challenge must be interpreted within the country's own demographic, social, geographic, and institutional circumstances. National evidence is essential because global estimates cannot fully describe variations in health behaviour, service accessibility, household resources, or the experiences of communities in different parts of the country. The 2023 national STEPS survey provides an important evidence base for examining behavioural and biological risk factors among adults aged 18–69 years, including tobacco use, alcohol consumption, diet, physical activity, blood pressure, blood glucose, and blood lipids (Ministry of Health, Timor-Leste, 2023; WHO, 2023).

This chapter establishes the conceptual and analytical foundations for the book. It first clarifies the meaning of chronic diseases and NCDs, then examines global epidemiological changes and their associated risk factors, before considering how these developments relate to the health priorities of Timor-Leste. The central argument is that effective responses must combine reliable evidence, prevention, equitable access to health services, and long-term institutional commitment.

1.1. Concepts, Definitions, and the Global Burden of Chronic Diseases and NCDs

Understanding Chronic Diseases and Noncommunicable Diseases

Chronic diseases are generally understood as health conditions that persist over an extended period and frequently require continuing medical attention, long-term management, or adjustments to everyday life. Their clinical course varies considerably. Some conditions develop gradually, while others may involve acute episodes followed by prolonged recovery, persistent impairment, or continuing risk of complications. Chronic diseases may be controlled successfully for many years, but their duration can create cumulative demands on individuals, families, health professionals, and health systems.

Noncommunicable diseases, by contrast, are classified primarily according to their noncommunicable nature and underlying patterns of causation. They are not ordinarily transmitted directly from one person to another as infectious diseases are. Their development commonly involves interactions among genetic predisposition, physiological processes, environmental exposures, behavioural factors, and social conditions. The WHO (2024) identifies cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes as the four major NCD groups.

Although the concepts overlap substantially, chronic disease and noncommunicable disease should not be treated as perfectly interchangeable terms. Many NCDs are chronic, but the concept of chronic illness can also encompass certain long-lasting infectious conditions. Conversely, some noncommunicable conditions may have an acute presentation or course. This distinction is important for academic precision because the scope of a study determines which diseases, populations, outcomes, and interventions should be included in its analysis.

For this book, the term chronic diseases and NCDs is used to describe the broad public health challenge of long-term illness, with the principal epidemiological emphasis placed on NCDs and their associated risk factors. The discussion concentrates on cardiovascular diseases, diabetes, cancers, and chronic respiratory diseases, while also recognizing other relevant chronic conditions, including chronic kidney disease, where appropriate evidence is available. This operational approach provides a coherent basis for examining disease patterns, disability, premature mortality, and the demands placed on health services in Timor-Leste.

Major Categories of Noncommunicable Diseases

Cardiovascular diseases include disorders affecting the heart and blood vessels, such as ischemic heart disease, stroke, and other vascular conditions. Their development is associated with multiple factors, including hypertension, tobacco exposure, abnormal blood lipids, diabetes, unhealthy diets, and physical inactivity. Because cardiovascular events can result in sudden death or substantial long-term disability, their public health consequences extend beyond the immediate clinical episode to rehabilitation, family caregiving, and continuing health service needs (Roth et al., 2020).

Diabetes mellitus is a metabolic disorder characterized by elevated blood glucose resulting from impaired insulin production, impaired insulin action, or a combination of these mechanisms. The major forms include type 1 and type 2 diabetes, which differ in their underlying mechanisms and treatment requirements. When diabetes is not adequately controlled, it may contribute to cardiovascular disease, kidney damage, visual impairment, nerve damage, and other complications. Effective management therefore requires more than the initial diagnosis: it depends on continuity of care, access to appropriate medication, patient education, monitoring, and timely management of complications.

Cancer comprises a diverse group of diseases characterized by the abnormal growth and spread of cells. Different cancers have different causes, natural histories, preventive opportunities, and treatment requirements. Tobacco exposure, certain infections, alcohol consumption, environmental exposures, and other risk factors contribute to particular cancer types, although not every cancer is preventable. Population-based prevention, appropriate screening where recommended, timely diagnosis, and access to effective treatment are therefore important components of cancer control.

Chronic respiratory diseases include asthma and chronic obstructive pulmonary disease, among other persistent respiratory conditions. Their burden may be influenced by tobacco smoke, air pollution, occupational exposures, indoor air quality, and other environmental or individual factors. Depending on severity, these conditions may restrict physical activity, reduce productivity, require repeated clinical consultations, and contribute to emergency admissions.

These four major groups are interconnected through shared risk factors and, in many cases, common social determinants. An individual may experience hypertension, diabetes, and cardiovascular disease simultaneously, creating complex treatment needs that cannot be addressed effectively through fragmented disease-specific services. Such multimorbidity reinforces the importance of integrated primary health care, coordinated follow-up, and person-centred management (Atun et al., 2013; WHO, 2020a).

The Global Burden and Its Public Health Significance

The global burden of NCDs is expressed through several complementary indicators, including incidence, prevalence, mortality, premature mortality, disability, and years of healthy life lost. Each indicator answers a different question. Incidence describes the occurrence of new cases, prevalence indicates how many people are living with a condition, and mortality measures deaths attributed to a disease. Disability and healthy life expectancy indicators provide additional insight into the consequences experienced by people who survive.

According to the WHO (2024), NCDs account for a substantial majority of deaths worldwide, with the burden disproportionately affecting low- and middle-income countries. This unequal distribution reflects differences in exposure to risk factors, socioeconomic circumstances, opportunities for prevention, and access to timely diagnosis and treatment. Many premature deaths could be reduced through effective public health policies, early detection, appropriate clinical management, and stronger health systems.

The Global Burden of Disease (GBD) framework provides a systematic approach to comparing diseases and risk factors across populations and over time. By combining mortality estimates with measures of nonfatal health loss, GBD studies allow researchers to identify conditions that contribute most substantially to lost healthy life years. The GBD 2019 analysis, for example, assessed 369 diseases and injuries

across 204 countries and territories, illustrating the breadth of conditions that can be examined within a common comparative framework (GBD 2019 Diseases and Injuries Collaborators, 2020).

Nevertheless, international estimates must be interpreted carefully when applied to a particular country. Estimates derived from statistical models may be necessary where routine data are incomplete, but they are not equivalent to direct measurement of every outcome in the population. Differences in data availability, disease registration, diagnostic capacity, and estimation methods can affect comparisons. For Timor-Leste, global estimates should therefore be interpreted alongside national surveys, routine health information, mortality data, and other locally relevant evidence whenever available.

The significance of the global NCD burden is consequently both epidemiological and institutional. It demonstrates the need to prevent avoidable illness while ensuring that people who already live with chronic conditions can obtain appropriate and continuous care. For countries with limited resources, this requires careful prioritization, investment in primary health care, reliable essential medicines, effective referral pathways, and public health policies that address the conditions in which people live and work.

Chronic Diseases, Disability, and Human Development

The consequences of chronic diseases cannot be fully understood by counting deaths alone. Many people live for years with conditions that reduce their mobility, independence, capacity to work, or ability to participate in family and community life. Others experience intermittent complications that require repeated hospital visits, ongoing medication, or support from caregivers. These outcomes demonstrate why disease burden should include both premature mortality and nonfatal health loss.

The disability-adjusted life year (DALY) provides a framework for combining these dimensions. It is calculated as the sum of years of life lost due to premature mortality (YLLs) and years lived with disability (YLDs). A DALY represents one year of healthy life lost. This measure makes it possible to compare conditions that cause early death with those that produce prolonged illness or disability, although its interpretation depends on the underlying data and estimation methods (Murray & Lopez, 1996; GBD 2019 Diseases and Injuries Collaborators, 2020).

Chronic disease also has implications for household finances and social development. Continuing treatment may involve expenses for medication, diagnostic tests, transportation, and time away from work. Where a family member develops substantial disability, relatives may assume additional caregiving responsibilities, potentially reducing their own employment or educational opportunities. These effects can reinforce existing inequalities when people with fewer resources face greater barriers to preventive services and long-term treatment.

The relationship between chronic disease and human development is therefore multidirectional. Poor health can restrict education, employment, income, and social participation, while disadvantageous social and economic conditions can increase exposure to risk factors and reduce access to care. Marmot et al. (2008) emphasize that health inequalities are closely linked to the distribution of social opportunities and resources. This perspective is particularly relevant when examining how national policies can reduce preventable illness while improving fairness in access to health services.

For Timor-Leste, this broader understanding is important because disease prevention and management should not be considered exclusively as clinical responsibilities. They are also connected to household well-being, community resilience, the protection of vulnerable populations, and the long-term development of the country. A health system that detects disease earlier, supports effective long-term treatment, and reduces avoidable complications can help preserve both individual functioning and social participation.

Implications for the Study of NCDs in Timor-Leste

The concepts introduced in this section establish three principles for the remainder of the book. First, chronic diseases and NCDs must be defined consistently so that the population, conditions, and outcomes under investigation remain clear. Second, the burden of disease should be examined through multiple indicators, including mortality, prevalence, disability, and DALYs, rather than relying on a single measure. Third, international evidence must be interpreted in relation to national data and the circumstances of the communities whose health is being studied.

The Timor-Leste STEPS survey provides a particularly important starting point for examining population-level risk factors. Conducted in 2023 among adults aged 18–69 years, it collected information on behavioural risks and included physical and

biochemical measurements relevant to NCD prevention (Ministry of Health, Timor-Leste, 2023; WHO, 2023). These data can support a more evidence-informed discussion of national priorities, provided that their sampling design, age coverage, definitions, and limitations are considered carefully.

At the same time, a survey of risk factors does not by itself establish the full national burden of every chronic disease. Estimates of disease incidence, mortality, disability, treatment outcomes, and financial consequences require additional sources and appropriate analytical methods. Recognizing this distinction is essential to avoid overstating what any single dataset can demonstrate.

The central task, therefore, is to connect international knowledge with reliable national evidence and to translate both into practical approaches for prevention, diagnosis, treatment, and health system improvement. This conceptual foundation leads directly to the next section, which examines global trends, epidemiological transition, and the emerging risk factors shaping the distribution of chronic diseases across populations.

1.2. Global Trends, Epidemiological Transition, and Emerging Risk Factors

Global Trends in Chronic Diseases and Noncommunicable Diseases

The global distribution of chronic diseases and noncommunicable diseases (NCDs) has changed substantially over recent decades, reflecting transformations in population structure, living conditions, economic development, food systems, urban environments, and health care. Improvements in sanitation, vaccination, infectious disease control, and medical treatment have contributed to longer life expectancy in many parts of the world. However, longer lives do not automatically translate into longer periods of healthy living. As populations survive into older age, the number of people requiring long-term management for cardiovascular diseases, diabetes, cancer, and chronic respiratory conditions may increase, particularly when risk factors remain widespread and health systems are unable to provide continuous preventive and clinical care.

NCDs have become a major global public health concern because they contribute substantially to premature mortality and long-term disability. The World Health Organization (WHO, 2024) identifies cardiovascular diseases as the leading group of NCD-related causes of death, followed by cancers, chronic respiratory diseases, and

diabetes. These categories are not independent of one another. Common risk factors, including tobacco use, unhealthy diets, physical inactivity, and harmful alcohol consumption, can contribute to multiple conditions simultaneously. Metabolic risks such as elevated blood pressure, high blood glucose, excess body weight, and abnormal blood lipids further increase the likelihood of developing serious complications.

Global Burden of Disease research provides a broader picture of these changes by assessing mortality, nonfatal health loss, and risk factors across countries and over time. The GBD 2019 analysis documented the substantial contribution of chronic conditions to the worldwide burden of disease and demonstrated the importance of examining both fatal and nonfatal outcomes (GBD 2019 Diseases and Injuries Collaborators, 2020). Such analyses help identify priority conditions, but their findings should be interpreted alongside country-specific evidence because national patterns vary according to age structure, socioeconomic circumstances, environmental exposures, and the availability of preventive and curative services.

An important feature of the global NCD burden is its unequal distribution. Although these diseases affect populations in high-income countries as well as low- and middle-income countries, the consequences can be particularly severe where access to early diagnosis, essential medicines, specialist services, and rehabilitation is limited. People may reach health facilities only after complications have developed, increasing the likelihood of premature death, prolonged disability, and costly treatment. The WHO's global NCD policy framework therefore emphasizes prevention, early detection, effective treatment, and coordinated action across sectors rather than relying exclusively on hospital-based care (WHO, 2013, 2023a).

For Timor-Leste, these international trends provide a framework for asking important national questions. Which risk factors are most prevalent among adults? Which diseases create the greatest demands on primary and hospital services? Are people in rural and remote communities able to obtain timely diagnosis and continuous treatment? What evidence is available to distinguish directly observed national findings from modelled estimates? Addressing these questions is necessary to ensure that global knowledge supports, rather than substitutes for, an accurate understanding of local health needs.

Epidemiological Transition and Changing Population Health Needs

The concept of epidemiological transition describes broad changes in the distribution of disease and causes of death as populations experience demographic, social, environmental, and economic transformations. In simplified historical accounts, populations move from periods in which infectious diseases, malnutrition, and high child mortality dominate toward situations in which chronic diseases and age-related conditions account for a growing share of illness and death. However, this framework should not be interpreted as a universal or irreversible sequence. Different diseases may rise or decline simultaneously, and infectious diseases can remain important even as NCDs become increasingly prominent.

Countries may therefore experience a double burden of disease, in which communicable diseases and maternal, neonatal, and nutritional conditions coexist with a growing burden of NCDs. This situation is especially important for health systems that must address established public health needs while expanding their capacity for long-term chronic disease prevention and management. It requires careful planning because the resources needed for acute treatment, vaccination, maternal health, and infectious disease control may overlap with those required for diabetes monitoring, cardiovascular risk reduction, cancer diagnosis, and chronic respiratory care.

Demographic change is one factor contributing to this transition. As mortality declines and life expectancy increases, more people may survive into ages at which chronic diseases become more common. Population growth can also increase the absolute number of people requiring care even when age-specific disease rates remain stable. Consequently, policymakers must distinguish between changes in disease risk, changes in population size, and changes in the age distribution of the population. Age-standardized rates are useful for comparing populations with different age structures, while absolute case counts are essential for estimating the actual volume of services required.

Epidemiological transition is also influenced by changes in the environments in which people live. Urbanization, changes in occupational activity, motorized transport, access to processed foods, tobacco availability, and shifts in consumption patterns can alter exposure to risk factors. Yet the direction and magnitude of these effects depend on local circumstances. Urban residence does not inevitably produce poor health, and rural residence does not guarantee protection from NCDs. Rural populations may experience distinct challenges, including limited access to preventive services, longer

travel distances, and delayed diagnosis. These differences must be investigated empirically rather than assumed.

The social determinants of health are central to understanding these processes. Marmot et al. (2008) argue that health inequalities are closely related to the conditions in which people are born, grow, live, work, and age, as well as the distribution of power and resources that shapes those conditions. For chronic diseases, this means that health outcomes cannot be explained solely through individual choices. Income, education, food availability, housing, working conditions, health literacy, and access to quality services all influence exposure to risk and the ability to respond to illness.

In Timor-Leste, epidemiological transition should therefore be examined as a context-specific process rather than a predetermined progression. The country's population structure, geographic diversity, household circumstances, and health service capacity may produce patterns that differ from those observed in other countries. National surveillance, routine health information, and carefully interpreted international estimates are needed to understand these patterns and to anticipate future service requirements.

Behavioural risk factors are important because they can be modified through individual support, community interventions, public policy, and changes in the environments that influence everyday choices. Four major behavioural risks commonly emphasized in NCD prevention are tobacco use, harmful alcohol consumption, unhealthy diets, and insufficient physical activity. These factors are associated with several major NCDs, although their effects differ according to the intensity and duration of exposure, individual susceptibility, and interactions with other determinants (WHO, 2013, 2017).

Tobacco use is a major preventable cause of illness and premature death. Exposure to tobacco smoke increases the risk of cardiovascular disease, multiple cancers, and chronic respiratory conditions. Prevention requires more than providing information about health risks. Effective strategies may include tobacco taxation, smoke-free environments, restrictions on advertising and promotion, accessible cessation support, and enforcement of relevant regulations. The WHO Framework Convention on Tobacco Control and associated policy measures provide an international foundation for these interventions, although national implementation and monitoring remain essential.Unhealthy diets can contribute to elevated blood pressure, excess body weight, high blood glucose, and abnormal blood lipids. Dietary risk is shaped by food availability, affordability, marketing, household resources, cultural practices, and the

time available for food preparation. Consequently, nutrition interventions should not focus exclusively on telling individuals what to eat. They should also consider whether healthier foods are affordable and accessible, whether consumers have adequate information, and whether public policies support healthier food environments. Reducing excessive salt intake, limiting industrial trans fats, and improving access to nutritious foods are among the approaches recommended in global NCD prevention frameworks (WHO, 2017).

Insufficient physical activity is another important risk factor for cardiovascular disease, type 2 diabetes, and other adverse health outcomes. Regular physical activity supports cardiovascular function, metabolic health, physical capacity, and psychological well-being. However, opportunities for activity depend partly on the built environment, transportation systems, workplace practices, personal safety, and the availability of appropriate recreational spaces. Public health strategies can therefore encourage active transport, community exercise, and school-based activity while also addressing environmental barriers that make movement difficult or unsafe.

Harmful alcohol consumption contributes to a range of health and social consequences, including certain cancers, liver disease, injuries, and cardiovascular complications. The magnitude of harm depends on consumption patterns, individual circumstances, and the wider social environment. Prevention strategies may include appropriate regulation, measures to reduce harmful consumption, public education, and accessible treatment for alcohol-related disorders (WHO, 2014a).

These behavioural factors frequently interact rather than operate in isolation. A person exposed to several risk factors may face a greater overall probability of developing disease than someone exposed to only one. In addition, behavioural risks may cluster in populations experiencing economic disadvantage, limited health information, or restricted access to preventive services. This makes it important to examine both individual exposure and the social conditions that shape patterns of risk.

For Timor-Leste, the national STEPS survey offers a basis for describing selected behavioural and biological risk factors among adults. Its findings should be used to establish an evidence-based picture of the population rather than to generalize automatically to every age group or community. Interventions should also be adapted to local languages, cultural practices, food systems, and the realities of daily life. Community participation can help ensure that prevention messages are understandable, appropriate, and practical.

Metabolic Risk Factors and the Pathways to Chronic Disease

Behavioural exposures can contribute to metabolic changes that increase the likelihood of chronic disease. Four particularly important metabolic risk factors are raised blood pressure, elevated blood glucose, excess body weight, and abnormal blood lipids. These conditions can develop over time and may remain undetected until complications arise. Their significance lies not only in their direct clinical effects but also in their role as intermediate pathways linking social and behavioural conditions to cardiovascular disease, diabetes, kidney disease, and other long-term health outcomes.

Raised blood pressure is a major risk factor for stroke, ischemic heart disease, heart failure, and kidney damage. Because hypertension may produce few noticeable symptoms, individuals can live with elevated blood pressure without recognizing the danger. Screening, accurate measurement, appropriate treatment, and regular follow-up are therefore important components of primary health care. Effective management also requires reliable access to prescribed medicines and continuity of monitoring, rather than diagnosis during a single consultation without subsequent care.

Elevated blood glucose may indicate diabetes or an increased risk of developing it, depending on the diagnostic criteria and clinical context. Diabetes management aims to maintain appropriate glucose control, reduce cardiovascular risk, and prevent or delay complications. This requires individualized care, patient education, access to diagnostic services, and assessment for associated conditions. Where services are fragmented, patients may receive treatment for one problem while other important risks remain unrecognized.

Excess body weight can contribute to insulin resistance, elevated blood pressure, abnormal lipid levels, and other metabolic disturbances. However, body mass index alone does not fully characterize metabolic health or individual risk. Assessment should be interpreted alongside other clinical information, including waist circumference where appropriate, blood pressure, blood glucose, and lipid measurements. Public health responses should avoid stigmatizing individuals and instead address the social and environmental factors that influence diet, physical activity, and access to health-promoting resources.

Abnormal blood lipids, including elevated levels of certain atherogenic lipoproteins, can contribute to the development of atherosclerosis and increase cardiovascular risk.

Effective management may involve dietary modification, physical activity, smoking cessation, and lipid-lowering medication when clinically indicated. The appropriate approach depends on an individual's overall risk profile, existing disease, and clinical guidelines.

These metabolic risks often coexist, creating a cumulative burden that can be overlooked when health services address each risk factor independently. Integrated cardiovascular risk assessment can help identify individuals who may benefit from coordinated prevention and treatment. The WHO's package of essential NCD interventions for primary health care provides guidance for strengthening essential prevention and management services in settings with constrained resources (WHO, 2020a).

In Timor-Leste, the practical implications include ensuring that primary care facilities can measure blood pressure accurately, assess diabetes risk, provide appropriate laboratory testing where available, initiate evidence-based treatment, and arrange follow-up. The feasibility of these services depends on trained personnel, equipment, medicines, referral arrangements, and functioning information systems. National survey findings can help identify population-level needs, but clinical records and service data are also needed to understand how effectively detected risks are being managed.

Social, Commercial, and Environmental Determinants

The distribution of chronic diseases is influenced by factors that extend beyond individual behaviour and biological susceptibility. Social determinants include income, education, employment, housing, food security, social protection, and access to health services. These conditions influence both exposure to risk and the capacity to prevent, detect, and manage illness. People who face financial insecurity may have difficulty obtaining nutritious foods, maintaining healthy living conditions, travelling to health facilities, or paying indirect costs associated with repeated consultations.

Commercial determinants are also relevant. The availability, pricing, promotion, and distribution of tobacco, alcohol, highly processed foods, and other products can influence population exposure to health risks. Commercial practices interact with regulation, public information, household purchasing power, and cultural preferences. As a result, prevention strategies may need to combine individual education with measures that shape healthier environments. The WHO's recommended NCD

interventions emphasize the importance of policies that reduce population exposure to established risk factors rather than relying only on clinical treatment after disease has developed (WHO, 2017).

Environmental conditions can further influence chronic disease risk. Air pollution, indoor smoke, occupational exposures, limited opportunities for safe physical activity, and changes in local food environments may affect different populations in different ways. The contribution of any specific exposure must be assessed using appropriate evidence rather than assumed from geographic or economic characteristics alone.

Health systems themselves are part of this broader context. A population may have a high prevalence of hypertension or diabetes, yet the consequences will depend partly on whether people can obtain screening, diagnosis, medicines, follow-up, and treatment for complications. Kruk et al. (2018) emphasize that health systems must be evaluated not only by their existence or coverage but also by the quality of care they deliver and the outcomes people experience.

For Timor-Leste, these determinants should be studied across diverse geographic and social settings. Differences between Dili and more remote communities may affect travel time, service availability, continuity of care, and access to health information. However, such differences require evidence from national surveys, facility assessments, and community research. A rigorous analysis should distinguish measured disparities from hypotheses that require further investigation.

The policy implication is clear: chronic disease prevention cannot be achieved through clinical services alone. It requires coordination among health institutions, education, agriculture, local government, transport, finance, and other sectors that influence living conditions. Community-based approaches can complement national policy by ensuring that interventions reflect local priorities and are adapted to the needs of different populations.

Implications for Prevention and Health Policy in Timor-Leste

The global trends and risk factors examined in this section demonstrate that the future burden of chronic diseases is not determined by demographic change alone. It is also shaped by the prevalence of modifiable risk factors, the distribution of social resources, the quality of health services, and the capacity of governments and communities to implement effective prevention strategies. Although the relative importance of individual risk factors differs between populations, a comprehensive public health

response should address behavioural risks, metabolic conditions, and the structural circumstances that influence health.

For Timor-Leste, the first priority is to strengthen the use of reliable national evidence. The 2023 STEPS survey provides an important foundation for describing selected risk factors among adults aged 18–69 years. Its findings should be interpreted together with routine health information, mortality records, facility-level data, and relevant international estimates. This combined approach can help distinguish widespread risk factors from emerging concerns, identify gaps in service delivery, and establish priorities for future research.

The second priority is to strengthen prevention and early detection through primary health care. Screening for hypertension and diabetes, appropriate cardiovascular risk assessment, tobacco cessation support, nutrition education, and counselling on physical activity can help reduce avoidable illness when supported by reliable treatment and follow-up. Screening without the capacity to confirm diagnoses, provide treatment, and monitor outcomes may create limited benefit. Therefore, prevention programmes should be planned together with the services required to respond to the risks they identify (WHO, 2020a).

The third priority is to recognize that equitable prevention requires attention to differences in social circumstances and geographic access. Programmes designed for urban facilities may not be sufficient for people living in remote communities. Community health workers, culturally appropriate health education, outreach services, and effective referral systems can help connect national strategies with local needs. Their effectiveness, however, depends on appropriate training, supervision, resources, and integration into the wider health system.

Finally, national policy should establish measurable objectives and mechanisms for monitoring progress. Indicators may include tobacco use, raised blood pressure, diabetes detection and control, access to essential medicines, continuity of treatment, avoidable hospital admissions, and premature mortality from major NCDs. The selection of indicators should reflect national priorities, data availability, and the capacity to collect information consistently over time.

The central lesson of this section is that epidemiological transition should not be regarded as an inevitable movement toward increasing disease burden. Although demographic and social changes create new health needs, evidence-based

prevention, effective primary care, and coordinated public policy can alter the trajectory of chronic diseases. Timor-Leste's challenge is to translate available evidence into interventions that are practical, equitable, and sustainable.

Concluding Perspective

Global trends demonstrate the growing importance of chronic diseases and NCDs, but they also reveal substantial opportunities for prevention and improved care. Tobacco control, healthier food environments, physical activity, management of metabolic risk factors, and stronger primary health care can contribute to reducing premature mortality and disability. These measures are most effective when individual support is reinforced by public policy and equitable access to services.

For Timor-Leste, understanding global evidence is a starting point rather than an endpoint. National data, community experiences, health system assessments, and careful interpretation of disease estimates are necessary to determine which interventions should receive priority. The country's response must account for the interaction between behavioural risks, social determinants, geographical realities, and the capacity to deliver continuing care.

The next section turns directly to this national context. Section 1.3 will examine the demographic and socioeconomic circumstances of Timor-Leste, the available evidence on population health, the challenges of interpreting national disease patterns, and the priorities that should guide a context-sensitive response to chronic diseases and NCDs.

1.3. The Timor-Leste Context: Population Health, Epidemiological Change, and National Priorities

1.3.1. Introduction: A Young Nation Facing Changing Health Needs

Timor-Leste faces a complex public health transition in which the demands of preventing infectious diseases and improving maternal and child health coexist with the growing importance of chronic diseases and noncommunicable diseases (NCDs). This transition is particularly significant for a young nation whose future development depends on the health, education, productivity, and social participation of its population. Although a relatively young population may provide important demographic opportunities, it does not protect a country from the long-term consequences of

hypertension, cardiovascular diseases, diabetes, cancer, chronic respiratory diseases, and other conditions requiring sustained prevention and care.

The challenge for Timor-Leste is therefore not simply to respond to a new category of disease. It is to adapt the health system to a changing pattern of population health needs without neglecting existing priorities. Health services must continue to address communicable diseases, maternal and child health, nutrition, and other essential needs while developing stronger capacities for NCD prevention, early diagnosis, treatment, rehabilitation, and long-term follow-up. This requires coordination across primary health care, hospitals, public health programmes, local communities, and national policy institutions (Ministry of Health, Timor-Leste, 2011; World Health Organization [WHO], 2019).

Understanding this transition requires more than examining individual diseases. It also requires attention to demographic change, living conditions, health-related behaviour, geographical access to services, household resources, and the institutional capacity to provide continuous care. These factors influence not only who develops chronic disease but also whether individuals receive a timely diagnosis, can maintain treatment, and avoid preventable complications.

For this reason, the Timor-Leste context should be understood through three connected dimensions: the changing epidemiological profile of the population, the distribution of risk and vulnerability across communities, and the capacity of the health system to respond equitably.

1.3.2. Demographic Change and the Epidemiological Transition

The epidemiological transition describes changes in the major causes of illness, disability, and death as populations experience social, economic, demographic, and environmental transformations. In many countries, improvements in sanitation, vaccination, nutrition, maternal care, and the treatment of infectious diseases have contributed to longer life expectancy and changes in the pattern of mortality. As more people survive into adulthood and older age, the number of individuals living with chronic conditions may increase, creating additional demands for preventive and long-term health services (Murray & Lopez, 1996; WHO, 2023b).

For Timor-Leste, this transition should not be interpreted as a complete replacement of infectious diseases by NCDs. Different forms of disease burden may coexist within the same population, household, or health facility. A community may require services

for infectious disease control and maternal health while also needing reliable access to blood pressure measurement, diabetes screening, cardiovascular risk assessment, and long-term medication. Such coexistence creates a double burden for health systems that must respond to multiple priorities with finite financial, human, and material resources.

Demographic change also influences the time horizon of health planning. Chronic diseases frequently develop over many years, and their complications may become more apparent as people age. However, the foundations of chronic disease prevention are often established much earlier in life through nutrition, physical activity, tobacco avoidance, reduced harmful alcohol consumption, and the prevention and management of metabolic risk factors. Consequently, a strategy focused exclusively on older adults or on treating advanced disease would overlook opportunities to reduce future illness among younger generations.

The implications extend beyond clinical services. A healthy population contributes to educational participation, workforce capacity, household stability, and national development. Conversely, premature illness and disability may interrupt education, reduce employment opportunities, increase household expenditure, and place additional demands on family members who provide unpaid care. Preventing chronic disease is therefore both a health priority and a long-term investment in human development.

1.3.3. The Importance of National Epidemiological Evidence

Effective policy requires reliable information about the distribution of diseases and their risk factors. Without such evidence, governments may find it difficult to determine which conditions deserve immediate attention, which population groups face the greatest risks, and where prevention and treatment resources should be allocated.

The World Health Organization's STEPwise approach to NCD risk-factor surveillance, commonly known as the STEPS approach, provides a framework for collecting comparable information on behavioural and biological risk factors. The Timor-Leste STEPS survey conducted in 2023 is an important source for understanding adult NCD risk factors in the country. Its findings can inform national discussions about prevention, screening, health promotion, and the development of primary health care services (Ministry of Health, Timor-Leste, 2023).

The analytical value of a national survey depends, however, on careful interpretation. Risk-factor prevalence is not identical to the prevalence of diagnosed disease, and neither measure alone describes the complete burden of disability, premature mortality, or unmet treatment need. A survey may identify individuals with elevated blood pressure or other risk factors, but additional information may be required to establish whether they have previously been diagnosed, whether they receive treatment, and whether their condition is adequately controlled.

Similarly, estimates from population surveys must be interpreted according to their sampling methods, age range, definitions, and measurement procedures. Comparisons between years or countries are meaningful only when the underlying indicators are sufficiently comparable. A national policy document should therefore report exact prevalence estimates, confidence intervals, and subgroup differences directly from the original survey report rather than rely on general summaries or assume that global estimates precisely represent the Timor-Leste population.

National survey data should also be complemented by routine health-service information, mortality statistics, hospital records, population-based research, and international comparative estimates. Each source contributes a different perspective. Routine records may reveal patterns among people who access services, while population surveys can identify risk factors among individuals who may not have sought care. Global Burden of Disease estimates can support comparisons across conditions and locations, but modelled estimates should not be treated as interchangeable with direct national measurements (Institute for Health Metrics and Evaluation [IHME], 2024).

The central priority is to develop an integrated evidence base that allows decision-makers to move from identifying health problems to measuring service gaps and evaluating the results of interventions.

1.3.4. Geographic Access, Social Conditions, and Health Inequality

The burden of chronic diseases is shaped not only by individual behaviour and biological vulnerability but also by the social and geographical conditions in which people live. In Timor-Leste, national planning must consider the needs of urban populations, rural communities, and people living in geographically isolated areas. Distance from health facilities, transportation availability, household income, health

literacy, and the availability of trained health professionals can influence whether individuals receive preventive services and maintain long-term treatment.

Chronic disease management differs from the management of many acute conditions because it requires repeated contact with health services. A person living with hypertension may need regular blood pressure monitoring, medication adjustments, counselling, and assessment for complications. A person with diabetes may require continued access to medicines, blood glucose monitoring, dietary guidance, and screening for eye, kidney, cardiovascular, and foot complications. If these services are difficult to reach, treatment may be interrupted even when an individual understands the importance of managing the condition.

Geographical barriers can therefore become clinical risks. Delayed diagnosis may allow a condition to progress without adequate intervention, while inconsistent follow-up can reduce the effectiveness of treatment. The burden may be particularly severe when households must absorb transport costs, time away from employment, and other expenses associated with repeated visits. These costs can affect the ability of people with limited financial resources to remain engaged in care.

Social determinants also influence the development of disease before a person enters the health system. Access to nutritious food, safe places for physical activity, education, secure employment, and reliable health information can shape exposure to risk factors over a lifetime. These determinants are not distributed equally, and individual choices are often constrained by the opportunities available within a person's community. Public health strategies should therefore avoid explaining chronic disease exclusively in terms of personal responsibility.

The social determinants of health framework emphasizes that health inequalities emerge from the conditions in which people are born, grow, live, work, and age, as well as from the distribution of power, resources, and opportunities (Marmot et al., 2008). Applied to Timor-Leste, this perspective supports interventions that combine clinical services with community engagement, health education, accessible screening, and wider efforts to improve the conditions that make healthy living possible.

Equity should consequently be assessed not only by whether a service exists nationally, but also by whether people with different geographical, social, and economic circumstances can use that service effectively. A health programme may be formally available to everyone while remaining difficult to access for communities facing the

greatest barriers. Measuring this difference is essential for designing an inclusive response to chronic disease.

1.3.5. Primary Health Care and the Continuity of Chronic Disease Management

Primary health care is a central platform for addressing the long-term nature of NCDs. Its role extends beyond providing an initial consultation: it includes health promotion, risk assessment, early diagnosis, treatment, follow-up, referral, and coordination with other services. A strong primary health care system can help prevent avoidable complications and reduce unnecessary dependence on hospital-based treatment (WHO, 2019).

For Timor-Leste, strengthening this platform means ensuring that essential chronic disease services are incorporated into routine care rather than treated as separate activities available only through specialised programmes. Opportunities to measure blood pressure, assess diabetes risk, provide tobacco-cessation advice, and communicate practical information about nutrition and physical activity can be integrated into appropriate health encounters. Such integration may help identify risk earlier and make preventive care more accessible.

However, integration requires more than adding new tasks to an already demanding clinical workload. Health professionals need suitable training, clear clinical protocols, essential equipment, reliable medicine supplies, functioning referral pathways, and access to supervision. Facilities also need appropriate information systems to identify patients requiring follow-up and to monitor whether treatment is effective. Without these supporting conditions, a policy commitment to integrated care may not translate into consistent services.

Continuity is particularly important for people who live with more than one chronic condition. A patient may require management of hypertension, diabetes, and cardiovascular risk simultaneously. If each condition is addressed through disconnected services, care may become fragmented, confusing, and inefficient. A person-centred approach instead considers the individual's overall health status, treatment needs, family circumstances, and preferences.

The quality of a health system should therefore be judged not only by its ability to deliver individual clinical interventions but also by its capacity to provide safe, effective, timely, equitable, and coordinated care. Kruk et al. (2018) emphasize the importance of high-quality health systems that improve population health rather than merely

expand service coverage. For chronic disease management, this distinction is fundamental: access to a clinic does not automatically guarantee accurate diagnosis, appropriate treatment, or sustained improvement in health.

A practical national approach should connect community-level prevention with facility-based diagnosis and treatment, referral to higher levels of care when necessary, and follow-up after referral. Such continuity can help ensure that patients do not become disconnected from the health system as their needs change over time.

1.3.6. National Priorities: Prevention, Equity, and Sustainable Development

The development of an effective NCD response requires clear national priorities. Timor-Leste's National Health Sector Strategic Plan 2011–2030 provides an important policy context for strengthening health services and improving population health (Ministry of Health, Timor-Leste, 2011). These broader objectives should be connected to specific, measurable actions for chronic disease prevention and management.

First, prevention should be treated as a long-term national investment. Tobacco use, harmful alcohol consumption, unhealthy diets, physical inactivity, and exposure to air pollution contribute to preventable health risks. Policies addressing these factors can complement clinical care by reducing exposure across entire populations. WHO's global action plan for the prevention and control of NCDs highlights the need for coordinated action across health systems and wider public policy (WHO, 2013).

Second, early detection should be linked to treatment capacity. Screening programmes may identify individuals at elevated risk, but screening alone cannot ensure improved health outcomes. A successful programme requires confirmatory assessment where appropriate, communication of results, access to treatment, and continued monitoring. Expanding screening without planning for follow-up risks creating a gap between identifying a health problem and responding to it.

Third, essential medicines and diagnostic services must be accessible and reliable. For chronic diseases, interruptions in medication supply, inadequate monitoring, or delays in referral can undermine treatment over time. Procurement, distribution, workforce planning, and service organisation should therefore be considered integral parts of disease prevention and control rather than merely administrative concerns.

Fourth, national strategies should incorporate equity into their objectives and indicators. National averages may conceal differences between age groups, sexes,

geographical areas, and socioeconomic groups. Disaggregated information can help determine whether improvements reach people who face the greatest barriers. This is particularly important when resources are limited and decisions must balance competing health priorities.

Fifth, community participation should be recognised as a necessary component of sustainable prevention. Families, community leaders, schools, civil society organisations, and local health workers can contribute to health education and the development of locally appropriate interventions. Community participation should not, however, become a substitute for the government's responsibility to provide adequately resourced and equitable health services.

Finally, NCD prevention should be understood within the broader framework of sustainable development and universal health coverage. The United Nations' 2030 Agenda for Sustainable Development connects health improvement with poverty reduction, education, equality, and inclusive development (United Nations, 2015). For Timor-Leste, reducing preventable chronic disease can help protect household well-being, strengthen human capacity, and support the country's long-term social and economic aspirations.

1.3.7. Strengthening Research, Surveillance, and Accountability

A sustainable response to chronic diseases requires a national capacity to generate, interpret, and apply health evidence. Surveillance should not be limited to collecting information about risk factors; it should also support the evaluation of prevention programmes, the monitoring of treatment outcomes, and the identification of inequalities in access to care. Reliable information enables policymakers to assess whether investments are improving population health and whether the benefits are reaching the communities that need them most.

Timor-Leste would benefit from stronger connections between population surveys, routine health information systems, mortality registration, facility records, and locally conducted research. These sources should be used together while recognising their different strengths and limitations. Data quality, consistent definitions, appropriate protection of personal information, and regular reporting are essential to ensure that findings can guide decisions responsibly.

Research should also investigate questions that national statistics alone cannot fully answer. These include why some individuals delay seeking care, what barriers prevent

patients from maintaining treatment, how families manage the financial and practical consequences of chronic illness, and which community-based interventions are feasible in different geographical settings. Research conducted in partnership with local institutions and communities can help ensure that recommendations reflect Timor-Leste's circumstances rather than simply reproduce strategies developed elsewhere.

Accountability is equally important. National goals should be translated into measurable indicators, clear institutional responsibilities, and periodic reviews of progress. Relevant indicators may include the availability of essential medicines, the proportion of eligible adults receiving appropriate screening, continuity of treatment, control of selected risk factors, avoidable hospital admissions, and differences in service coverage between population groups. The selection of indicators should reflect the quality and availability of national data, with baseline values established before progress is assessed.

International estimates, including those produced through the Global Burden of Disease study, can provide useful comparative perspectives on mortality, disability, and population health loss (IHME, 2024). Nevertheless, international comparisons should complement rather than replace national evidence. The strongest policy decisions emerge when global knowledge is critically assessed alongside local epidemiology, professional experience, community perspectives, and the practical capacity of health services.

1.3.8. Concluding Perspective: Building a Healthier Future for Timor-Leste

Chronic diseases and NCDs present Timor-Leste with a challenge that extends beyond clinical medicine. Their prevention and management are connected to demographic change, social inequality, geographical access, national development, and the capacity of institutions to provide sustained and equitable care. The coexistence of communicable diseases and chronic conditions makes it necessary to strengthen the health system without abandoning existing public health responsibilities.

The country's response should be guided by reliable national evidence, strong primary health care, accessible essential medicines, a competent health workforce, community participation, and policies that address preventable risk factors. It should also recognise that prevention begins long before a disease becomes clinically apparent and that effective treatment depends on continuity rather than isolated encounters.

For a young nation, these priorities carry particular significance. Investment in prevention and early management can help protect the health and productive capacity of current and future generations. Conversely, delays in developing effective chronic disease services may allow avoidable complications to accumulate, increasing demands on families, health facilities, and public resources.

The central policy question is therefore not whether Timor-Leste can afford to strengthen its response to chronic diseases, but how it can organise its available resources to achieve the greatest and fairest improvements in population health. Answering this question requires an integrated approach that combines epidemiological evidence, social understanding, health system planning, and long-term political commitment.

This chapter has established the conceptual and contextual foundations for examining chronic diseases in Timor-Leste. The next chapter moves from the broad landscape of epidemiological change to the specific disease patterns, risk factors, and population-level determinants that shape the national burden.

CHAPTER 2. EPIDEMIOLOGY, RISK FACTORS, AND DISEASE PATTERNS

2.1. Cardiovascular Diseases, Diabetes, Cancer, and Chronic Respiratory Diseases

2.1.1. Introduction: Understanding the Major NCD Groups

Noncommunicable diseases comprise a broad range of conditions that generally require long-term prevention, treatment, and management. Among them, cardiovascular diseases, diabetes, cancer, and chronic respiratory diseases are widely recognised as four major groups responsible for a substantial proportion of the global burden of NCD-related illness and premature death (World Health Organization [WHO], 2024). Although these conditions differ in their biological mechanisms and clinical manifestations, they share several important characteristics: their development is often influenced by multiple risk factors, their consequences may accumulate over many years, and their management frequently requires sustained interaction between individuals and health services.

Understanding these diseases is essential for developing an effective response in Timor-Leste. Each condition presents distinct clinical requirements, yet all depend on several common health system capacities, including prevention, early recognition, accurate diagnosis, access to essential medicines, appropriate referral, and continuity of care. A fragmented approach that addresses each disease independently may overlook opportunities to integrate services and reduce the burden on patients, families, and health facilities.

Epidemiological analysis also requires distinguishing between disease occurrence, risk-factor exposure, disability, and mortality. The presence of a risk factor does not mean that every exposed individual will develop a particular disease. Similarly, a high number of diagnosed cases may reflect both a genuine disease burden and improvements in screening or case detection. Reliable interpretation therefore requires attention to the definitions, methods, and limitations of the available evidence.

Global comparative studies provide a broad understanding of the scale and distribution of NCDs, while national surveys and health-service records help explain how these conditions affect particular populations (GBD 2019 Diseases and Injuries Collaborators, 2020; Roth et al., 2020). For Timor-Leste, these sources should be interpreted together with locally collected evidence, including the 2023 national STEPS

survey, while avoiding the assumption that global averages precisely represent national disease patterns (Ministry of Health, Timor-Leste, 2023).

This section examines the four major NCD groups, focusing on their clinical characteristics, epidemiological significance, risk factors, and implications for health services in Timor-Leste.

2.1.2. Cardiovascular Diseases: A Major Challenge for Prevention and Long-Term Care

Cardiovascular diseases (CVDs) refer to disorders of the heart and blood vessels. Major forms include ischaemic heart disease, cerebrovascular disease such as stroke, hypertensive heart disease, and other conditions affecting cardiovascular function. Their development is influenced by a combination of biological, behavioural, environmental, and social factors. High blood pressure, abnormal blood lipids, tobacco use, diabetes, unhealthy diets, physical inactivity, and harmful alcohol consumption are among the important modifiable risk factors associated with cardiovascular disease.

Cardiovascular diseases are especially significant because they may cause sudden, life-threatening events while also producing long-term disability. A heart attack or stroke can lead to lasting functional limitations, dependence on family caregivers, and reduced participation in education, employment, and community life. Their consequences therefore extend beyond survival and must be assessed in terms of disability, quality of life, and the resources required for continuing care.

The epidemiological importance of cardiovascular disease is reflected in global comparative research. Roth et al. (2020) examined the worldwide burden of cardiovascular diseases between 1990 and 2019, documenting the substantial role of these conditions in global mortality and morbidity. Such findings underline the importance of combining primary prevention with effective acute treatment and long-term secondary prevention.

Hypertension is a particularly important entry point for cardiovascular prevention because it can remain undiagnosed for years while increasing the risk of stroke, heart disease, kidney disease, and other complications. Regular blood pressure measurement can identify elevated readings, but diagnosis requires appropriate clinical assessment. Effective management then depends on counselling, lifestyle

support, access to suitable medication when indicated, monitoring, and treatment adjustment.

For Timor-Leste, cardiovascular prevention should be integrated into routine primary health care rather than reserved exclusively for hospital settings. Blood pressure assessment, identification of other cardiovascular risk factors, patient education, and appropriate follow-up can help detect risk before serious complications develop. However, implementation must be supported by trained personnel, functioning equipment, reliable medicine supplies, and clear referral pathways.

The management of acute cardiovascular events also requires health system preparedness. Suspected heart attacks and strokes demand timely recognition, urgent assessment, and access to appropriate treatment. Where specialised diagnostic and treatment services are concentrated in particular facilities, transport and referral delays may influence outcomes. These challenges make coordination between community-level services, primary care facilities, and hospitals an important part of cardiovascular disease control.

Cardiovascular disease should therefore be approached as a continuum of prevention and care: reducing risk before disease develops, identifying early disease, responding effectively to acute events, preventing recurrence, and supporting rehabilitation. This integrated perspective is particularly relevant in settings where resources must be allocated across several competing health priorities.

2.1.3. Diabetes Mellitus: A Growing Challenge for Lifelong Health

Diabetes mellitus is a chronic metabolic condition characterised by elevated blood glucose levels resulting from problems with insulin production, insulin action, or both. The principal forms include type 1 diabetes, type 2 diabetes, and gestational diabetes. Although their causes and clinical characteristics differ, all forms require appropriate recognition and management to reduce the risk of complications.

Type 1 diabetes generally results from autoimmune destruction of the pancreatic cells responsible for insulin production and requires lifelong insulin treatment. Type 2 diabetes involves a combination of insulin resistance and progressive impairment of insulin secretion. It is associated with multiple influences, including genetic susceptibility, age, body composition, physical activity, dietary patterns, and wider social and environmental conditions. Gestational diabetes is identified during

pregnancy and has implications for both maternal and child health, including future risk of type 2 diabetes.

Diabetes is clinically significant because persistent hyperglycaemia can damage blood vessels and nerves over time. Potential complications include cardiovascular disease, chronic kidney disease, diabetic retinopathy, peripheral neuropathy, and diabetic foot problems. Some complications may progress without obvious symptoms during their early stages, making regular assessment and appropriate follow-up important components of care.

The burden of diabetes cannot be understood solely through the number of people diagnosed. Epidemiological assessment should also consider undiagnosed disease, access to diagnostic services, the availability and affordability of medicines, the quality of glucose monitoring, and the occurrence of preventable complications. A population may have individuals living with diabetes who remain unaware of their condition because screening opportunities are limited or symptoms have not prompted them to seek care. Consequently, reported diagnoses may not capture the full underlying burden.

Diabetes management requires a sustained partnership between health professionals and patients. Depending on the type of diabetes and the individual's clinical circumstances, care may include dietary guidance, physical activity, glucose monitoring, oral medicines, insulin, blood pressure management, and assessment of cardiovascular and kidney risks. Education should help patients understand their condition and recognise situations requiring urgent medical attention. Treatment plans must also account for individual resources, cultural practices, household circumstances, and the practical availability of recommended foods and medicines.

For Timor-Leste, diabetes prevention and management should be incorporated into broader primary health care development. Screening should be directed according to appropriate clinical guidance and risk assessment, with clear pathways for confirming diagnoses and initiating treatment. Once diabetes is identified, patients need reliable access to follow-up and complication screening rather than occasional consultations without continuity.

A further priority is to avoid presenting diabetes as the inevitable result of personal behaviour. Although diet and physical activity influence type 2 diabetes risk, these behaviours are shaped by food availability, household income, working conditions,

education, and the environments in which people live. Prevention is more likely to be equitable when individual counselling is accompanied by efforts to improve access to healthy food, opportunities for physical activity, and accessible health information.

Diabetes also demonstrates why chronic disease services must be coordinated. People with diabetes frequently require assessment of blood pressure, cardiovascular risk, kidney function, and other potential complications. Integrating these services can improve the overall quality of care and reduce unnecessary duplication of appointments and procedures. Global Burden of Disease research provides a comparative framework for examining diabetes and its contribution to mortality and disability, but national estimates should be interpreted alongside Timor-Leste's own surveillance and clinical data (GBD 2019 Diseases and Injuries Collaborators, 2020; Institute for Health Metrics and Evaluation [IHME], 2024).

2.1.4. Cancer: Early Detection, Diagnosis, and Continuity of Treatment

Cancer is a group of diseases characterised by the abnormal growth and spread of cells that can invade surrounding tissues or spread to other parts of the body. The term includes many distinct diseases with different causes, clinical presentations, treatment options, and prognoses. Cancer should therefore not be treated as a single epidemiological entity when planning prevention, screening, diagnosis, or treatment services.

Cancer risk is influenced by several factors, including age, inherited susceptibility, tobacco use, alcohol consumption, certain infections, exposure to carcinogens, and other environmental and behavioural influences. Some cancers can be prevented through risk reduction and vaccination against relevant infections. Others may be detected earlier through evidence-based screening or prompt investigation of symptoms. The appropriate approach depends on the specific cancer, the available evidence, the characteristics of the population, and the capacity of the health system to provide timely follow-up.

A critical distinction must be made between screening and early diagnosis. Screening generally involves testing people who do not have symptoms to identify those who may need further assessment. Early diagnosis involves recognising possible disease in people who present with symptoms and ensuring that they receive timely investigation. Both approaches can contribute to better outcomes when appropriately implemented, but screening is not automatically beneficial simply because a test is available. It

requires suitable selection criteria, diagnostic confirmation, referral, treatment, and monitoring.

The consequences of cancer can be extensive. Patients may experience pain, fatigue, nutritional difficulties, psychological distress, and loss of functional capacity. Treatment can involve surgery, radiotherapy, systemic medicines, or combinations of these approaches, depending on the diagnosis and stage of disease. Where specialised services are limited or concentrated geographically, the process of obtaining a diagnosis and completing treatment may involve substantial logistical and financial challenges for patients and their families.

In Timor-Leste, cancer control should be developed through a realistic assessment of national needs and service capacity. Important priorities include improving awareness of warning signs, strengthening the diagnostic pathway, ensuring appropriate referral, and establishing reliable systems for recording diagnoses and outcomes. National planning should identify which services can be delivered locally, which require referral to specialised facilities, and how patients can be supported throughout the process.

Prevention and early detection should be informed by evidence rather than by an assumption that all cancers can be addressed through the same intervention. For example, vaccination against human papillomavirus can contribute to cervical cancer prevention, while tobacco control is important for reducing the risk of several cancers and other chronic diseases. Cervical cancer screening and treatment of precancerous lesions, where feasible and supported by appropriate follow-up, can also form part of an organised prevention strategy.

The epidemiology of cancer depends heavily on the quality of population data. Reliable estimates of incidence, mortality, stage at diagnosis, and survival require consistent recording and, where possible, population-based cancer registration. Hospital records alone may not represent all people with cancer, particularly those who cannot reach diagnostic services or who receive care outside the reporting system. Where data are incomplete, the limitations should be explicitly acknowledged rather than interpreting the absence of recorded cases as evidence of low disease occurrence.

Cancer control also requires attention to dignity and supportive care. When cure is not possible or treatment cannot eliminate all symptoms, appropriate pain relief, psychological support, rehabilitation, and palliative care remain essential. A health

system's responsibility extends beyond attempts to prolong life; it also includes reducing suffering and supporting patients and families throughout serious illness.

2.1.5. Chronic Respiratory Diseases: Persistent Illness and Environmental Risk

Chronic respiratory diseases affect the airways and other structures of the respiratory system. Major conditions include chronic obstructive pulmonary disease (COPD) and asthma, although other chronic respiratory disorders also contribute to long-term illness and disability. These conditions differ in their underlying mechanisms and treatment requirements, but both may interfere with daily activities, sleep, work, and participation in family and community life.

COPD is commonly associated with persistent respiratory symptoms and airflow limitation. Tobacco smoking is an important risk factor, while exposure to household and outdoor air pollution, occupational hazards, and other harmful airborne substances can also contribute to respiratory disease. Asthma involves chronic airway inflammation and variable respiratory symptoms, including wheezing, breathlessness, chest tightness, and coughing. Symptoms may be triggered or aggravated by allergens, respiratory infections, smoke, air pollution, and other individual or environmental factors.

The relationship between respiratory health and the environment is especially important for public health planning. Exposure to smoke from household fuels, tobacco, traffic, and other pollution sources may affect respiratory health, although the level of risk varies according to the source, intensity, duration, and individual susceptibility. Prevention therefore requires both clinical management and attention to the environmental conditions that influence exposure.

Chronic respiratory diseases may be under-recognised when persistent cough, wheezing, or breathlessness is attributed to temporary illness without appropriate assessment. Accurate diagnosis is important because respiratory symptoms can have multiple causes and treatment differs between conditions. Depending on the clinical presentation, assessment may require a detailed history, physical examination, and appropriate lung-function testing where available.

Management varies according to the disease and its severity. It may include avoidance of relevant triggers, smoking cessation support, inhaled medicines, patient education, vaccination where indicated, and an action plan for worsening symptoms. For COPD, selected patients may also benefit from pulmonary rehabilitation and other

interventions based on clinical assessment. For asthma, appropriate use of inhaled corticosteroid-containing treatment is important in reducing the risk of severe exacerbations; treatment should follow current clinical guidance and individual needs.

For Timor-Leste, improving respiratory care requires attention to the availability of appropriate diagnostic equipment, trained personnel, essential inhaled medicines, and patient education. Inhaler technique is particularly important because incorrect use can reduce the amount of medicine reaching the airways. Patients also need to understand how to use prescribed treatment, recognise deterioration, and seek urgent care when symptoms become severe.

The burden of chronic respiratory disease should be evaluated through more than hospital admissions. Many people may live with persistent symptoms that limit daily functioning without receiving a formal diagnosis. Population surveys, clinical records, and environmental information can help establish a more complete picture of respiratory health and identify opportunities for prevention.

Respiratory disease also illustrates the interconnected nature of NCD risk factors. Tobacco control can reduce respiratory risk while also helping prevent cardiovascular disease and several cancers. Measures to improve air quality may provide benefits across multiple health outcomes. These connections strengthen the case for coordinated policies that address common risks rather than designing separate interventions for every disease category (WHO, 2021).

2.1.6. Shared Risk Factors and the Interconnected Nature of Major NCDs

Although cardiovascular diseases, diabetes, cancer, and chronic respiratory diseases have different clinical characteristics, their risk factors overlap considerably. Tobacco use, harmful alcohol consumption, unhealthy dietary patterns, physical inactivity, and air pollution contribute to several forms of NCDs. Metabolic factors, including elevated blood pressure, high blood glucose, and abnormal blood lipids, further increase the risk of particular diseases and complications (GBD 2019 Risk Factors Collaborators, 2020; WHO, 2023a).

This overlap has important implications for health policy. If each disease is addressed through a separate programme, health services may duplicate activities, patients may need to attend multiple appointments, and opportunities for coordinated prevention may be missed. An integrated approach can address several risks during the same

appropriate health encounter, provided that screening and treatment decisions follow clinical guidance.

For example, tobacco cessation can contribute to the prevention of cardiovascular disease, chronic respiratory disease, and several cancers. Healthy dietary patterns and physical activity can help reduce the risk of type 2 diabetes and cardiovascular disease while supporting broader health and well-being. Measures that reduce exposure to harmful air pollution can benefit respiratory and cardiovascular health. These shared benefits make prevention potentially more efficient when interventions are planned across disease categories.

However, shared risk factors do not mean that all NCDs can be prevented in the same way. Genetic susceptibility, age, infections, occupational exposures, and other condition-specific factors remain important. Public health programmes must combine broad population-level measures with interventions designed for particular diseases and vulnerable groups.

The relationship between risk factors and disease also involves interactions over time. An individual's risk may reflect cumulative exposure across childhood, adolescence, adulthood, and older age. This life-course perspective emphasises the importance of early prevention, sustained health promotion, and opportunities to reduce risk at different stages of life. It also recognises that people already living with chronic disease can benefit from risk reduction and appropriate treatment.

For Timor-Leste, a coordinated strategy could connect health education, risk assessment, clinical management, tobacco control, nutrition initiatives, and community-based prevention. Its effectiveness would depend on local evidence, available resources, and the ability of health services to deliver interventions consistently. National planning should therefore prioritise measures that are evidence-based, feasible, equitable, and capable of producing benefits across more than one health outcome.

2.1.7. Implications for Disease Surveillance and Health Service Planning in Timor-Leste

The four major NCD groups require different forms of clinical expertise, diagnostic capacity, and treatment, but they also depend on common health system foundations. These include an adequately trained workforce, reliable access to essential medicines,

functional diagnostic services, appropriate referral systems, and information systems capable of supporting long-term follow-up.

A key priority for Timor-Leste is to strengthen the connection between disease surveillance and service planning. National risk-factor surveys can identify patterns of exposure, while routine clinical data can help reveal the conditions treated within health facilities. Mortality records and population-based studies can contribute further information about premature death, disability, and gaps in care. When interpreted together, these sources can support more informed decisions about national priorities.

The evidence base should distinguish between four important questions:

- How common is the disease? This concerns prevalence, incidence, and the distribution of disease across population groups.
- What contributes to its occurrence? This concerns behavioural, metabolic, environmental, social, and other determinants.
- What are its consequences? This includes mortality, disability, reduced quality of life, and household and societal costs.
- How effectively does the health system respond? This includes timely diagnosis, treatment availability, continuity of care, and the prevention of avoidable complications.

These questions are related but cannot be answered by a single indicator. For example, a high number of recorded diagnoses may reflect a substantial disease burden, improved detection, or both. Similarly, low recorded prevalence may reflect genuinely lower occurrence or limited access to diagnosis. Interpretation must therefore consider the completeness and limitations of the available data.

The Timor-Leste STEPS survey provides an important foundation for analysing adult NCD risk factors, but risk-factor findings should not automatically be interpreted as direct estimates of diagnosed cardiovascular disease, diabetes, cancer, or chronic respiratory disease (Ministry of Health, Timor-Leste, 2023). Exact national estimates should be drawn from the relevant survey tables, disease registries, clinical records, or appropriately documented comparative studies. Where national data are incomplete, that limitation should be reported transparently.Service planning should also account for differences in the complexity of care. Blood pressure measurement and selected risk assessments can often be incorporated into primary health care,

whereas advanced cancer diagnostics, specialised cardiovascular procedures, and certain respiratory investigations may require referral to facilities with additional expertise and equipment. Effective planning must identify how these levels of care can work together, including how patients can access services when travel distances and other barriers are substantial.

An integrated approach should not be confused with providing every intervention at every facility. Rather, it involves establishing a coordinated network in which patients receive appropriate care at the right level, essential information follows them through the referral process, and responsibility for follow-up is clearly defined. This principle is consistent with broader evidence that health system quality depends on the effective delivery and coordination of care, not simply the existence of facilities or formal service coverage (Kruk et al., 2018).

2.1.8. Concluding Perspective

Cardiovascular diseases, diabetes, cancer, and chronic respiratory diseases represent distinct but interconnected challenges for population health. Their prevention and management require attention to specific disease mechanisms while recognising the shared influence of behavioural, metabolic, environmental, and social determinants.

For Timor-Leste, the implications extend from community-level prevention to the organisation of specialised clinical services. Reliable surveillance, early recognition, accessible diagnosis, appropriate treatment, and sustained follow-up are all essential to reducing avoidable illness and disability. These efforts should be supported by policies that address health inequalities and by investments that strengthen the capacity of primary health care to manage chronic conditions over time.

A sound national response must also remain evidence-based. International research offers valuable comparative perspectives, but national priorities should be informed by reliable local data and an explicit assessment of the country's service capacity. Where evidence is incomplete, strengthening surveillance and research should be part of the response rather than a reason to delay all preventive action.

Ultimately, the objective is not merely to treat four categories of disease independently. It is to build a health system capable of preventing avoidable risks, identifying illness earlier, providing effective and continuous care, and protecting the dignity and well-being of people living with chronic conditions.

2.2. Epidemiological Measures, Risk Distribution, and Population-Level Determinants

2.2.1. Introduction: Why Epidemiological Measurement Matters

Epidemiology provides the scientific foundation for understanding how diseases are distributed across populations, which factors influence their occurrence, and how prevention and treatment strategies can improve health outcomes. In the context of chronic diseases and noncommunicable diseases (NCDs), epidemiological analysis is particularly important because these conditions often develop over long periods, involve multiple interacting risk factors, and produce consequences that extend beyond mortality to disability, reduced quality of life, and economic hardship.For Timor-Leste, epidemiological measurement is essential for identifying national health priorities and allocating limited resources according to population needs. Decision-makers require reliable information about how frequently chronic diseases occur, which groups experience the greatest risks, how risk factors vary across communities, and whether existing health services are reaching the people who need them. Without such information, policy decisions may rely excessively on assumptions or global averages that do not adequately reflect local circumstances.

Epidemiological indicators must also be interpreted carefully. Prevalence, incidence, mortality, case fatality, and risk-factor prevalence measure different aspects of population health. Each indicator answers a particular question, and none independently provides a complete account of disease burden. International comparative studies, such as those produced through the Global Burden of Disease (GBD) study, combine multiple sources and analytical methods to estimate mortality, morbidity, and population health loss across diseases and geographical settings (GBD 2019 Diseases and Injuries Collaborators, 2020).This section examines the principal epidemiological measures used in chronic disease research, the distribution of risk across population groups, the social and environmental determinants of disease, and the implications of these concepts for public health planning in Timor-Leste.

2.2.2. Prevalence and Incidence: Measuring the Distribution of Disease

Prevalence and incidence are two fundamental measures used to describe disease occurrence. Although they are related, they provide different information and should not be used interchangeably.

Prevalence refers to the proportion of a population that has a particular disease at a specified point in time or during a defined period. It includes people with previously diagnosed disease who continue to live with the condition, as well as newly identified cases that meet the relevant definition. Prevalence is particularly useful for chronic diseases because many affected individuals require ongoing treatment and monitoring for years.

For example, estimating the prevalence of diabetes can help health authorities understand the potential demand for glucose monitoring, medicines, clinical consultations, and complication screening. Similarly, the prevalence of hypertension can inform the planning of blood pressure assessment, counselling, treatment, and long-term follow-up. However, prevalence estimates depend on the population studied, the diagnostic criteria applied, and the completeness of case identification.

Incidence measures the occurrence of new cases in a population over a specified period. Depending on the study design, incidence may be expressed as the number of new cases arising in a population at risk during a period or as an incidence rate based on person-time of observation. Incidence is particularly useful for understanding the emergence of disease and evaluating whether preventive interventions are associated with changes in the occurrence of new cases.

The distinction is important for interpreting national health trends. A disease's prevalence can increase even when its incidence remains stable or declines, particularly when people survive longer with the condition. Conversely, prevalence may remain relatively low despite substantial incidence if affected individuals experience high mortality or if diagnosis and reporting are incomplete.

For chronic diseases, the relationship between incidence, duration, survival, and diagnosis is especially important. Improved treatment may extend survival and increase the number of people living with a disease, creating a greater need for long-term services even when the rate of new cases does not rise. Consequently, health planning should consider both the development of new disease and the number of people who require continuing care.

In Timor-Leste, the interpretation of prevalence and incidence must take into account the available data sources. Population surveys can estimate selected risk factors and, where appropriate measurements are included, indicators of undiagnosed disease. Routine clinical records can provide information about people who use health services,

but they may not represent everyone living with a condition. A sound assessment should therefore document the data source, population covered, case definition, and principal limitations of each estimate (Bonita et al., 2006).

2.2.3. Mortality, Case Fatality, and Premature Death

Mortality measures describe deaths in a population and provide important evidence about the consequences of disease. The crude mortality rate expresses the number of deaths during a specified period relative to the population. Cause-specific mortality focuses on deaths attributed to a particular disease, while age-specific mortality examines mortality within a defined age group.

Age standardisation is often necessary when comparing populations with different age structures. Because many chronic diseases become more common as people grow older, a population with a larger proportion of older adults may have a higher crude disease or mortality rate even if its age-specific risks are similar to those of a younger population. Age-standardised measures help reduce this distortion and support more meaningful comparisons, although they do not eliminate all differences in data quality or population circumstances.

Case fatality describes the proportion of people identified as having a particular disease or event who die from it within a specified period. It is distinct from population mortality because it focuses on outcomes among affected individuals rather than deaths in the entire population. For acute conditions such as stroke or heart attack, case fatality may be influenced by the severity of the event, the time taken to obtain care, the availability of effective treatment, and the quality of clinical management.

Premature mortality is also important in NCD analysis. Deaths occurring earlier than expected can have substantial consequences for individuals, families, and national development. When people die during their working years or while supporting dependent family members, the consequences may include lost income, disrupted caregiving, and increased financial vulnerability.

Reliable mortality analysis depends on accurate death registration and cause-of-death classification. Where registration is incomplete or medical certification is inconsistent, the recorded distribution of causes may not fully reflect the actual pattern of mortality. Modelled estimates can help address information gaps, but they introduce uncertainty and should be distinguished from directly observed national statistics (GBD 2021 Causes of Death Collaborators, 2024).

For Timor-Leste, strengthening mortality information would support more accurate assessments of the contribution of NCDs to premature death and help identify where improvements in prevention, emergency response, and continuing care may produce the greatest benefits.

2.2.4. Risk Factors and Population-Level Exposure

Risk factors are characteristics, behaviours, exposures, or circumstances associated with an increased probability of developing a disease or experiencing an adverse health outcome. In chronic disease epidemiology, risk factors are commonly classified into behavioural, metabolic, environmental, and social categories. These categories interact, and their influence may accumulate throughout an individual's life.

Behavioural risk factors include tobacco use, harmful alcohol consumption, physical inactivity, and unhealthy dietary patterns. Metabolic risk factors include elevated blood pressure, high blood glucose, abnormal blood lipids, and excess adiposity. Environmental exposures, such as air pollution, may contribute to respiratory and cardiovascular disease, while social determinants influence opportunities for prevention, diagnosis, and treatment.

The distinction between a risk factor and a disease is essential. Elevated blood pressure increases cardiovascular risk but is not equivalent to a stroke or heart attack. Similarly, a person may have an elevated risk of diabetes without meeting the diagnostic criteria for diabetes itself. Epidemiological reports should identify whether an indicator measures a behaviour, a biological condition, a diagnosed disease, or a clinical outcome.

Population-level risk-factor prevalence helps estimate the scale of prevention needs. If a substantial proportion of adults are exposed to a modifiable risk factor, even a moderate reduction in exposure may yield important public health benefits. Nevertheless, the magnitude of benefit depends on the strength of the association, the distribution of exposure, the effectiveness of interventions, and the time required for health outcomes to change.

The GBD 2019 Risk Factors Collaborators (2020) provide a framework for examining the contribution of major risk factors to disease burden across populations. Such analyses can inform priority setting, but their estimates depend on available evidence, methodological assumptions, and the comparability of underlying data.

In Timor-Leste, the 2023 STEPS survey provides an important foundation for understanding selected adult NCD risk factors (Ministry of Health, Timor-Leste, 2023). Its findings should be used with attention to the survey's definitions, sampling design, age range, and measurement procedures. Exact prevalence estimates should be taken from the original report rather than inferred from global patterns or general descriptions of the national health situation.

2.2.5. The Distribution of Risk Across Population Groups

Population averages are useful for describing broad trends, but they can conceal substantial differences between groups. Epidemiological analysis should therefore examine how disease and risk factors are distributed according to age, sex, geographical location, socioeconomic circumstances, education, occupation, and other relevant characteristics.

Age is a major determinant of chronic disease patterns. The probability of developing many NCDs increases with age, while the effects of earlier exposures may become evident only after many years. However, younger adults may also experience important risk factors or develop chronic conditions. Prevention should therefore not be restricted to older people, and clinical assessment should reflect individual risk rather than age alone.

Sex and gender may influence disease patterns through biological differences, social roles, occupational exposures, health-related behaviour, and access to services. Epidemiological studies should distinguish biological factors from socially determined influences where possible. They should also avoid assuming that all members of a group share the same risk profile.

Geography affects exposure and access to care. Urban and rural populations may differ in their food environments, opportunities for physical activity, occupational circumstances, exposure to pollution, and proximity to health facilities. These differences should be investigated empirically rather than assumed to follow a single pattern. Rural residence does not automatically imply greater risk for every NCD, just as urban residence does not guarantee better access to effective care.

Socioeconomic position can influence both disease occurrence and disease outcomes. People with fewer financial resources may have less access to nutritious food, preventive services, diagnostic tests, and regular treatment. Once a chronic condition

develops, transport expenses, lost working time, and continuing treatment costs may further increase household vulnerability.

These dimensions may interact. For example, a person living in a geographically isolated community with limited financial resources may face several overlapping barriers to obtaining care. Analysing only one characteristic at a time may therefore underestimate the complexity of health inequality.

For Timor-Leste, disaggregated epidemiological analysis can help identify whether particular communities or population groups face greater exposure to risk factors or greater barriers to services. Such analysis should be based on reliable data and should respect privacy, especially when subgroup sizes are small. Its purpose is not to label communities as inherently unhealthy, but to identify where additional support and more accessible services may be needed.

2.2.6. Social Determinants and the Life-Course Perspective

Chronic diseases do not develop in isolation from social conditions. Income, education, housing, employment, food security, family circumstances, and access to health care can influence exposure to risk factors and the ability to prevent or manage illness. These factors also shape the resources available to individuals when a disease is diagnosed.

The social determinants of health framework emphasises that health inequalities arise partly from the distribution of opportunities and resources within society (Marmot et al., 2008). This perspective challenges explanations that attribute chronic disease exclusively to personal choices. Behaviour matters, but choices are made within environments that can either support or constrain healthier living.

A life-course perspective further explains why prevention must begin early. Nutrition, physical activity, tobacco exposure, and other influences during childhood and adolescence may shape health risks later in life. During adulthood, work patterns, household responsibilities, food access, and environmental exposures continue to influence health. In older age, the accumulation of previous exposures may interact with frailty, multiple chronic conditions, and changing care needs.

This perspective supports a combination of universal prevention and targeted assistance. Population-wide measures can reduce exposure to common risks, while targeted interventions can support people facing particularly high risks or substantial

barriers to care. The appropriate balance depends on national epidemiological evidence, service capacity, and the resources available.

In Timor-Leste, applying a life-course approach means linking health promotion with maternal and child health services, schools, community programmes, workplaces, and primary health care. It also means recognising that people already living with chronic disease need effective management and protection from avoidable complications. Prevention and treatment are complementary responsibilities rather than competing alternatives.

2.2.7. Interpreting Evidence: Data Quality, Uncertainty, and Comparability

Epidemiological measures are only as reliable as the data and methods used to produce them. National surveys, routine health records, hospital data, disease registries, mortality statistics, and international estimates differ in their purposes and limitations. Understanding these differences is essential before drawing conclusions about disease trends or comparing Timor-Leste with other countries.

Population surveys can estimate the prevalence of selected risk factors among the population covered by their sampling design. However, survey results may be affected by non-response, measurement error, recall bias, and limitations in the range of variables collected. A survey focused on adults within a particular age range should not automatically be treated as representative of children, adolescents, or older adults outside that range.

Routine health information systems provide information about people who access services, but their completeness may depend on facility reporting, diagnostic capacity, and patterns of service utilisation. If access to diagnosis is limited, recorded disease prevalence may underestimate the true number of people affected. Conversely, improvements in case detection may increase recorded diagnoses even when underlying disease occurrence has not changed substantially.

International estimates can help fill some information gaps and facilitate comparisons across countries and years. Nevertheless, modelled estimates may combine national observations with statistical assumptions and information from other settings. They should therefore be presented as estimates, with appropriate attribution and attention to uncertainty, rather than as direct measurements of every disease in Timor-Leste (Institute for Health Metrics and Evaluation, 2024).

Comparisons over time require particular care. Changes in case definitions, survey methods, diagnostic technology, population structure, or reporting completeness can affect apparent trends. Before concluding that a disease has increased or declined, analysts should determine whether the indicators being compared are sufficiently consistent.

A transparent epidemiological assessment should identify the source of each estimate, its year, the population covered, the indicator definition, and any important limitations. Where uncertainty is substantial, that uncertainty should be acknowledged explicitly. Responsible interpretation strengthens rather than weakens the usefulness of health evidence.

2.2.8. Implications for Health Planning and Priority Setting in Timor-Leste

Epidemiological measurement should ultimately support decisions that improve population health. For Timor-Leste, the practical task is to translate information about disease occurrence and risk distribution into feasible and equitable prevention and treatment strategies.

First, national priorities should be informed by both disease burden and the potential to prevent avoidable illness. Conditions associated with substantial premature mortality, disability, or preventable complications may warrant particular attention. However, decisions must also consider intervention effectiveness, implementation costs, workforce requirements, and the consequences of diverting resources from other essential health services.

Second, planning should distinguish population-level prevention from clinical care. Measures that reduce exposure to tobacco, harmful alcohol consumption, unhealthy diets, physical inactivity, and air pollution can benefit many people. Clinical services are needed to identify and manage individuals with elevated risk or established disease. These approaches reinforce one another and should be coordinated.

Third, data collection should be connected to operational decisions. If surveillance identifies a risk factor as a priority, health authorities should consider whether appropriate prevention and treatment services are available, whether communities can access them, and how progress will be measured. Collecting data without a clear process for applying the findings limits its value.Fourth, equity should be incorporated into the evaluation of programmes. Improvements in national averages may conceal persistent gaps between geographical areas or socioeconomic groups. Monitoring

should therefore assess not only overall coverage but also whether services reach people who face greater barriers.

Finally, epidemiological evidence should be reviewed periodically. Disease patterns, risk factors, technologies, and health system capacity change over time. Regular analysis enables policymakers to revise priorities, identify emerging needs, and evaluate whether interventions are achieving their intended outcomes.The goal is not to pursue perfect information before taking action. Rather, it is to make the best possible decisions using available evidence, clearly identify remaining uncertainties, and strengthen the information systems needed for continuous improvement.

2.2.9. Concluding Perspective

Epidemiological measurement provides an essential foundation for understanding chronic diseases and NCDs. Prevalence and incidence describe disease occurrence; mortality and case fatality illuminate fatal outcomes; risk-factor indicators reveal opportunities for prevention; and disaggregated analysis helps identify inequalities between population groups. These measures become most useful when interpreted together and supported by transparent methods.

For Timor-Leste, the challenge is to build a reliable evidence base that connects national surveillance with health service planning, community needs, and the evaluation of interventions. The 2023 STEPS survey, routine health information, mortality data, and appropriately interpreted international estimates can each contribute to this task. Their limitations must remain visible, and numerical claims should be grounded in the relevant source.

Ultimately, epidemiology is not merely the technical measurement of disease. It is a practical means of identifying avoidable suffering, directing resources responsibly, and evaluating whether health policies are improving people's lives. Its value lies in helping ensure that prevention and care respond to the actual distribution of health needs rather than to assumptions about the population.

2.3. NCD Risk Factors and Health Inequalities in Timor-Leste

2.3.1. Introduction: Understanding Risk Within the Timor-Leste Context

Noncommunicable diseases (NCDs) emerge through complex interactions between individual characteristics, behavioural patterns, biological processes, environmental exposures, and social conditions. Understanding these interactions is particularly

important for Timor-Leste, where health policy must respond to chronic disease risks while continuing to address other essential population health needs. An effective national strategy requires more than identifying risk factors in isolation; it must examine why these risks occur, how they are distributed, and what determines people's ability to prevent or manage their consequences.

Major modifiable NCD risk factors include tobacco use, harmful alcohol consumption, unhealthy dietary patterns, physical inactivity, elevated blood pressure, abnormal blood glucose, and abnormal blood lipids. Exposure to air pollution and other environmental hazards can also contribute to particular chronic conditions. These risks often coexist, increasing the likelihood of cardiovascular disease, diabetes, cancer, and chronic respiratory disease (GBD 2019 Risk Factors Collaborators, 2020; World Health Organization [WHO], 2023a).

However, risk is not distributed uniformly throughout a population. Differences in income, education, employment, geography, food availability, health literacy, and access to clinical services can influence both disease development and disease outcomes. People facing several disadvantages may encounter greater difficulty obtaining preventive care, receiving an early diagnosis, or maintaining long-term treatment.

For Timor-Leste, the 2023 STEPS survey provides an important national source for understanding selected adult NCD risk factors. Its results should be interpreted according to the survey's methods, definitions, and population coverage rather than assumed to represent every age group or every aspect of the national disease burden (Ministry of Health, Timor-Leste, 2023).

This section examines the principal risk factors and their relationship with health inequality, with particular attention to the implications for prevention, primary health care, and national policy.

2.3.2. Tobacco Use and the Prevention of Avoidable Disease

Tobacco use is a major preventable risk factor for several NCDs. Exposure to tobacco smoke is associated with cardiovascular disease, chronic respiratory disease, and multiple forms of cancer. The health consequences are not limited to people who smoke: exposure to second-hand smoke can also harm the health of others, including children and adults who do not use tobacco themselves.

Tobacco-related harm develops through several biological pathways. Harmful substances in tobacco smoke can damage blood vessels, contribute to inflammation, increase cardiovascular risk, and injure respiratory tissues. The cumulative nature of exposure means that the benefits of prevention extend across the life course. Preventing initiation among young people and supporting cessation among current users are therefore complementary priorities.

An effective tobacco-control strategy requires more than communicating that smoking is harmful. Public policy can influence tobacco exposure through taxation, restrictions on advertising and promotion, smoke-free environments, health warnings, enforcement of relevant legislation, and access to cessation support. WHO identifies comprehensive tobacco control as an important component of NCD prevention (WHO, 2020b).

In Timor-Leste, tobacco prevention should be informed by reliable national evidence about the prevalence and distribution of tobacco use. Where survey findings identify particular population groups with higher exposure, interventions can be designed to reach those groups without assuming that all communities face identical risks. Research should also consider the social and commercial circumstances that influence tobacco use, particularly among young people.

Primary health care facilities can contribute by asking patients about tobacco use, providing brief advice, supporting cessation attempts, and referring individuals to appropriate services where available. Such interventions should be integrated into routine care rather than offered only when patients present with respiratory or cardiovascular symptoms.

Tobacco control also illustrates the importance of policy coherence. Public health education is unlikely to achieve its full potential if commercial practices, social environments, and weak enforcement continue to encourage tobacco use. Coordinated action across health services, education, regulatory institutions, and communities can help reduce exposure and protect future generations.

The relevant policy objective is not to stigmatise individuals who use tobacco. It is to reduce preventable harm, support people who wish to stop, protect others from second-hand smoke, and create environments that make tobacco-free living more achievable.

2.3.3. Alcohol Consumption and Related Health Risks

Alcohol consumption can contribute to several adverse health outcomes, including liver disease, selected cancers, cardiovascular conditions, injuries, and social harms. The level and pattern of consumption, individual characteristics, and the wider circumstances in which alcohol is consumed influence the nature and magnitude of these risks. Harmful alcohol use may also interact with other NCD risk factors, including tobacco use, unhealthy dietary patterns, and difficulties maintaining treatment for existing chronic conditions (WHO, 2014a).

The public health implications extend beyond individual clinical outcomes. Alcohol-related harm can affect family relationships, household finances, workplace participation, and community safety. Where harmful consumption contributes to injuries or other acute events, it may also increase demand for emergency and hospital services. These consequences demonstrate why alcohol-related harm should be considered within a broad population health framework.

For Timor-Leste, national policy should be grounded in reliable evidence about alcohol consumption patterns and associated harms. Survey data can help describe reported consumption, while clinical and other relevant records may contribute information about alcohol-related illness and injury. These sources should be interpreted carefully because self-reported behaviour may be affected by recall, social expectations, and differences in how survey questions are understood.

Prevention strategies can include public education, appropriate regulation of availability and marketing, measures to reduce harmful consumption, and accessible support for individuals experiencing alcohol-related problems. The selection and implementation of specific interventions should reflect national law, available resources, local evidence, and the needs of affected communities.

Health professionals can contribute through respectful communication, appropriate screening when indicated, brief interventions, and referral to suitable services. Such care should protect confidentiality and avoid treating alcohol-related problems solely as moral failures. Individuals may need clinical support, counselling, or other assistance to reduce harm, and services should respond to those needs without stigma.

Policy should also recognise that the consequences of alcohol consumption are not limited to the person drinking. Protecting children, reducing injury, supporting families,

and preventing violence are relevant considerations within a comprehensive response. The aim is to reduce avoidable harm while ensuring that prevention and care remain evidence-based, proportionate, and respectful of human dignity.

2.3.4. Unhealthy Diets, Food Environments, and Nutritional Risk

Dietary patterns influence the risk of cardiovascular disease, type 2 diabetes, certain cancers, and other chronic conditions. Diets high in sodium, free sugars, and unhealthy fats can contribute to elevated blood pressure, excess energy intake, and metabolic risk. Inadequate consumption of fruits, vegetables, and other nutritious foods may also reduce the quality of the overall diet. These relationships are influenced by the total dietary pattern rather than by a single food considered in isolation (WHO, 2023a).

Nutrition-related risks must be understood in the context of food availability, affordability, cultural practices, household resources, and changing food environments. Advice to consume healthier foods may be difficult to follow when households have limited purchasing power, nutritious options are not consistently available, or work and family responsibilities restrict the time available for food preparation. Health promotion that ignores these constraints may place responsibility on individuals without addressing the conditions that shape their choices.

Timor-Leste's nutrition priorities should also account for the possibility that undernutrition and diet-related NCD risks can coexist within the same population. Efforts to prevent obesity or excessive consumption of certain foods should not undermine the need to ensure adequate nutrition, particularly for children, pregnant women, and other groups with specific nutritional requirements. The goal is a healthy and balanced dietary pattern, not simply the reduction of body weight or total food intake.

Food environments influence dietary behaviour through prices, product availability, marketing, portion sizes, and the convenience of different choices. Public policy can support healthier diets by improving nutrition education, strengthening food and nutrition programmes, considering the nutritional quality of food provided in public institutions, and supporting access to nutritious foods. Measures should be adapted to local circumstances and evaluated for their feasibility and impact.

At the clinical level, dietary counselling should be individualised and practical. People with hypertension, diabetes, or cardiovascular disease may benefit from guidance that reflects their clinical condition, cultural food practices, financial circumstances, and

access to suitable foods. Recommendations should be clear enough to support everyday decisions and should avoid presenting healthy eating as an option available only to households with substantial financial resources.

For Timor-Leste, nutrition policy should therefore connect health education with wider food security and development priorities. Such coordination can help prevent chronic disease while supporting the nutritional needs of the population across the life course.

2.3.5. Physical Inactivity and the Built Environment

Regular physical activity contributes to cardiovascular health, metabolic function, musculoskeletal strength, and psychological well-being. Inadequate physical activity is associated with an increased risk of several chronic diseases, including cardiovascular disease and type 2 diabetes. It can also contribute to reduced functional capacity over time. WHO guidance emphasises the importance of physical activity as a component of population health promotion (WHO, 2010a).

Physical activity should not be understood exclusively as participation in organised sport or structured exercise. Walking, active travel, household tasks, occupational activity, recreation, and other forms of movement may contribute to overall activity levels. The health benefits depend on the type, duration, intensity, and frequency of activity, as well as individual health status and functional ability.

Opportunities for physical activity are shaped by the environments in which people live and work. Safe walking routes, accessible public spaces, school facilities, working conditions, transport systems, and community infrastructure can influence how easily people incorporate movement into daily life. Conversely, unsafe roads, limited recreational space, long working hours, disability-related barriers, or concerns about personal safety may restrict participation.

In Timor-Leste, prevention strategies should be designed around locally appropriate opportunities rather than assuming that every community requires the same intervention. Community walking initiatives, school-based physical activity, accessible recreation, and health promotion through primary care may be relevant options, depending on local needs and available resources. These initiatives should be assessed for safety, accessibility, cultural acceptability, and sustainability.

Clinical advice should also reflect the needs of people already living with chronic disease. Physical activity recommendations may require adaptation for individuals with

cardiovascular complications, mobility limitations, or other health conditions. Where appropriate, health professionals should help patients identify safe and realistic ways to increase activity rather than provide general instructions without considering their circumstances.

An equitable approach to physical activity promotion combines individual education with supportive environments. It recognises that healthier behaviour is more achievable when communities have practical opportunities to adopt it.

2.3.6. Metabolic Risk Factors: Hypertension, Elevated Blood Glucose, and Abnormal Blood Lipids

Metabolic risk factors are important links between behavioural and environmental exposures and the development of chronic disease. Elevated blood pressure, high blood glucose, and abnormal blood lipid levels can increase the risk of cardiovascular disease and other complications. Excess adiposity is also associated with increased risk of several metabolic conditions, although its relationship with health varies across individuals and populations.

Hypertension is a major cardiovascular risk factor because it can remain undetected for long periods while contributing to damage to blood vessels and organs. Regular measurement and appropriate clinical assessment are therefore important. However, a single elevated reading should not automatically be treated as a definitive diagnosis in every circumstance. Diagnosis and management should follow appropriate clinical protocols.

Elevated blood glucose may indicate increased risk of diabetes or the presence of diabetes, depending on the test used and the clinical context. Appropriate testing and interpretation are required to distinguish these possibilities. Early identification can create opportunities for preventive intervention and timely treatment, but the value of detection depends on whether individuals can access follow-up care.

Abnormal blood lipid levels, including elevated levels of certain atherogenic lipoproteins, contribute to cardiovascular risk. Their assessment and management should be considered alongside other risk factors rather than interpreted in isolation. For individuals with multiple risk factors, integrated cardiovascular risk assessment can help guide preventive decisions according to current clinical recommendations and available resources.

These factors often cluster within the same individual. A person may have hypertension, elevated blood glucose, and abnormal blood lipids simultaneously, increasing the importance of coordinated assessment. Treating each risk factor through disconnected services can make care more complicated and place additional demands on patients. Integrated primary health care can provide a more coherent approach to assessment, counselling, treatment, and monitoring.

For Timor-Leste, the practical priority is to ensure that risk-factor detection is connected to appropriate clinical action. Health facilities need reliable equipment, trained personnel, standardised procedures, access to essential medicines, and systems for following patients over time. Where resources are limited, screening and treatment strategies should be prioritised according to clinical guidance, population needs, and the capacity to deliver effective follow-up.

The presence of a metabolic risk factor should not be interpreted as evidence of personal failure. Biological susceptibility, family history, food environments, stress, income, and access to preventive care may all influence risk. Prevention is most effective when individual support is combined with policies and services that make healthy living and appropriate treatment more accessible.

2.3.7. Air Pollution, Household Environments, and Occupational Exposures

Environmental exposures are important determinants of chronic disease because people encounter them through the air they breathe, the places where they live and work, and the conditions in which they perform everyday activities. Air pollution is associated with adverse respiratory and cardiovascular outcomes, while some occupational exposures can contribute to chronic respiratory disease and cancer. The nature of the risk depends on the pollutant or hazard, the intensity and duration of exposure, and the susceptibility of the exposed individual (WHO, 2021).

Household air pollution may arise from the use of polluting fuels and cooking technologies in inadequately ventilated spaces. Outdoor air pollution can originate from transport, industrial activities, waste burning, and other sources, depending on local conditions. Occupational exposures may include dust, fumes, chemicals, and other hazardous substances. These risks require appropriate assessment rather than assumptions about the exposure experienced by every household or worker.

In Timor-Leste, environmental health assessment should investigate relevant exposure patterns and identify where preventive action is feasible. Research may need

to consider household cooking practices, ventilation, urban air quality, occupational conditions, and the availability of safer alternatives. Findings should inform interventions appropriate to the exposure identified, rather than applying a single solution to all environmental risks.

Environmental prevention is particularly important because individuals may have limited control over the conditions to which they are exposed. Advice to avoid smoke or hazardous substances may be insufficient when households lack access to cleaner energy or workers cannot readily change their working conditions. Effective prevention may therefore require coordination between health authorities, environmental agencies, labour institutions, local governments, and communities.

Health professionals can contribute by identifying relevant exposure histories, recognising possible occupational or environmental causes of symptoms, and providing advice suited to individual circumstances. Public health surveillance can also help identify patterns that warrant further investigation. Where evidence is incomplete, targeted research and environmental monitoring can improve the basis for policy decisions.

Reducing harmful environmental exposure can produce benefits across multiple NCD categories. Improvements in air quality may support respiratory and cardiovascular health, while safer occupational environments can reduce the risk of selected chronic conditions. Environmental prevention should therefore be integrated into the wider national strategy for chronic disease control.

2.3.8. Health Inequalities: Why Risk and Outcomes Differ Between Communities

Health inequalities arise when health risks, disease outcomes, or access to effective care differ systematically between population groups. Some differences may reflect biological characteristics or age distributions, while others are shaped by social and economic conditions that could be changed through public policy. The distinction is important because not every difference represents an avoidable injustice, but many persistent inequalities reflect unequal opportunities to achieve good health.

In Timor-Leste, geographical accessibility is an important consideration in assessing health service equity. People living far from facilities may face transport difficulties, longer travel times, and additional costs when seeking routine consultations. These challenges are particularly consequential for chronic diseases because care frequently

requires repeated visits rather than a single encounter. A service that is accessible once may still be difficult to use consistently over months or years.

Financial circumstances can create further barriers. Although direct consultation costs may vary, patients and families may incur expenses related to transport, food, accommodation, and time away from employment or household responsibilities. These indirect costs can influence whether patients attend follow-up appointments, complete diagnostic investigations, or continue treatment. For households with limited resources, chronic disease can thus become both a health problem and a source of financial vulnerability.

Health literacy also matters. People need understandable information about risk factors, symptoms, treatment options, and the importance of continuity of care. However, health literacy should not be interpreted simply as an individual's ability to understand instructions. The communication practices of health services, language accessibility, cultural relevance, and the time available for consultation also influence whether information is understood and applied.

Gender, age, disability, and social circumstances may create additional barriers in particular contexts. These factors should be examined using local evidence rather than assumed to affect all communities in the same way. Effective policy requires identifying which groups experience specific obstacles and determining what changes would make services more accessible and responsive.

The social determinants of health framework emphasises that unequal health outcomes are linked to the distribution of social resources and opportunities (Marmot et al., 2008). Applied to NCD prevention, this perspective supports action on both immediate clinical needs and the broader conditions that influence risk. Health services remain essential, but they cannot independently eliminate all the social and environmental factors that contribute to chronic disease.

An equitable strategy therefore combines universal access to essential preventive and clinical services with additional support for populations facing greater barriers. This approach aims not to provide identical assistance regardless of need, but to ensure that people have a fair opportunity to benefit from effective prevention and treatment.

2.3.9. Community-Based Prevention and the Role of Primary Health Care

Community-based prevention can help connect national NCD priorities with the daily experiences of individuals and families. Community health workers, local health professionals, schools, civil society organisations, and community leaders may contribute to health education, risk awareness, referral, and follow-up, depending on their roles and training. Their involvement can improve communication between health services and communities, particularly where access to formal facilities is challenging.

Primary health care is central to this process because it can provide a continuing point of contact for prevention, diagnosis, treatment, and referral. Integrating NCD services into routine care can create opportunities to identify risk factors, provide counselling, initiate appropriate management, and monitor progress. WHO's primary health care and essential NCD intervention frameworks support strengthening these functions as part of broader health system development (WHO, 2019; WHO, 2020a).

Community engagement should be designed as a partnership rather than a one-way transfer of information. Programmes are more likely to be relevant when communities can identify local barriers, contribute to the design of health messages, and provide feedback about service accessibility. Local participation may also help health authorities understand why apparently straightforward recommendations are difficult to implement in particular circumstances.

Nevertheless, community-based prevention cannot replace clinical services or adequate health system resources. Identifying an individual with a possible chronic condition is only the first step. Appropriate assessment, diagnosis, treatment, and follow-up must be available. Community programmes also require clear responsibilities, adequate supervision, appropriate training, and referral arrangements so that their activities are safe and effective.

For Timor-Leste, an integrated approach could link community health promotion with facility-based risk assessment, medicine availability, referral, and monitoring. Such a system would allow prevention to extend beyond occasional campaigns and become part of ongoing population health practice. The exact model should be adapted to local geography, workforce capacity, community preferences, and available resources.

Evaluation is necessary to determine whether community interventions improve knowledge, increase access to preventive services, support treatment continuity, or reduce relevant risk factors. Programmes should be adjusted according to evidence

rather than continued solely because activities have been delivered. The ultimate measure of success is not the number of educational sessions conducted, but whether interventions produce meaningful and equitable improvements in health.

2.3.10. Policy Priorities for Reducing NCD Risk and Inequality

A national strategy for NCD prevention should combine population-wide measures with targeted interventions for people at increased risk. This requires coordination between health services and other sectors because many determinants of chronic disease lie outside the direct control of clinical care.

Several priorities are particularly relevant to Timor-Leste.

First, strengthen national surveillance and evidence use. The findings of the 2023 STEPS survey should be interpreted alongside routine health information, mortality data, and other relevant research. National reporting should clearly identify the year, population, definition, and source of each indicator. Where data gaps remain, the country should strengthen the systems needed to address them.

Second, integrate prevention into primary health care. Appropriate risk assessment, health promotion, early diagnosis, and follow-up should be incorporated into routine services. Integration must be supported by training, clinical protocols, equipment, medicines, referral pathways, and systems for tracking patients who require continuing care.

Third, reduce common modifiable risk factors. Evidence-based measures addressing tobacco use, harmful alcohol consumption, unhealthy diets, physical inactivity, and environmental exposure can reduce risk across several NCD categories. Policies should be adapted to national circumstances and implemented in ways that are feasible, equitable, and measurable.

Fourth, address barriers to continuous care. Geographic distance, indirect costs, communication difficulties, and limited service availability can interrupt long-term management. Health planning should identify the most significant barriers in different communities and consider appropriate responses, including improved referral coordination, outreach where feasible, and reliable access to essential medicines.

Fifth, incorporate equity into monitoring and accountability. National averages should be supplemented, where data permit, by analysis across age, sex, geographical area,

and socioeconomic circumstances. This can reveal whether interventions are reaching people who face the greatest barriers and whether disparities are narrowing over time.

Sixth, strengthen cooperation across sectors. Health authorities cannot independently determine all aspects of food availability, transport safety, environmental exposure, education, or working conditions. Coordination with relevant public institutions and community organisations can support prevention measures that extend beyond health facilities.

Finally, evaluate implementation and adapt policy. Programmes should have clear objectives, measurable indicators, and periodic review. Evaluation should examine not only whether activities were delivered, but whether they improved access, reduced exposure to risk, strengthened treatment continuity, or produced better health outcomes.

These priorities are consistent with the broader international emphasis on integrated NCD prevention, strong primary health care, and action on social determinants (WHO, 2013; WHO, 2017; WHO, 2023a). Their implementation in Timor-Leste should be guided by national evidence, institutional capacity, community participation, and a realistic assessment of available resources.

2.3.11. Concluding Perspective: Equity as a Foundation of NCD Prevention

The distribution of NCD risk factors in Timor-Leste must be understood through the interaction of individual behaviour, biological susceptibility, environmental exposure, and social conditions. Tobacco use, harmful alcohol consumption, unhealthy diets, physical inactivity, metabolic risks, and air pollution contribute to the development of chronic diseases, but their effects are shaped by the opportunities and constraints people experience throughout their lives.

An effective national response must therefore go beyond advising individuals to change their behaviour. It should make prevention practical, strengthen primary health care, improve access to appropriate diagnosis and treatment, and address the social and geographical barriers that prevent people from benefiting from available services. Community participation and intersectoral cooperation are essential to ensure that interventions respond to local needs and remain sustainable.

Reliable evidence is equally important. The Timor-Leste STEPS survey and other national data sources provide a foundation for understanding risk, but estimates must

be interpreted according to their methods and limitations. Transparent reporting, continued surveillance, and further locally relevant research can help identify priorities and evaluate whether interventions are reducing preventable illness and health inequalities.

Ultimately, the objective of NCD prevention is not simply to reduce statistical indicators. It is to protect the ability of individuals to live healthy and dignified lives, participate in their communities, support their families, and contribute to national development. For Timor-Leste, achieving this objective requires a health system that recognises differences in need and responds with effective, accessible, and equitable care.

CHAPTER 3. DISEASE BURDEN, DALYs, AND THE SOCIOECONOMIC CONSEQUENCES OF CHRONIC DISEASES

3.1. Measuring Disease Burden: Mortality, Disability, YLLs, YLDs, and DALYs

3.1.1. Introduction: Beyond Counting Diseases

Understanding the burden of chronic diseases requires more than counting the number of people diagnosed or the number of deaths attributed to a particular condition. Chronic diseases can affect individuals for many years, reduce their functional capacity, limit their participation in society, and create continuing demands on families and health services. A comprehensive assessment must therefore consider both premature mortality and the loss of healthy life associated with illness and disability.

Disease burden measurement provides a framework for bringing these different consequences into a more systematic analysis. It enables researchers and policymakers to compare health conditions, identify important contributors to population health loss, and assess where preventive and clinical interventions may produce substantial benefits. In this context, the disability-adjusted life year (DALY) is a particularly important measure because it combines years of life lost due to premature mortality with years lived with disability (Murray & Lopez, 1996; GBD 2019 Diseases and Injuries Collaborators, 2020).

For Timor-Leste, measuring disease burden can support decisions about the balance between prevention, early diagnosis, long-term treatment, rehabilitation, and other health priorities. A condition that causes relatively few deaths may still create substantial disability, while a condition associated with premature mortality may generate significant loss of potential years of life. Considering these outcomes together provides a broader understanding of population health needs.

However, disease burden indicators must be interpreted carefully. They do not independently explain why a disease occurs, whether health services are meeting patients' needs, or which intervention is most appropriate. Their value depends on the quality of the underlying evidence, the definitions used, and the policy questions they are intended to answer.

This section introduces the principal measures used in burden-of-disease analysis and explains their relevance to chronic disease research and health planning in Timor-Leste.

3.1.2. Mortality as an Indicator of Population Health Loss

Mortality remains one of the most important indicators in public health because it measures the occurrence of death within a population. Information about the number, causes, and age distribution of deaths can reveal the severity of particular health problems and identify opportunities to prevent avoidable loss of life.

Crude mortality rates describe the number of deaths relative to the size of a population during a specified period. Cause-specific mortality rates focus on deaths attributed to a particular disease, while age-specific rates describe mortality within particular age groups. These measures answer different questions and should be selected according to the purpose of the analysis.

When comparing populations, age standardisation is often necessary. Chronic diseases become more common with increasing age, so populations with different age structures may have different crude mortality rates even when their age-specific risks are similar. Age-standardised measures help improve comparability by adjusting for differences in population age composition.

Mortality analysis is especially important for cardiovascular diseases, cancer, diabetes-related complications, and chronic respiratory diseases. These conditions can cause premature death through different pathways, and their mortality patterns may be influenced by prevention, diagnostic capacity, treatment availability, and the quality of clinical care. Examining mortality by age, sex, disease category, and geographical area can help identify patterns that deserve further investigation.

Nevertheless, recorded mortality does not always provide a complete account of disease-related deaths. The accuracy of cause-of-death statistics depends on the completeness of death registration, the availability of medical certification, and the consistency of diagnostic and coding practices. Where these systems are incomplete, some deaths may be unregistered or assigned to broad or uncertain causes.

Global Burden of Disease research uses multiple sources and modelling methods to estimate cause-specific mortality where direct observations are incomplete or inconsistent (GBD 2021 Causes of Death Collaborators, 2024). Such estimates can support comparative analysis, but they should be distinguished from directly observed national statistics. For Timor-Leste, transparent reporting should identify whether mortality figures come from national registration systems, research studies, or modelled international estimates.

3.1.3. Years of Life Lost: Measuring Premature Mortality

The concept of years of life lost (YLLs) extends mortality analysis by considering not only how many people die but also how early those deaths occur. A death at a younger age generally represents a greater loss of potential years of life than a death occurring later in life. YLLs provide a systematic way to account for this difference.

In standard burden-of-disease analysis, YLLs are calculated by multiplying the number of deaths at a particular age by the standard remaining life expectancy at that age. The calculation is then summed across relevant ages to estimate the total years of life lost from a disease or cause of death.

Conceptually, the calculation is:

$$\text{YLLs} = \sum_{a} D_a \times L_a$$

where:

- \(D_a\) is the number of deaths at age \(a\).
- \(L_a\) is the standard remaining life expectancy at that age.

This formulation illustrates why the age distribution of deaths matters. Two conditions may cause the same number of deaths but produce different totals of YLLs if one causes deaths at substantially younger ages.

For chronic diseases, YLLs can help reveal the consequences of premature mortality from conditions that might otherwise be viewed primarily as diseases of older age. Cardiovascular disease, certain cancers, and diabetes-related complications can affect people during working and family-forming years, with consequences extending beyond the loss of life itself.

In Timor-Leste, YLL analysis could help assess the contribution of premature NCD mortality to national health loss. It may also inform comparisons between disease categories and support discussion about prevention, early diagnosis, and timely treatment. Such analysis requires reliable information about the number and age distribution of deaths, together with clearly documented assumptions about the standard life expectancy used.

YLLs should not, however, be interpreted as a direct measure of the quality of a particular health service. Premature mortality may reflect multiple influences, including disease incidence, social determinants, access to care, and the effectiveness of prevention and treatment. Further investigation is required to understand the mechanisms behind observed patterns and identify appropriate responses.

3.1.4. Years Lived with Disability: Recognising Non-Fatal Health Loss

Mortality alone cannot capture the full impact of chronic diseases. Many people live for years with conditions that cause pain, fatigue, reduced mobility, impaired vision, emotional distress, or limitations in their ability to perform everyday activities. Years lived with disability (YLDs) provide a way to quantify this non-fatal component of population health loss.

In burden-of-disease analysis, YLDs are generally calculated by combining the prevalence of a health condition or its relevant health states with disability weights that represent the severity of the associated health loss. Disability weights are derived using standardised methods to reflect the relative health impact of different conditions. They are not measures of a person's social worth, productivity, or dignity.

A simplified prevalence-based formulation is:

\[\text{YLDs}=\sum_s P_s \times DW_s \]

where:

- \(P_s\) represents the number of people living in health state \(s\).
- \(DW_s\) represents the disability weight assigned to that health state.

The calculation may distinguish between different severity levels within the same disease. For example, the effects of a mild condition may differ substantially from those of an advanced condition involving serious functional limitations. The precise methodology depends on the burden-of-disease framework and the data available.

YLDs are especially relevant to chronic disease analysis because survival does not necessarily mean full recovery. A person who survives a stroke may experience lasting difficulties with movement, speech, memory, or independent living. Someone with chronic respiratory disease may experience recurring breathlessness and reduced exercise tolerance. Diabetes may contribute to visual impairment, kidney disease, neuropathy, or other complications that affect daily functioning.

These consequences may require rehabilitation, assistive devices, long-term medication, home adaptations, and support from family members. They can also influence whether individuals are able to work, study, care for relatives, or participate fully in community life. YLDs help make these non-fatal consequences more visible in population-level analysis.

For Timor-Leste, disability measurement is important because an exclusive focus on deaths could underestimate the demands chronic diseases place on individuals, families, and health services. However, reliable estimation requires information about disease prevalence, severity, and the functional consequences experienced by affected populations. Where national data are limited, international modelled estimates may provide context, but their limitations should be clearly acknowledged (GBD 2019 Diseases and Injuries Collaborators, 2020).

It is also important to distinguish the statistical concept of disability used in burden-of-disease analysis from the broader lived experience of disability. A disability weight summarises a health-state valuation for comparative analysis; it does not fully capture discrimination, social exclusion, inaccessible environments, or the personal meaning of living with a chronic condition. These dimensions require complementary social, qualitative, and service-access research.

3.1.5. Disability-Adjusted Life Years: Combining Mortality and Disability

The disability-adjusted life year is a summary measure that combines premature mortality and non-fatal health loss into a common metric. One DALY represents one year of healthy life lost relative to the reference standard used in the analysis. The measure enables researchers to assess the combined contribution of fatal and non-fatal outcomes associated with different diseases.

The standard relationship is:

$$\text{DALYs}=\text{YLLs}+\text{YLDs}$$

In this formulation, YLLs represent healthy years lost because of premature death, while YLDs represent healthy years lost while living with a health condition or disability. Adding these components provides an estimate of total health loss attributable to a disease or risk factor within the defined population and period (Murray & Lopez, 1996; GBD 2019 Diseases and Injuries Collaborators, 2020).

Consider a hypothetical example. Suppose a population experiences 800 YLLs and 350 YLDs from a particular chronic disease during a given year. The estimated total burden would be 1,150 DALYs. These figures are illustrative only and do not represent observed data from Timor-Leste.

DALYs allow comparisons across conditions that produce different combinations of mortality and disability. One disease may cause substantial premature death, while another may result in fewer deaths but a large number of people living with long-term impairment. By combining these outcomes, DALYs help prevent the analysis from focusing exclusively on one dimension of disease burden.

The measure is widely used in comparative global health research, including the Global Burden of Disease study, to examine differences across diseases, risk factors, populations, and time periods (GBD 2019 Diseases and Injuries Collaborators, 2020; GBD 2021 Diseases and Injuries Collaborators, 2024).

Nevertheless, DALYs have limitations. They depend on methodological choices, the quality of mortality and morbidity data, and assumptions used to estimate disability. They also cannot fully describe the social and economic consequences of disease, the distribution of suffering within households, or the fairness of access to care. A DALY estimate is therefore a valuable epidemiological indicator, not a complete measure of human well-being.

For Timor-Leste, DALYs can support priority setting when interpreted alongside national health objectives, intervention effectiveness, financial constraints, equity considerations, and community needs. The measure should inform decisions rather than determine them automatically.

3.1.6. Comparing Diseases Through DALYs and Related Indicators

One of the principal advantages of DALYs is that they make it possible to compare health loss across different disease categories using a common unit. However, comparisons must be designed carefully to ensure that the figures refer to compatible populations, time periods, and methods.

Several indicators are particularly useful:

- Total DALYs: The overall number of healthy years lost in a population. This helps describe the total scale of health loss associated with a disease.

- DALY rate: DALYs relative to population size, commonly expressed per 100,000 people. This supports comparisons between populations of different sizes.
- Age-standardised DALY rate: A rate adjusted to a standard age structure. This helps comparisons between populations with different demographic profiles.
- YLL and YLD contributions: Separate estimates showing whether total burden is driven more by premature mortality or non-fatal health loss.
- DALYs attributable to risk factors: Estimates of the portion of health loss associated with specified exposures, subject to the assumptions and methods used.

These indicators answer different questions. Total DALYs may be especially relevant when estimating the overall scale of services and support required. Age-standardised rates are often more appropriate when comparing underlying disease burden between populations with different age structures. The proportions attributable to YLLs and YLDs can help identify whether prevention of premature death, reduction of disability, or both require particular attention.

Risk-attributable DALYs require additional care. Estimating the burden attributable to tobacco use, high blood pressure, or another risk factor involves comparing observed or estimated health outcomes with a theoretical minimum-risk exposure distribution. The resulting estimates depend on evidence about the relationship between exposure and disease, the distribution of exposure, and the assumptions used in the model (GBD 2019 Risk Factors Collaborators, 2020).

A risk factor may contribute to several diseases, and the same disease may be associated with multiple interacting risks. Consequently, attributable burden estimates for different risk factors should not simply be added together without considering overlap and the specific methodology used. Otherwise, the total may be misinterpreted.

For national planning, these indicators are most useful when presented with clear definitions and appropriate interpretation. Policymakers should know whether a figure represents an absolute number, a population rate, an age-standardised estimate, or a modelled attribution to a particular risk factor.

3.1.7. DALYs and the Interpretation of Global Burden of Disease Estimates

The Global Burden of Disease (GBD) study provides a systematic framework for estimating mortality, non-fatal health loss, and risk-attributable disease burden across countries and territories. Its comparative approach is particularly useful where national data sources differ in completeness, measurement methods, or reporting practices. By combining available evidence with statistical modelling, the study produces estimates that can help researchers examine health trends and compare the relative contributions of different conditions (GBD 2019 Diseases and Injuries Collaborators, 2020; GBD 2021 Diseases and Injuries Collaborators, 2024).

However, the existence of an international estimate does not mean that the underlying national data are complete. Modelled estimates may be necessary precisely because direct observations are limited. Their interpretation should therefore consider uncertainty intervals, source data, methodological assumptions, and the degree to which the available evidence represents the population being studied.

Three distinctions are especially important.

First, observed data and modelled estimates are not identical. National surveys, death registrations, and clinical records contain information collected directly from particular populations or services. GBD estimates may integrate such information with other evidence and modelling procedures to produce comparable results. Both can be useful, but their different origins should be made explicit.

Second, absolute burden and comparative rates answer different questions. A country may have a relatively small total number of DALYs because its population is small, while still experiencing a substantial burden relative to its population size. Conversely, a large country may have a high total burden even when its population rate is lower. Interpretation should match the policy question being addressed.

Third, uncertainty must remain visible. An estimate should not be presented as an exact count when its value is derived from a model with a range of uncertainty. Comparing small differences between estimates may be misleading when their uncertainty ranges are broad or their underlying data are not equally reliable.

For Timor-Leste, international estimates can provide a useful starting point for understanding the relative contribution of chronic diseases to mortality and disability. Nevertheless, these estimates should be assessed alongside national surveillance,

mortality information, clinical records, and local research. Where differences occur between data sources, the appropriate response is to investigate their definitions, coverage, and methods rather than automatically assume that one source is correct in every respect.

The GBD framework is therefore best understood as a tool for structured comparison and priority setting, not as a substitute for strengthening national health information systems. Timor-Leste's long-term capacity to plan effective NCD services will depend on improving both the availability of reliable local data and the ability to interpret international evidence critically.

3.1.8. Using Disease Burden Measures to Set Health Priorities

Disease burden measures can support public health priority setting by identifying conditions associated with substantial mortality, disability, or total health loss. Yet the highest DALY estimate does not automatically identify the intervention that should receive the greatest share of national resources. Policy decisions must also consider the preventability of the burden, intervention effectiveness, implementation costs, health workforce capacity, and the distribution of benefits across population groups.

For example, a disease associated with a large number of DALYs may require complex specialist services that cannot be expanded rapidly. At the same time, population-level interventions addressing common risk factors may prevent health loss across several disease categories. A balanced strategy should therefore consider both the existing burden and the opportunities to reduce future burden.

For Timor-Leste, several complementary questions can guide this process:

1. Which chronic diseases account for substantial mortality, disability, or premature loss of healthy life?
2. Which modifiable risk factors contribute to these outcomes, and what evidence supports effective interventions?
3. Which interventions can be delivered through existing primary health care services, and which require additional diagnostic or specialist capacity?
4. Which population groups experience the greatest burden or face the most significant barriers to prevention and treatment?

5. What resources, workforce capabilities, and information systems are required to implement and evaluate the proposed interventions?

These questions connect epidemiological measurement with the practical realities of national health planning. They also help prevent a narrow focus on disease rankings from overlooking the feasibility and equity of potential responses.

DALYs should be used alongside other indicators, including disease prevalence, incidence, premature mortality, treatment coverage, medicine availability, financial protection, and patient-reported outcomes. A country may reduce mortality while many people continue to experience substantial disability. Alternatively, an intervention may improve access to care before a measurable reduction in DALYs becomes apparent. Monitoring multiple indicators can reveal these different dimensions of progress.

Equity must remain central to priority setting. A national programme may produce substantial aggregate benefits while leaving remote or disadvantaged communities with limited access. Decision-makers should therefore consider not only how much health loss an intervention may prevent but also who is likely to benefit and whether additional support is required to reach underserved populations.

In this way, burden-of-disease measurement becomes part of a wider decision-making framework that combines epidemiological evidence, clinical knowledge, economic analysis, ethical considerations, and national development priorities.

3.1.9. Limitations of Burden Measurement and the Importance of Human Experience

Despite their value, mortality, YLLs, YLDs, and DALYs cannot fully capture the experience of living with chronic disease. They provide structured indicators of population health loss, but they do not describe every aspect of pain, uncertainty, family responsibility, social exclusion, or the personal meaning of illness.

For example, two individuals with the same diagnosis and similar clinical indicators may experience very different circumstances. One may have access to regular treatment, transport, family support, and suitable living conditions. Another may face financial hardship, long travel distances, interrupted treatment, or limited assistance with daily activities. Standardised disease burden indicators may not fully reveal these differences.

Similarly, the impact of chronic disease extends to family members and informal caregivers. Relatives may provide transportation, financial assistance, medication support, and daily care. These responsibilities can reduce their available time for employment, education, and rest. Although some economic and social consequences can be measured through complementary methods, they are not fully represented by DALYs alone.

For this reason, quantitative epidemiology should be complemented by research on patient experiences, household costs, access to services, and the social determinants of health. Qualitative studies can help explain why people delay seeking care, what prevents treatment continuity, and how families adapt to long-term illness. Household surveys and health financing studies can provide further information about the economic burden of disease.

These complementary approaches are especially relevant to Timor-Leste, where geographical accessibility, household resources, and the availability of continuing services may influence the experience of chronic disease. Their contribution is not to replace epidemiological measurement but to make its interpretation more complete.

A humane approach to public health recognises that people are more than the statistical categories used to describe their illnesses. Disease burden measures are most valuable when they help identify preventable suffering and support decisions that improve the lives of individuals, families, and communities.

3.1.10. Concluding Perspective

Measuring the burden of chronic diseases requires a combination of indicators that capture premature mortality, non-fatal health loss, and the broader distribution of disease across populations. Mortality statistics describe deaths; YLLs account for the loss of potential years of life; YLDs estimate health loss associated with disability; and DALYs combine these components into a common measure of total healthy life lost.

These measures provide a valuable foundation for comparing diseases, identifying major contributors to population health loss, and supporting evidence-informed health planning. Their usefulness, however, depends on reliable data, transparent methods, and careful interpretation of uncertainty. They should be complemented by information about service coverage, social inequalities, household costs, and the lived experience of chronic illness.

For Timor-Leste, the priority is to connect disease burden analysis with practical decisions about prevention, primary health care, diagnosis, treatment, rehabilitation, and long-term follow-up. International estimates can provide important comparative insights, but national surveillance and research remain essential for understanding local needs and evaluating the effectiveness of policy responses.

Ultimately, the purpose of measuring disease burden is not merely to rank diseases or calculate years lost. It is to help identify where preventable mortality and disability can be reduced, where resources can achieve meaningful improvements, and how health services can respond more fairly to the needs of the population.

3.2 The Global and Timor-Leste Burden of NCDs: Comparative Analysis and Priority Setting

3.2.1 Introduction: From Global Evidence to National Priorities

Noncommunicable diseases (NCDs) represent a major challenge for health systems worldwide because of their long-term consequences for mortality, disability, health expenditure, and human development. Cardiovascular diseases, cancers, diabetes, and chronic respiratory diseases account for a substantial proportion of the global burden of illness. Their effects are intensified by population ageing, persistent exposure to preventable risk factors, social inequalities, and uneven access to timely and effective health care (World Health Organization [WHO], 2024; GBD 2021 Causes of Death Collaborators, 2024).

However, the global distribution of NCDs does not automatically explain the specific health priorities of every country. National disease burdens are shaped by demographic structures, historical conditions, living environments, socioeconomic development, patterns of health-related behaviour, and the capacity of health services to prevent, diagnose, and manage chronic conditions. For this reason, global estimates should provide a comparative framework rather than a substitute for national epidemiological assessment.

For Timor-Leste, understanding the burden of NCDs requires consideration of both the country's changing health profile and the continuing importance of infectious diseases, maternal and child health, malnutrition, and other public health priorities. The challenge is not simply to replace one set of health priorities with another. Rather, it is to develop

a health system capable of responding to overlapping needs while ensuring that the prevention and treatment of chronic diseases receive appropriate attention.

This section examines the global burden of NCDs, the interpretation of comparative disease estimates, the implications for Timor-Leste, and the principles that can guide national priority setting. It emphasizes that reliable decisions require a combination of epidemiological evidence, health system analysis, equity considerations, and an understanding of the circumstances in which people live.

3.2.2 The Global Burden of Noncommunicable Diseases

The global burden of NCDs reflects several interconnected processes. Many people are living longer, while changes in food systems, urban environments, employment patterns, transportation, tobacco exposure, alcohol consumption, and physical activity have influenced the distribution of risk factors. At the same time, advances in diagnosis and treatment have enabled many people to survive conditions that previously caused earlier death, increasing the need for long-term disease management and rehabilitation.

Global Burden of Disease analyses demonstrate the importance of examining both fatal and non-fatal health outcomes. Deaths reveal the consequences of diseases that lead to premature mortality, while disability estimates help describe the substantial health loss experienced by people who survive with chronic conditions. DALYs combine these dimensions and permit comparisons across diseases, populations, and periods (GBD 2019 Diseases and Injuries Collaborators, 2020; GBD 2021 Diseases and Injuries Collaborators, 2024).

Cardiovascular diseases remain a particularly important component of global NCD burden. Ischaemic heart disease and stroke can cause premature death, long-term disability, and substantial demand for emergency treatment, rehabilitation, and continuing clinical care. Their consequences are associated with multiple interacting factors, including elevated blood pressure, tobacco use, unhealthy dietary patterns, physical inactivity, diabetes, and environmental and social conditions (Roth et al., 2020; GBD 2019 Risk Factors Collaborators, 2020).

Diabetes also contributes to the global burden through both immediate and long-term complications. Although diabetes can often be managed through sustained clinical care and appropriate lifestyle support, inadequate control may increase the risk of cardiovascular disease, kidney disease, visual impairment, nerve damage, and lower-

limb complications. These outcomes create a need for regular monitoring, essential medicines, health education, and timely access to higher-level services.

Cancer presents a different but equally important set of challenges. The burden varies substantially according to cancer type, age, sex, exposure to risk factors, screening opportunities, and access to diagnosis and treatment. The consequences may include premature mortality, prolonged treatment, disability, psychological distress, and significant costs for households and health systems. Effective responses therefore require prevention, early recognition of warning signs, appropriate diagnostic pathways, treatment capacity, and palliative care when needed.

Chronic respiratory diseases, including chronic obstructive pulmonary disease and asthma, also contribute to non-fatal illness, premature mortality, and recurring demand for clinical services. Tobacco smoke, household and ambient air pollution, occupational exposures, and other environmental factors can influence respiratory health. Effective management depends on reducing exposure to preventable risks while ensuring access to diagnosis, appropriate medication, patient education, and follow-up care (WHO, 2021, 2024).

These disease categories should not be treated as entirely separate problems. Their risk factors frequently overlap, and a person may live with more than one chronic condition. This coexistence, commonly described as multimorbidity, can complicate diagnosis, increase treatment demands, and require coordination across multiple health services. Consequently, public health strategies that address shared risk factors and strengthen primary health care may offer benefits across several disease categories rather than focusing exclusively on one condition at a time (Atun et al., 2013; WHO, 2020a).

3.2.3 Interpreting Global Trends Without Overlooking National Differences

Global statistics are useful for identifying broad patterns, but their interpretation requires attention to the indicators being compared. The total number of deaths from a disease may increase as populations grow, even if the underlying age-specific risk remains stable or declines. Similarly, the number of people living with a chronic condition may increase because people survive longer, because diagnosis improves, or because the incidence of the disease rises. These explanations have different implications for policy.

Absolute numbers are particularly important for service planning. They help estimate the number of people who may require consultations, medicines, diagnostic investigations, rehabilitation, or continuing follow-up. Rates, by contrast, support comparisons between populations of different sizes. Age-standardized rates are especially useful when the age structures of the populations differ substantially, although they cannot eliminate all differences in data quality, socioeconomic conditions, or health system capacity.

The distinction between prevalence and incidence is also important. Prevalence measures the proportion of a population living with a condition, whereas incidence measures the occurrence of new cases over a defined period. A condition may have high prevalence because people survive for many years, even when the incidence is relatively stable. Conversely, a disease that causes rapid death may have a lower prevalence than expected from its incidence because affected individuals remain in the prevalent population for a shorter period.

These distinctions matter when comparing countries or assessing changes over time. An increase in recorded diabetes prevalence, for example, could reflect a genuine increase in disease occurrence, improved detection, changes in population age structure, or a combination of these factors. Without examining the underlying methods and data, it would be inappropriate to attribute the entire change to worsening population health.

The Global Burden of Disease framework helps address some of these challenges by applying standardized methods to available information. Nevertheless, estimates remain dependent on the quality, completeness, and comparability of the underlying evidence. The appropriate interpretation is therefore not that international estimates provide perfect measurements, but that they offer structured evidence that can be evaluated alongside national surveillance and local research (Institute for Health Metrics and Evaluation [IHME], 2024).

For Timor-Leste, this distinction is essential. International comparisons can help identify potentially important conditions and risk factors, but decisions about staffing, medicines, diagnostic services, and community interventions should also reflect local epidemiological evidence, service accessibility, and the practical needs of the population.

3.2.4 The NCD Burden in Timor-Leste: An Evolving Public Health Challenge

Timor-Leste's public health priorities must be understood within the context of its demographic, social, and health system development. The country continues to address a broad range of health needs, including infectious diseases, maternal and child health, nutrition, and the improvement of access to essential services. At the same time, NCD prevention and management require increasing attention as chronic conditions and their risk factors become important considerations for population health planning.

The country's National Health Sector Strategic Plan 2011–2030 provides a broader policy context for strengthening health services and improving population health. Its relevance to NCDs lies in the importance of accessible primary health care, prevention, health workforce development, referral systems, and the continuity of essential services (Ministry of Health, Timor-Leste, 2011). A comprehensive response to chronic disease should build on these health system foundations rather than establish disconnected services that are difficult to sustain.

The 2023 national STEPS survey is an important source of information on NCD-related risk factors among adults in Timor-Leste. STEPS surveys use a standardized approach to collecting information on behavioural and biological risk factors, helping countries identify prevention needs and monitor population-level risks. The findings can inform national discussions about tobacco use, alcohol consumption, dietary practices, physical activity, and relevant metabolic indicators, subject to the definitions and results reported in the survey (Ministry of Health, Timor-Leste, 2023).

However, risk-factor evidence should not be confused with direct measurements of total disease burden. A survey that identifies elevated blood pressure or other risk factors does not, by itself, establish the national number of deaths, DALYs, or cases attributable to each condition. Estimating those outcomes requires additional information, appropriate methods, and careful consideration of uncertainty. The distinction is important because a sound national assessment must connect risk-factor surveillance with disease occurrence, clinical outcomes, and the performance of health services.

Timor-Leste's geography and settlement patterns also influence the practical burden of chronic disease. People living in remote or rural communities may face greater difficulty reaching facilities that provide diagnostic testing, specialist assessment, or

continuing treatment. Travel time, transportation costs, medicine availability, and the need to attend repeated appointments can influence whether a person receives care consistently. These barriers can have serious consequences for conditions that require lifelong management rather than a single episode of treatment.

In addition, household resources, education, health literacy, food availability, employment conditions, and the built environment may influence both exposure to risk factors and the ability to respond to illness. These determinants mean that NCD prevention cannot be addressed exclusively through individual advice. It also requires supportive environments, accessible services, effective public policies, and community participation.

The central policy issue is therefore not whether Timor-Leste should prioritize chronic diseases at the expense of other health needs. It is how the country can respond to NCDs while continuing to strengthen essential services across the health system. This requires reliable data, realistic planning, coordinated prevention, and a commitment to ensuring that the benefits of health system development reach communities with different levels of access and resources.

3.2.5 Comparing Global and National Evidence

A meaningful comparison between global estimates and national information requires a structured approach. Different data sources may measure different outcomes, cover different populations, or refer to different years. A national risk-factor survey, a hospital information system, a mortality database, and a modelled international estimate should not be assumed to measure the same phenomenon.

The following framework illustrates how different sources can contribute to a more comprehensive assessment.

Evidence source	Main contribution	Key limitation to consider
National STEPS survey	Population-level information on selected NCD risk factors	Does not independently measure the full burden of disease or all clinical outcomes
Routine health service records	Information on diagnoses, consultations, treatment, and service use	May exclude people who cannot access or do not attend health facilities

Mortality registration and cause-of-death data	Evidence about deaths and their attributed causes	Completeness and diagnostic accuracy may vary
Disease-specific registries and clinical studies	Detailed information on selected diseases and outcomes	May not represent the entire national population
Global Burden of Disease estimates	Comparable estimates of deaths, YLLs, YLDs, DALYs, and risk attribution	Depend on available data, modelling assumptions, and statistical uncertainty
Household and qualitative research	Information on access barriers, financial pressures, and lived experience	Findings may be context-specific and not directly generalizable to the whole population

Source: Conceptual synthesis based on Bonita et al. (2006), GBD 2021 Diseases and Injuries Collaborators (2024), IHME (2024), and WHO (2023b).

The value of this framework is that each source contributes a different dimension of understanding. National surveys can identify important exposures; clinical records can reveal service utilization and diagnosed conditions; mortality information can clarify fatal outcomes; and international estimates can provide a comparative perspective. Qualitative research can then help explain why particular groups experience difficulties accessing care or maintaining treatment.

When evidence sources appear to disagree, the differences should be investigated rather than resolved by selecting whichever estimate appears most convenient. Researchers should examine the population covered, the period of observation, the case definition, the method of data collection, and the uncertainty associated with each result. Such scrutiny is necessary to avoid misleading conclusions about the scale or direction of change.

For Timor-Leste, strengthening the connections between these sources could improve both research and policy. Better integration of population surveys, routine health information, referral records, and mortality data would support more reliable assessment of disease patterns and service needs. At the same time, data systems must protect confidentiality and be designed to support legitimate public health purposes.

3.2.6 Identifying Priority Diseases and Risk Factors

Priority setting should begin with an assessment of the magnitude and distribution of health loss, but it should not end there. Decision-makers must also consider preventability, intervention effectiveness, implementation feasibility, equity, and the opportunity costs of using limited resources for one activity rather than another.

A disease may warrant urgent attention because it causes many premature deaths. Another may require priority because it produces extensive disability or because effective interventions are available but underused. A third may affect a smaller population but impose severe hardship on people who already experience substantial barriers to care. A comprehensive framework should be able to recognize all three situations.

In the context of Timor-Leste, several complementary priorities can guide the assessment.

First, shared risk-factor prevention can reduce exposure to determinants associated with multiple NCDs. Tobacco control, reduction of harmful alcohol use, healthier dietary environments, opportunities for physical activity, and measures to address relevant environmental exposures can contribute to the prevention of cardiovascular disease, diabetes, cancers, and chronic respiratory conditions. Population-level policies should be combined with culturally appropriate health education and practical support that enables people to adopt healthier behaviours (WHO, 2013, 2017).

Second, early detection and appropriate treatment are necessary to identify people who already have chronic conditions or elevated risk. Screening and testing should be linked to reliable follow-up, clinical assessment, and treatment. Detection alone is insufficient if people cannot obtain confirmatory tests, essential medicines, or continuing care. Programmes should therefore be evaluated according to their ability to improve health outcomes rather than simply the number of people screened.

Third, continuity of care is central to chronic disease management. Many conditions require regular monitoring, adherence support, adjustment of treatment, and referral when complications arise. Strengthening primary health care can help provide ongoing management closer to where people live, while clear referral pathways connect patients with diagnostic and specialist services when necessary. WHO's package of essential NCD interventions provides a framework for organizing selected essential services in primary care and resource-constrained settings (WHO, 2020a).

Fourth, equity-oriented service planning should identify populations whose needs may be obscured by national averages. Differences in access between urban and remote communities, between socioeconomic groups, and among people with different levels of health literacy should be considered when distributing staff, medicines, diagnostic resources, and outreach services. Equity does not mean providing identical services in every location; it means responding fairly to differences in need and barriers to access.

Finally, health system feasibility and sustainability must inform every proposed intervention. Strategies should account for workforce availability, medicine supply chains, diagnostic capacity, information systems, financing, and the ability of local services to maintain activities over time. A programme that depends on resources unavailable outside major centres may have limited national impact unless its implementation plan addresses those constraints.

Taken together, these priorities support a balanced approach in which disease burden estimates inform decisions, while the practical conditions required to deliver effective and equitable care determine how interventions should be implemented.

3.2.7 DALYs and Resource Allocation in a Resource-Constrained Health System

DALYs can support resource allocation by providing a common measure of health loss across diseases. They can help policymakers compare the contribution of premature mortality and disability to the overall burden and assess how that burden changes over time. However, DALYs should not be interpreted as a direct formula for distributing the national health budget.

Resource allocation involves ethical, economic, and operational considerations beyond the size of the estimated burden. An intervention may prevent a substantial number of DALYs but require infrastructure or specialist personnel that cannot be made available quickly. Another intervention may produce a smaller immediate reduction in measured health loss but be highly feasible, affordable, and effective in reaching underserved communities. Both considerations may be relevant to a balanced national strategy.

Cost-effectiveness analysis can complement burden estimates by examining the additional health benefits achieved relative to the resources required. Budget impact analysis can assess whether a proposed intervention is financially manageable within

the available health budget. Equity analysis can help determine who receives the benefits and whether the intervention reduces or widens existing inequalities. These approaches should be considered together, with explicit attention to assumptions and uncertainties.

For Timor-Leste, the implications are practical. Strengthening prevention and primary health care may help reduce the future need for costly treatment of avoidable complications. Reliable supplies of essential medicines and appropriate monitoring can support people living with hypertension and diabetes. Clear referral pathways can improve the management of patients who need more specialized assessment. Community outreach and health education can complement facility-based services, particularly where distance and transport costs make repeated visits difficult.

Nevertheless, the potential benefits of prevention should not be used to justify neglecting people who already have chronic diseases. Public health systems have responsibilities both to reduce future illness and to provide appropriate care for those currently affected. A strategy focused exclusively on long-term prevention may fail people with immediate treatment needs, while a strategy focused exclusively on hospital treatment may overlook opportunities to prevent avoidable disease and complications.

The appropriate balance is therefore a combination of population-level prevention, accessible primary care, timely referral, essential treatment, rehabilitation, and ongoing support. DALYs can inform this balance, but decisions must also reflect clinical effectiveness, feasibility, financial protection, and the ethical principle that people should not be denied essential care because of where they live or their socioeconomic circumstances.

3.2.8 Integrating NCDs into Universal Health Coverage

Universal health coverage (UHC) means that people can obtain the health services they need without suffering financial hardship. For chronic diseases, this requires more than the availability of isolated diagnostic or treatment services. It requires a functioning pathway through prevention, early detection, diagnosis, treatment, long-term follow-up, rehabilitation, and, where appropriate, palliative care. If one part of this pathway is inaccessible, the effectiveness of the entire system may be reduced.

The relationship between NCD management and UHC is especially important because chronic diseases often generate recurring costs for households and health services.

Patients may need regular consultations, laboratory tests, medicines, and transportation over many years. Even where services are formally available without direct fees, indirect expenses and the opportunity cost of time away from work or family responsibilities may create barriers to continuing care.

A people-centred approach should therefore examine not only whether services exist but whether individuals can use them consistently and receive care of adequate quality. High-quality health systems require competent health workers, reliable supplies, effective information systems, respectful interactions, and appropriate clinical processes. The existence of a health facility alone does not guarantee that patients will receive the services they need (Kruk et al., 2018).

For Timor-Leste, integrating NCD services into primary health care can help reduce fragmentation and strengthen continuity. Routine assessment of relevant risk factors, appropriate diagnosis, access to essential medicines, patient education, and scheduled follow-up can be incorporated into service delivery where resources and clinical guidelines permit. Health workers should also have clear criteria for referral and mechanisms for communicating information between community-based services and higher-level facilities.

This integration should be accompanied by appropriate financing and workforce planning. Health workers require training and supervision to identify and manage common chronic conditions, while facilities need reliable medicine procurement, diagnostic capacity, and information systems that support follow-up. Without these foundations, formal policies may not translate into consistent care.

Monitoring UHC should consequently include indicators of service coverage, quality, continuity, and financial protection. The WHO and World Bank framework for monitoring UHC emphasizes the importance of examining both service coverage and financial hardship rather than treating access as a single dimension (WHO & World Bank, 2023). For NCDs, these measures can help determine whether people are receiving effective care and whether the costs of chronic illness are undermining household well-being.

3.2.9 Strengthening Surveillance and Evidence for National Decision-Making

A sustainable response to chronic diseases depends on the ability to measure changing health needs and assess whether interventions are producing the intended results. Surveillance should therefore be understood as an ongoing public health

function rather than an occasional research activity. Periodic population surveys, routine health information, mortality data, disease registries, and targeted research each contribute to this function.

National risk-factor surveys, such as the Timor-Leste STEPS survey conducted in 2023, can establish important baselines for monitoring behavioural and biological risks. Repeated surveys using sufficiently comparable methods can help identify changes over time. However, survey results should be interpreted alongside clinical and mortality data because changes in risk-factor prevalence do not translate automatically into immediate changes in disease incidence or deaths (Ministry of Health, Timor-Leste, 2023).

Routine health information systems can contribute by recording diagnoses, treatment, follow-up, complications, and referral outcomes. To be useful, these systems require consistent definitions, adequate staff training, appropriate data quality checks, and clear procedures for using information in service planning. Data collection should not become an administrative burden disconnected from clinical care; it should help health workers and managers identify unmet needs and improve service delivery.

Disaggregation is particularly important. National averages may conceal substantial differences between municipalities, urban and rural settings, age groups, sexes, and socioeconomic groups. Where sample sizes and data quality permit, disaggregated information can reveal which populations face the greatest risk or the greatest barriers to care. Where the evidence is insufficient, this limitation should be stated openly and used to guide future data collection.

Research partnerships can strengthen national capacity, provided that they respond to locally defined priorities and contribute to sustainable skills development. Collaboration among the Ministry of Health, universities, health facilities, communities, and international partners can support epidemiological studies, health services research, economic evaluation, and implementation research. Such partnerships should promote appropriate data governance, transparent reporting, and the use of findings in national planning.

The objective is not to collect the largest possible volume of data. It is to develop a reliable evidence system that answers practical questions: Which populations are not receiving effective care? Which risk factors require stronger prevention? Where are essential medicines and diagnostic services unavailable? Which interventions are

feasible and produce meaningful benefits? How can progress be measured fairly across different communities?

A stronger evidence system would enable Timor-Leste to refine its NCD priorities over time and assess whether policies are reducing premature mortality, disability, and avoidable inequalities.

3.2.10 A Framework for NCD Priority Setting in Timor-Leste

Based on the preceding analysis, a practical framework for NCD priority setting should combine epidemiological magnitude with feasibility, equity, and long-term sustainability. The framework should not produce an automatic ranking that disregards national circumstances. Instead, it should make the reasoning behind policy decisions explicit and enable decision-makers to compare alternatives transparently.

The following dimensions provide a basis for structured assessment.

Priority dimension	Guiding question	Possible evidence
Magnitude of health loss	How much mortality, disability, or overall health loss is associated with the condition?	Mortality, incidence, prevalence, YLLs, YLDs, DALYs
Preventability and treatability	Are effective interventions available to reduce risk or improve outcomes?	Clinical guidelines, intervention reviews, WHO recommendations
Population equity	Which groups experience the greatest burden or barriers to care?	Disaggregated surveys, service records, qualitative research
Health system feasibility	Can the intervention be delivered reliably with available or realistically developable capacity?	Workforce, infrastructure, medicine supply, referral capacity

Financial sustainability	Can the intervention be funded without undermining other essential services?	Budget impact and economic evaluation
Continuity and quality	Can people receive appropriate care over the full course of the condition?	Follow-up, treatment availability, clinical outcomes, patient experience
Accountability	Can progress be monitored and the results used to improve policy?	Defined indicators, reporting schedules, review mechanisms

Source: Author's conceptual framework informed by WHO (2013, 2017, 2020a), Kruk et al. (2018), and WHO and World Bank (2023).

In practical terms, this framework could be applied through a recurring national review process. The review would examine available epidemiological evidence, identify major gaps, assess the feasibility of proposed interventions, consult health workers and communities, and agree on measurable priorities. The resulting decisions should identify responsibilities, resource requirements, timelines, and indicators for monitoring implementation.

Importantly, priority setting should be revisited as new evidence becomes available. A programme that initially appears feasible may encounter workforce or supply constraints during implementation. Conversely, improvements in primary health care or diagnostic capacity may create opportunities to expand services that were previously difficult to deliver. Regular review allows policy to respond to these developments without abandoning long-term strategic goals.

The framework should also maintain a distinction between national priorities and local implementation needs. A national strategy can establish common objectives and minimum service standards, while municipalities and health facilities adapt delivery arrangements to their populations, geography, workforce, and available infrastructure. Such flexibility can improve responsiveness while preserving consistency in clinical quality and public accountability.

Ultimately, the success of priority setting should be assessed by its effects on population health and people's ability to obtain appropriate care. A priority is not successful merely because it appears in a strategic document or receives a budget

allocation. Its value depends on whether implementation reduces avoidable illness and death, improves quality of life, and reaches people who would otherwise remain underserved.

3.2.11 Conclusion: Comparative Evidence as a Foundation for Equitable Action

Comparative analysis of global and national NCD burden provides an essential foundation for public health planning, but its value depends on careful interpretation. Global estimates identify broad patterns and support comparisons, while national surveys, routine health information, mortality records, and local research are needed to understand the circumstances and service needs of Timor-Leste.

The country's response to NCDs should recognize the combined importance of premature mortality, long-term disability, shared risk factors, and the social conditions that influence exposure and access to care. Cardiovascular diseases, diabetes, cancers, and chronic respiratory diseases require sustained prevention and management, supported by reliable information systems and effective primary health care. At the same time, these efforts must be integrated with other essential health priorities rather than treated as a separate programme isolated from the wider health system.

DALYs and related indicators can help decision-makers assess the magnitude of health loss, but they should be considered alongside intervention effectiveness, financial sustainability, service quality, and equity. In Timor-Leste, where geographical and socioeconomic circumstances may influence access to diagnosis and long-term treatment, the distribution of health benefits is as important as the national total.

A strong national strategy will therefore combine prevention, early detection, accessible treatment, continuity of care, rehabilitation, and evidence-based policy review. It will also strengthen the collection and use of reliable data so that decisions can be adjusted as population needs change. The ultimate purpose of comparative epidemiology is not merely to rank diseases, but to guide action that prevents avoidable suffering and improves the health and dignity of the population.

The next section examines how chronic diseases affect household finances, social relationships, productivity, and human development, extending the analysis from measured health loss to its broader socioeconomic consequences.

3.3 Economic, Social, and Human Development Consequences

3.3.1 Introduction: Chronic Diseases Beyond the Clinical Dimension

Chronic diseases are not merely medical conditions affecting individual bodies; they are also social and economic challenges that influence households, communities, institutions, and national development. Their consequences extend beyond diagnosis and treatment to include reduced functional capacity, loss of income, financial pressure, changes in family responsibilities, and unequal opportunities to participate in social and economic life. These effects may accumulate over many years, making chronic diseases a continuing concern for public health policy and human development.

The socioeconomic consequences of chronic diseases are closely connected to their clinical characteristics. Unlike many acute illnesses that may be resolved through a relatively short period of treatment, chronic conditions frequently require sustained monitoring, long-term medication, lifestyle adjustments, and repeated contact with health services. When complications arise, patients may also require hospitalization, rehabilitation, or specialized care. These demands can affect household expenditure and the time available for employment, education, and caregiving.

The relationship between chronic diseases and socioeconomic disadvantage is bidirectional. Poverty and limited access to resources may increase exposure to risk factors and reduce opportunities for early diagnosis, while the costs and consequences of illness may push households into deeper financial hardship. This interaction is consistent with evidence emphasizing the importance of social determinants in shaping health inequalities across populations (Marmot et al., 2008).

For Timor-Leste, understanding these consequences is particularly important because health outcomes are closely connected to household resources, access to services, employment opportunities, and geographical circumstances. An effective response to NCDs must therefore consider not only the clinical burden of disease but also how illness affects people's ability to maintain livelihoods, support families, and participate in community life.

This section examines the economic, social, and human development consequences of chronic diseases, with particular attention to household financial protection,

productivity, caregiving, inequality, and the implications for national development in Timor-Leste.

3.3.2 Direct Medical Costs and Household Financial Pressure

One of the most immediate economic consequences of chronic disease is the cost of obtaining and maintaining treatment. Depending on the condition, patients may need consultations, diagnostic tests, medication, hospitalization, rehabilitation, and periodic specialist assessment. These costs may continue for many years and can become difficult to manage when household income is irregular or when several family members require health care.

Direct medical costs include payments for health services, medicines, laboratory investigations, and other clinically related expenses. The actual financial burden depends on the availability of publicly funded services, the cost of medicines and diagnostics, the location of health facilities, and the extent to which patients must obtain care outside the public system. Even when consultation or treatment is provided without direct fees, families may still face substantial indirect costs.

In Timor-Leste, the financial consequences of chronic disease should therefore be assessed through the broader experience of accessing care rather than through official service charges alone. A household may need to pay for transportation, food during travel, accommodation near a referral facility, or additional support for accompanying relatives. If a patient must travel repeatedly for follow-up, these expenses can accumulate even when the clinical services themselves are affordable.

The timing of expenditure also matters. Chronic disease often generates recurring costs rather than a single payment. A household may initially manage the expenses associated with diagnosis but struggle to sustain regular treatment over time. Financial pressure may become more severe when complications require emergency care or hospitalization, particularly if the illness affects the person who provides the household's main income.

These patterns highlight the importance of financial protection as a central component of universal health coverage. The objective is not only to make services available but also to ensure that people can obtain the care they need without experiencing financial hardship. The WHO and World Bank monitoring framework emphasizes the relationship between service coverage and protection against health-related financial difficulties (WHO & World Bank, 2023).

For national policy, the implication is clear: chronic disease programmes should be assessed partly by whether patients can continue treatment over time. The availability of medicines, reliable follow-up, and accessible services may help reduce avoidable complications and limit the financial consequences associated with fragmented care.

3.3.3 Indirect Costs: Lost Income, Time, and Productivity

The economic burden of chronic disease extends beyond expenditure on medical services. Indirect costs arise when illness reduces a person's ability to work, attend school, perform household activities, or participate in other productive activities. These consequences may affect patients, caregivers, employers, and the wider economy.

A person living with cardiovascular disease, diabetes, cancer, or chronic respiratory illness may experience fatigue, pain, mobility limitations, visual impairment, or other symptoms that interfere with daily responsibilities. Depending on the severity of the condition, the person may reduce working hours, change occupations, take repeated leave, or stop working entirely. Even when employment continues, symptoms and treatment requirements may reduce productivity or make sustained participation more difficult.

The effects are not limited to formal employment. In communities where households depend on subsistence agriculture, small-scale trade, informal work, or unpaid domestic labour, illness can reduce the ability to produce food, maintain livelihoods, care for children, or support older relatives. Such losses may not appear in conventional employment statistics, yet they can have important consequences for household welfare.

Caregivers may also lose productive time when they accompany patients to appointments, assist with daily activities, or provide support following hospitalization. If care requires repeated travel or extended periods away from home, family members may need to reduce working hours or transfer responsibilities to other relatives. These arrangements can create additional pressures for households with limited financial reserves.

The economic consequences may accumulate gradually. A person with a chronic condition may continue working for several years but experience declining capacity as complications develop. Alternatively, an acute event such as a stroke may abruptly interrupt employment and create an immediate need for rehabilitation and long-term assistance. The nature and scale of productivity loss therefore vary according to

disease severity, age, occupational demands, treatment access, and the availability of social support.

For Timor-Leste, the assessment of indirect costs should recognize the diversity of livelihoods and the importance of unpaid work. A comprehensive analysis should include changes in household production, caregiving responsibilities, and participation in community life, rather than focusing exclusively on wages or formal employment. Such an approach would provide a more realistic picture of how chronic diseases influence economic security and social well-being.

3.3.4 Chronic Diseases and Household Poverty

Chronic diseases can contribute to poverty through several interconnected pathways. Households may face recurring medical and transportation expenses, lose income when a family member becomes unable to work, or redirect resources from food, education, housing, and other essential needs toward health care. When these pressures occur together, families may have limited capacity to respond to additional economic shocks.

The risk is particularly significant when illness affects a household member whose work is essential to family income or food production. If that person cannot continue working, other members may need to assume additional responsibilities or seek alternative sources of income. Children and young adults may also experience indirect consequences if family resources become insufficient to support their educational needs.

However, the relationship between chronic disease and poverty should not be interpreted as inevitable. The financial consequences depend on the availability of effective prevention, timely diagnosis, affordable treatment, social support, and mechanisms that protect households from excessive health-related expenditure. Public policy can therefore influence whether illness results in manageable disruption or prolonged financial hardship.

Social determinants of health provide an important framework for understanding these relationships. Income, education, employment, housing, food security, and access to essential services shape both the risk of disease and the capacity to manage its consequences. Marmot et al. (2008) emphasize that health inequalities arise from the conditions in which people are born, grow, live, work, and age. Chronic disease policy

must therefore consider the broader circumstances that influence people's opportunities to remain healthy.

In Timor-Leste, poverty-sensitive health planning should examine whether households can obtain essential medicines and maintain follow-up without sacrificing other basic needs. It should also consider the costs of travelling to services, the availability of community-based care, and the support required by people who experience disability or loss of income.

Reducing the economic consequences of chronic disease requires coordinated action. Accessible primary health care, effective risk-factor prevention, appropriate medicine availability, and timely management of complications can reduce avoidable health loss. Financial protection and relevant social support can further reduce the likelihood that illness will translate into severe household deprivation.

3.3.5 The Burden on Families and Informal Caregivers

Families frequently provide essential support to people living with chronic diseases. This support may include reminding patients to take medication, preparing appropriate meals, accompanying them to health facilities, assisting with personal care, managing household responsibilities, and providing emotional reassurance. In many settings, such assistance is essential to maintaining treatment continuity and quality of life.

Informal caregiving can strengthen family relationships and help patients remain connected to their communities. However, caregiving can also create physical, emotional, and financial pressures, especially when the condition is severe or requires continuous assistance. Family members may experience fatigue, anxiety, reduced social participation, and competing responsibilities between care, employment, education, and household duties.

The consequences are influenced by the nature of the disease. A person whose diabetes is well controlled may need periodic monitoring and support with treatment routines, whereas a person with complications may require more intensive assistance. Stroke-related disability can create substantial demands for mobility support and daily care. Advanced cancer or severe respiratory disease may require frequent clinical visits and symptom management. Consequently, the needs of patients and caregivers can change considerably over time.

These demands should not be regarded solely as private family responsibilities. Although families play a central role in care, the health system has a responsibility to provide appropriate clinical guidance, education, rehabilitation, and support. Without these services, households may be expected to manage complex conditions with limited information and inadequate resources.

For Timor-Leste, community-based care offers opportunities to strengthen support closer to where people live. Primary health care workers can provide patient education, monitor treatment, identify complications, and help families understand when referral is necessary. Community health initiatives may also improve communication between health services and households, particularly when distance makes frequent facility visits difficult.

Caregiver support should be integrated into chronic disease planning where feasible. Practical measures may include clear treatment instructions, counselling, training in appropriate care tasks, referral guidance, and coordination between health workers and family members. Such measures can reduce avoidable uncertainty and help caregivers provide support without assuming responsibilities beyond their capacity.

Recognizing caregivers is also a matter of dignity and fairness. Their contributions sustain many aspects of long-term care, but these contributions should not become an excuse for inadequate formal services. A balanced system values family participation while ensuring that patients and caregivers receive the information and professional support they need.

3.3.6 Effects on Education, Children, and Intergenerational Well-Being

Although chronic diseases are often discussed in relation to adults and older populations, their consequences can extend across generations. When a parent or caregiver develops a serious chronic condition, children may experience changes in household income, family responsibilities, emotional security, and educational opportunities. These effects depend on the severity of illness, the availability of alternative caregivers, and the resources available to the household.

Children may need to assist with household work, accompany relatives, or assume responsibilities that would otherwise be managed by adults. In some circumstances, the financial consequences of illness may make school-related expenses more difficult to meet. Repeated disruption can affect attendance, concentration, and opportunities for learning, although the specific effects vary considerably between households.

Chronic disease can also influence the well-being of children through the conditions in which they grow up. Household food security, opportunities for physical activity, access to health information, and exposure to tobacco smoke or other environmental risks can influence health across the life course. Family routines and community environments may support healthy behaviours, but these influences are also shaped by wider economic and social conditions.

The concept of intergenerational well-being highlights why prevention and early intervention are important. Reducing avoidable chronic disease among adults can protect household stability and preserve the resources needed for children's health and education. At the same time, creating healthier environments for children and young people may reduce future exposure to preventable NCD risk factors.

For Timor-Leste, this perspective suggests that chronic disease policy should not be isolated from education, nutrition, social protection, and community development. Schools and community programmes can contribute to health promotion, while primary health care can provide accessible information and early assessment. Broader policies affecting food environments, tobacco exposure, and opportunities for physical activity can also support healthier life-course trajectories.

Nevertheless, intergenerational effects should be examined through evidence rather than assumed to occur uniformly. Household surveys, qualitative interviews, and longitudinal research can help identify how chronic illness affects children's education, family resources, and future opportunities in different settings. Such evidence would support more targeted and effective social and health interventions.

3.3.7 Chronic Diseases, Inequality, and Social Exclusion

The burden of chronic disease is not distributed equally across populations. Exposure to risk factors, access to prevention, opportunities for early diagnosis, and the quality of continuing care may vary according to income, education, geography, employment, and other social circumstances. As a result, some groups may face a greater likelihood of developing complications or experiencing avoidable health loss.

Inequalities can emerge at several stages. People with limited access to health information may be less likely to recognize early symptoms or understand the need for regular monitoring. Individuals living far from health facilities may delay seeking care because of travel time and cost. Households with limited financial resources may find it difficult to obtain recommended medicines or maintain repeated follow-up. When

these barriers accumulate, the consequences of disease can become more severe even where effective interventions exist.

Social exclusion may also arise from the effects of illness itself. People living with disabilities caused by stroke, diabetes complications, cancer, or chronic respiratory disease may experience difficulties participating in employment, education, and community activities. Stigma, inadequate accessibility, or limited opportunities for reasonable accommodation can further restrict participation.

These inequalities are not adequately described by national averages alone. A country may show progress in overall service coverage while particular populations continue to experience poor access or outcomes. For this reason, disease burden and health service indicators should be disaggregated whenever the data allow, and findings should be interpreted in light of the quality and representativeness of the evidence.

The social determinants perspective emphasizes that reducing health inequalities requires action beyond the clinical encounter. Health systems must provide appropriate care, but other sectors also influence the conditions that shape exposure to risk and the ability to maintain health. Education, transport, food systems, employment, housing, environmental management, and social protection can all affect the opportunities available to people living with chronic disease (Marmot et al., 2008).

For Timor-Leste, an equity-oriented response should pay particular attention to communities facing geographical barriers and households with limited resources. Outreach services, reliable medicine supply, appropriate referral arrangements, health education, and community participation can help reduce some barriers. However, the effectiveness of these measures should be monitored to determine whether they actually improve access and outcomes for underserved populations.

Equity also requires that people living with chronic conditions be treated as participants in their own care rather than as passive recipients of services. Respectful communication, understandable information, shared decision-making, and attention to cultural and linguistic circumstances can improve the experience of care and strengthen trust between communities and health professionals.

3.3.8 National Productivity and Human Capital Development

Human capital refers broadly to the knowledge, skills, health, and capabilities that enable people to participate in social and economic life. Health is an important

component because people who can maintain their physical and mental functioning are generally better positioned to learn, work, care for others, and contribute to their communities. Chronic diseases can undermine these capabilities through premature mortality, disability, recurring illness, and reduced participation.

At the population level, chronic disease may influence productivity by reducing the availability of workers, increasing absenteeism, limiting working capacity, and placing additional demands on family caregivers. Premature death may also interrupt the transfer of skills and experience within households, workplaces, and communities. These consequences can be especially significant when affected individuals occupy roles that are difficult to replace or are responsible for supporting several dependants.

However, the economic value of health should not be reduced to productivity alone. People who cannot engage in paid employment continue to possess dignity, rights, relationships, and social value. The contribution of older people, persons with disabilities, and those who provide unpaid care cannot be adequately represented through conventional economic indicators. A human development perspective therefore recognizes health both as a contributor to economic opportunity and as an essential dimension of human well-being.

For Timor-Leste, investment in NCD prevention and management can support the development of a healthier population over the life course. Effective prevention may reduce avoidable illness, while accessible treatment can help people remain active and maintain their independence for longer. Rehabilitation and appropriate workplace accommodations can support participation among people who experience lasting functional limitations.

These benefits depend on the effectiveness and sustainability of health interventions. It would be inappropriate to assume that every investment in chronic disease care automatically produces a specific economic return. Economic evaluations must consider the intervention, target population, costs, outcomes, and time horizon. Nevertheless, the broader principle remains that preventable illness and disability can limit human capabilities, while effective and equitable health services can help protect them.

A national development strategy should therefore recognize health as both an outcome of development and a foundation for it. The prevention and management of chronic

diseases can contribute to human development by reducing avoidable health loss and supporting people's ability to participate in family, community, and economic life.

3.3.9 Health Expenditure, Public Resources, and Sustainable Development

Chronic diseases create continuing demands on public health resources. Depending on the disease and stage of illness, expenditure may include prevention programmes, primary care consultations, laboratory testing, essential medicines, emergency treatment, hospitalization, specialist services, rehabilitation, and palliative care. These demands can compete with other priorities when the health budget, workforce, and infrastructure are limited.

The challenge is not simply that chronic diseases require expenditure. All effective health systems must invest resources to prevent illness and treat people who need care. The central question is how resources can be allocated efficiently, fairly, and sustainably to achieve meaningful improvements in health. Underinvestment in prevention and primary care may contribute to avoidable complications, while insufficient access to essential treatment may allow manageable conditions to progress to more serious illness.

An integrated approach can help reduce fragmentation and improve the use of available resources. Shared risk-factor interventions can address several NCDs simultaneously. Primary health care can provide ongoing management for common chronic conditions, while referral services can focus on patients who require specialized assessment. Reliable medicine supply and clinical information systems can support continuity and reduce disruptions in treatment.

However, the cost-effectiveness of individual interventions must be assessed within the local context. Interventions that are effective elsewhere may require adaptation to the workforce, infrastructure, geography, and service delivery arrangements of Timor-Leste. Economic analysis should therefore consider both the expected health benefits and the practical costs of implementation, including recurrent costs after an initial programme has been established.

Sustainable financing also requires attention to financial protection. When households are unable to afford continuing care, patients may delay treatment or experience interruptions that worsen health outcomes. Policies that reduce financial barriers can help ensure that access depends less on a person's ability to pay at the point of need. The WHO and World Bank monitoring framework provides a useful basis for examining

whether progress toward UHC is accompanied by protection against financial hardship (WHO & World Bank, 2023).

NCD investment should also be considered in relation to the Sustainable Development Goals, particularly the commitment to healthy lives and well-being for all at all ages. Chronic disease prevention and management are connected to broader objectives involving poverty reduction, education, gender equality, decent work, and reduced inequalities (United Nations, 2015). These connections reinforce the importance of coordinated action across sectors rather than treating NCDs solely as a responsibility of clinical services.

For Timor-Leste, sustainable development requires that health investments be planned over the long term. Reliable financing, workforce development, effective procurement, appropriate monitoring, and transparent accountability are essential to maintaining programmes and ensuring that their benefits reach the population.

3.3.10 The Role of Prevention in Reducing Socioeconomic Consequences

Prevention is central to reducing the future burden of chronic disease because it can lower exposure to risk factors, delay disease onset, and reduce the likelihood of complications. Effective prevention can also help limit the economic and social consequences associated with avoidable illness, although the magnitude and timing of these benefits depend on the intervention and population concerned.

Population-level interventions seek to create conditions that make healthier choices more accessible. Examples include tobacco control, measures to reduce harmful alcohol use, policies supporting healthier food environments, opportunities for physical activity, and action to reduce harmful environmental exposures. These interventions can influence the distribution of risk across the population rather than relying exclusively on individual motivation or clinical advice (WHO, 2013, 2017).

Individual-level prevention and early detection remain important. Health workers can help people understand their risk factors, recognize warning signs, and access appropriate assessment. For people with established hypertension, diabetes, or other chronic conditions, effective treatment and follow-up can reduce the risk of complications. However, prevention programmes should avoid placing responsibility entirely on individuals when the resources and environments needed to maintain health are not equally available.

In Timor-Leste, prevention strategies should be culturally appropriate, practical, and responsive to the diversity of communities. Health communication should use understandable language and account for local beliefs, food practices, household circumstances, and access to information. Community participation can help identify barriers and develop approaches that are more acceptable and sustainable.

Prevention must also be connected to service capacity. Identifying elevated risk is beneficial only when people can access appropriate advice, confirmatory assessment, and effective management. Similarly, public campaigns may have limited impact if healthier food options, safe opportunities for physical activity, or essential treatment remain inaccessible to the people most in need.

Evaluation should therefore examine more than programme activities. Measures such as the number of educational sessions delivered or people screened may describe implementation, but they do not independently establish that disease incidence, complications, or inequalities have declined. Monitoring should link programme delivery to relevant behavioural, clinical, and population health outcomes wherever feasible.

A comprehensive prevention strategy can help reduce future health loss while supporting household well-being and national development. Its success, however, depends on sustained implementation, reliable evidence, and coordination between health services, communities, and other sectors.

3.3.11 Toward a Human-Centred Approach to Chronic Disease Policy

The socioeconomic analysis of chronic diseases demonstrates that effective policy must recognize the person behind the diagnosis. Epidemiological indicators are essential for understanding patterns of health loss, yet they cannot fully capture the experience of living with pain, fatigue, disability, uncertainty, or dependence on continuing care. Nor can they independently measure the social and emotional contributions of family members who support people with chronic conditions.

A human-centred approach begins by recognizing that patients have needs extending beyond the immediate treatment of disease. They need understandable information, respectful communication, continuity of care, access to appropriate medicines, and support in managing the practical demands of illness. They should also be able to participate in decisions about their treatment to the extent appropriate to their circumstances and preferences.

Such an approach does not conflict with evidence-based medicine or economic evaluation. Rather, it complements them by ensuring that clinical effectiveness and resource efficiency are considered alongside dignity, accessibility, patient experience, and equity. A programme that improves a statistical indicator but leaves underserved communities unable to obtain care cannot be regarded as a complete public health success.

For Timor-Leste, human-centred chronic disease policy should connect clinical services with the realities of household life. This includes attention to travel requirements, treatment affordability, health literacy, family caregiving, cultural circumstances, and the practical challenges of maintaining regular follow-up. Community participation can help health services understand these realities and design interventions that are more responsive to people's needs.

The approach should also recognize the importance of social inclusion. People living with chronic diseases should not be excluded from education, employment, or community participation merely because they have a diagnosis or disability. Appropriate rehabilitation, accessible environments, supportive workplaces, and respectful public communication can help protect opportunities for participation.

Ultimately, chronic disease policy should be judged not only by how many deaths it prevents or how many DALYs it reduces, but also by whether it enables people to live with greater security, independence, and dignity. Quantitative indicators remain indispensable, but their purpose is to inform decisions that improve real human lives.

3.3.12 Conclusion: Chronic Disease, Equity, and the Future of Human Development

Chronic diseases have consequences that extend well beyond the clinical setting. They can generate recurring medical expenses, reduce income and productivity, increase caregiving responsibilities, disrupt education, deepen household poverty, and contribute to inequalities in health and social participation. These consequences are interconnected, and their severity depends partly on the availability of prevention, effective treatment, financial protection, and social support.

For Timor-Leste, the implications are significant for both health policy and national development. A comprehensive response must strengthen prevention and early detection while ensuring that people already living with chronic diseases can access continuous, appropriate, and affordable care. It must also recognize the contribution of

families and caregivers without transferring the full responsibility for long-term management onto households.

The socioeconomic burden of NCDs reinforces the importance of primary health care, universal health coverage, reliable medicine supply, equitable service distribution, and coordinated action across sectors. Policies addressing education, food environments, employment, transport, and social protection can complement clinical interventions by improving the conditions in which people live and manage illness.

At the same time, national decisions should be grounded in reliable evidence. Economic evaluations, population surveys, health service data, and qualitative research can help identify where interventions are most needed and whether they are reducing avoidable health loss and financial hardship. Such evidence should inform decisions transparently while preserving attention to dignity and fairness.

The central lesson is that chronic disease is not only a question of how long people live, but also of how well they can live, participate, and maintain meaningful relationships throughout their lives. Reducing its burden is therefore both a public health objective and a commitment to human development.

This conclusion completes Chapter 3 by connecting epidemiological measurement with the broader consequences of chronic diseases for households, communities, and national progress. The next chapter turns to the health system response, examining the capacity required to deliver effective, equitable, and continuous NCD prevention and care in Timor-Leste.

CHAPTER 4

HEALTH SYSTEM CHALLENGES, PREVENTION, AND STRATEGIC DIRECTIONS FOR TIMOR-LESTE

Chronic diseases and noncommunicable diseases (NCDs) require health systems to move beyond episodic treatment toward sustained, coordinated, and people-centred care. Unlike health conditions that may be resolved through a single clinical encounter, many chronic diseases demand repeated assessment, long-term medication, lifestyle support, early identification of complications, and continuing communication between patients and health professionals. These requirements place substantial demands on health workforce capacity, medicine availability, diagnostic services, referral systems, financing, and health information infrastructure.

The challenge is especially important for countries where health systems must respond simultaneously to infectious diseases, maternal and child health needs, nutritional challenges, and the growing requirements associated with chronic conditions. Timor-Leste must therefore strengthen its capacity to prevent and manage NCDs without weakening other essential health services. This requires strategic planning that recognizes existing constraints while building sustainable pathways toward universal health coverage and improved population health.

An effective response cannot rely exclusively on hospitals or specialized services. Primary health care, community engagement, preventive policies, and continuity of treatment are equally important. Health systems must be able to identify risk early, provide appropriate first-line management, refer patients when necessary, and maintain follow-up after diagnosis or hospitalization. These functions are interconnected, and weaknesses in one area can reduce the effectiveness of the entire system (Atun et al., 2013; Kruk et al., 2018).

In Timor-Leste, the National Health Sector Strategic Plan 2011–2030 provides an important policy foundation for strengthening health services and improving access to care (Ministry of Health, Timor-Leste, 2011). The continuing development of NCD prevention and management should be aligned with broader national objectives, including equitable service delivery, health workforce development, stronger primary care, and improved health information systems.

This chapter examines the institutional and operational challenges involved in responding to chronic diseases, the importance of prevention and integrated

community-based care, and the strategic directions that can support a more equitable and resilient health system in Timor-Leste. It emphasizes that sustainable progress depends not only on clinical interventions but also on the quality, accessibility, coordination, and accountability of the services through which those interventions are delivered.

4.1 Health System Capacity, Universal Health Coverage, and Continuity of Care

4.1.1 Introduction: From Episodic Treatment to Long-Term Care

The management of chronic diseases requires a different pattern of health service delivery from that traditionally associated with acute illness. Acute conditions often demand rapid assessment and immediate intervention, whereas chronic conditions require repeated interactions between patients and health professionals over extended periods. Effective management depends on early diagnosis, continuing monitoring, adherence support, appropriate treatment adjustment, and timely responses to complications.

For example, a person living with hypertension may require regular blood pressure measurement and ongoing medication, even when no symptoms are present. A person with diabetes may need periodic clinical assessment, laboratory monitoring, education, and evaluation for complications. Following a stroke, patients may require acute treatment, rehabilitation, secondary prevention, and continuing support to maintain functional independence. These examples demonstrate that chronic disease care is not a single intervention but a continuing process.

The capacity to provide such care depends on the interaction of multiple health system components. Health workers need appropriate knowledge and clinical skills; facilities require essential medicines and diagnostic equipment; referral systems must connect patients with more specialized services; and information systems should support follow-up and communication. Financing arrangements must also enable patients to obtain services without unnecessary financial hardship.

Atun et al. (2013) emphasize the importance of adapting health systems to the requirements of NCD prevention and control. This adaptation involves more than adding disease-specific activities to existing services. It requires reconsidering how care is organized, how resources are allocated, how patients move between levels of the system, and how health professionals coordinate their responsibilities.

For Timor-Leste, this transition should be pursued through a realistic and phased approach. Existing primary health care structures can provide a foundation for expanded chronic disease services, while referral hospitals and specialized providers contribute to the management of complex cases. The central objective is to establish a system in which people can obtain appropriate care throughout the course of their illness, regardless of where they live or their socioeconomic circumstances.

4.1.2 Health Workforce Capacity and Clinical Competence

A competent and adequately supported health workforce is fundamental to effective chronic disease prevention and management. Health professionals must be able to recognize risk factors, identify symptoms, conduct appropriate assessments, initiate treatment within their scope of practice, provide patient education, and recognize when referral is required. These responsibilities require both initial professional education and continuing development.

Chronic disease management presents several specific workforce challenges. First, many important risk factors, including hypertension and elevated blood glucose, may remain undetected for long periods because they do not necessarily produce obvious symptoms. Health workers need the skills and resources to identify these conditions through appropriate assessment and testing. Second, treatment decisions often require ongoing review rather than a one-time prescription. Third, patients may have multiple chronic conditions, making coordinated assessment and medication management increasingly important.

The development of clinical competence should therefore extend beyond disease-specific knowledge. Health workers also need communication skills, the ability to support behavioural change, an understanding of social determinants, and competence in shared decision-making. These skills can help patients understand their conditions and participate more effectively in long-term care.

For Timor-Leste, workforce planning should consider the distribution of health professionals across hospitals, community health centres, and other primary care settings. National staffing numbers alone may not reveal whether trained personnel are available where the need is greatest. Remote communities may face different service constraints from more accessible urban areas, and workforce policies should account for these differences.

Continuing professional development can strengthen clinical quality, particularly when it is connected to practical service needs. Training may include the assessment and management of hypertension and diabetes, cardiovascular risk evaluation, recognition of stroke symptoms, counselling on tobacco cessation, appropriate referral, and the monitoring of treatment outcomes. However, training should be accompanied by supportive supervision, clinical protocols, adequate equipment, and reliable supplies. Knowledge cannot be translated into effective practice if health workers lack the resources needed to provide recommended services.

Task-sharing may also contribute to service delivery when responsibilities are clearly defined, training is appropriate, and professional support is available. Nurses, midwives, community-based health workers, and other trained personnel can contribute to prevention, risk assessment, patient education, follow-up, and referral within their authorized roles. Such arrangements should strengthen coordinated care rather than create parallel responsibilities without adequate communication.

The ultimate objective is not simply to increase the number of trained personnel, but to ensure that health workers are competent, supported, appropriately distributed, and able to deliver safe and respectful care. Workforce development is therefore both a technical requirement and a foundation for equitable health system performance (Kruk et al., 2018).

4.1.3 Essential Medicines, Diagnostic Capacity, and Health Infrastructure

Effective chronic disease management depends on the continuous availability of essential medicines, reliable diagnostic services, and functioning health facilities. A diagnosis has limited practical value if the patient cannot obtain appropriate treatment, while a prescription cannot achieve its intended effect if medicines are unavailable or unaffordable.

People living with chronic diseases may need medicines over many years. Interruptions in treatment can contribute to poor disease control and increase the risk of complications. For example, inconsistent access to appropriate antihypertensive or diabetes medication may undermine efforts to prevent cardiovascular and renal complications. The consequences may be particularly serious for people who have already experienced a major cardiovascular event or who have multiple chronic conditions.

Medicine availability depends on several interconnected processes, including forecasting demand, procurement, storage, distribution, inventory management, and monitoring of shortages. Weaknesses at any stage can disrupt continuity of care. Reliable supply systems therefore require accurate information about patient needs, clear procurement procedures, appropriate storage, and mechanisms for identifying and resolving stock problems.

Diagnostic capacity is equally important. Blood pressure measurement, blood glucose testing, laboratory assessment, and other appropriate investigations can support the identification and monitoring of chronic conditions. More complex diagnostic needs may require referral to facilities with specialized equipment and expertise. A functioning referral system must ensure that patients can access these services and that results are communicated to the professionals responsible for their continuing care.

In Timor-Leste, geographic distance and differences in facility capacity should be considered when planning diagnostic services and medicine distribution. Not every facility can provide every test or specialist intervention. However, a coordinated system can establish which services should be available at each level, how patients should be referred, and how essential medicines and supplies should be maintained.

Infrastructure planning should also account for electricity, water, communications, equipment maintenance, and appropriate storage conditions. These operational requirements may appear separate from chronic disease management, but they influence whether clinical services can be delivered reliably. Equipment that is unavailable, unsuitable, or poorly maintained cannot support effective diagnosis and follow-up.

The WHO package of essential NCD interventions provides guidance for organizing selected evidence-based services in primary care, including in resource-constrained settings (World Health Organization [WHO], 2020a). Its application in Timor-Leste should be adapted to national clinical guidance, workforce capacity, supply availability, and referral arrangements.

The central principle is continuity. Medicines, diagnostics, and infrastructure should be planned as components of a connected service rather than as isolated resources. Their value depends on whether patients can obtain appropriate care when it is needed and maintain that care over time.

4.1.4 Primary Health Care as the Foundation of Chronic Disease Management

Primary health care is essential to chronic disease management because it provides a point of contact for prevention, early detection, treatment, and long-term follow-up. Many common NCD risk factors and conditions can be identified and managed through appropriately equipped primary care services, reducing the need for patients to rely exclusively on hospitals or specialist facilities.

A strong primary health care system can support routine risk assessment, patient education, treatment initiation within clinical protocols, medication monitoring, and referral when complications require additional expertise. It can also help identify patients who have missed appointments or experienced difficulties maintaining treatment. These functions are particularly important for conditions that may progress silently before producing serious complications.

Primary care should not be understood merely as a lower level of the health system. It is a central point for coordinating services and maintaining relationships with patients and communities. When primary care teams have access to appropriate clinical information and clear referral pathways, they can help ensure that treatment remains coherent as patients move between facilities.

For Timor-Leste, strengthening primary health care offers an opportunity to bring chronic disease services closer to communities. Community health centres and other established primary care platforms can provide a basis for prevention, screening where appropriate, clinical management, and continuing education. The precise range of services should reflect the competence of staff, available diagnostics, national guidelines, and mechanisms for referral.

Continuity is especially important when patients experience multiple health conditions. If services are organized entirely around individual diseases, patients may face repeated appointments, inconsistent advice, and fragmented medication management. A more integrated approach can help health workers consider the patient's overall clinical needs, treatment burden, functional capacity, and circumstances at home.

The WHO framework on integrated, people-centred health services emphasizes the importance of coordinating care around people's needs rather than organizing services solely around institutions or isolated interventions (WHO, 2016). For chronic diseases, this means that patients should experience a connected pathway across prevention, primary care, hospital services, rehabilitation, and community support.

Primary health care cannot operate effectively in isolation. It requires access to medicines, laboratory services, professional supervision, referral hospitals, and reliable health information. Investment in primary care should therefore be accompanied by measures that strengthen the wider health system. When these components function together, primary health care can become a practical foundation for more equitable and sustainable NCD management.

4.1.5 Referral Systems and Coordination Across Levels of Care

Chronic diseases sometimes require services beyond the capacity of primary care. Severe symptoms, suspected complications, diagnostic uncertainty, acute cardiovascular events, and other complex clinical situations may require urgent referral or specialist assessment. A well-functioning referral system enables patients to move to an appropriate level of care without unnecessary delay and ensures that information returns to the professionals responsible for continuing management.

Referral systems involve more than the physical transfer of patients. They require clear clinical criteria, communication between facilities, appropriate transport arrangements, documentation, and follow-up after specialist assessment. If patients are referred without adequate information or cannot return to primary care after treatment, the continuity of care may be compromised.

For example, a patient treated for a stroke may need acute hospital management followed by rehabilitation, risk-factor control, medication review, and long-term monitoring. If discharge information is not communicated to the primary care team, opportunities for secondary prevention may be lost. Similarly, a patient referred for a diabetes complication may need continued monitoring after specialist treatment. Without coordination, different providers may deliver disconnected services or fail to identify changes in the patient's condition.

Geographic circumstances can make referral particularly challenging in Timor-Leste. Distance, transport availability, road conditions, and the costs associated with travel may influence whether patients can reach higher-level services. These barriers should be incorporated into clinical planning rather than treated as problems outside the health system's responsibility.

Practical improvements may include standardized referral forms, clear communication protocols, scheduled follow-up after discharge, designated contact points between facilities, and systems for tracking whether referred patients complete the

recommended assessment. Where digital communication is feasible, electronic information exchange may support coordination. However, digital tools should complement rather than replace reliable basic referral procedures.

The system should also avoid unnecessary referrals caused by insufficient primary care support. Training, clinical guidance, access to basic diagnostics, and professional consultation can help primary care teams manage appropriate cases locally while recognizing when specialist care is necessary.

Effective referral and coordination are essential to quality health systems because the quality of an individual clinical encounter does not guarantee a good overall outcome. Patients need care that remains coherent across facilities and over time. For chronic diseases, this continuity can be as important as the initial diagnosis or treatment decision (Kruk et al., 2018).

4.1.6 Universal Health Coverage and Financial Protection

Universal health coverage is a central objective for health systems seeking to ensure that people can access the services they need without experiencing financial hardship. For chronic diseases, UHC requires attention to both service availability and the costs that patients and families face when obtaining continuing care.

Chronic disease management can generate recurring expenses through consultations, medicines, diagnostic tests, transport, and follow-up. These costs may become difficult to sustain when patients have limited income or need care over many years. The risk of financial hardship may be increased when treatment requires repeated travel or when complications lead to hospitalization and additional support needs.

Financial protection should therefore be assessed broadly. Official charges for clinical services are only one part of the picture. Indirect expenses, time away from employment, and the cost of accompanying relatives can influence whether patients seek care and continue treatment. A service may be geographically available but remain practically inaccessible if the total cost of obtaining it is too high.

For Timor-Leste, progress toward UHC should include attention to the accessibility and continuity of essential NCD services. This may involve improving the availability of appropriate medicines, strengthening primary care, reducing avoidable travel through suitable local service delivery, and ensuring that referral arrangements are practical for patients and families.

WHO and World Bank monitoring of UHC emphasizes the importance of assessing service coverage alongside financial protection (WHO & World Bank, 2023). This combined perspective is important because a health system cannot be considered fully responsive if people are unable to obtain effective care or if accessing treatment exposes households to severe financial strain.

The financial protection dimension also has implications for equity. People with greater financial resources may be more able to pay for transport, obtain services outside their locality, or absorb income losses during illness. Without appropriate safeguards, differences in household resources can translate into differences in diagnosis, treatment continuity, and health outcomes.

A sustainable approach must therefore balance the cost of providing services with the responsibility to ensure equitable access. Economic evaluation can help identify efficient interventions, but resource allocation should also consider the distribution of benefits and the needs of underserved populations.

For chronic disease care, UHC is not achieved through a single policy or financing mechanism. It requires a coordinated set of arrangements that make essential services available, ensure appropriate quality, and protect people from avoidable financial hardship throughout the course of illness.

4.1.7 Health Information Systems, Patient Records, and Continuity of Care

Reliable health information is essential for managing chronic diseases because clinical decisions often depend on changes observed over time. Health workers need to know a patient's previous diagnoses, medication history, relevant test results, treatment response, complications, and planned follow-up. Without accessible and accurate information, continuity becomes more difficult and the risk of duplicated tests, medication errors, or missed care may increase.

At the population level, health information systems help managers understand service utilization, medicine requirements, disease patterns, and gaps in care. Information about missed appointments, referral completion, treatment availability, and clinical outcomes can identify problems that are not visible through facility attendance totals alone.

In Timor-Leste, strengthening health information systems should include consistent documentation, appropriate data definitions, staff training, data quality review, and

procedures for using information in planning. The system should be practical enough for health workers to maintain while providing information that supports clinical care and public health decisions.

Patient records are particularly important when individuals receive services from multiple facilities. A patient may begin treatment at a community health centre, receive diagnostic assessment at another facility, and later attend a referral hospital. If relevant information is not communicated between providers, the patient may have to repeat the same history, and important clinical decisions may be made without access to previous findings.

Digital health tools may help address some of these challenges when infrastructure, connectivity, technical support, and data governance are adequate. Electronic records, appointment reminders, and remote communication can support follow-up and information exchange. However, digital systems should be selected according to demonstrated needs and operational feasibility rather than novelty alone.

Privacy and confidentiality must remain central. Chronic disease records can contain sensitive personal and clinical information, and access should be limited to legitimate purposes under appropriate safeguards. Data systems should also consider technical reliability, backup procedures, staff responsibilities, and the ability to operate when connectivity is limited.

Information systems should ultimately be evaluated by whether they improve care and decision-making. Collecting more data does not automatically improve health outcomes. Information becomes valuable when health workers can use it to identify risks, coordinate treatment, respond to missed follow-up, and improve service quality.

4.1.8 Quality of Care, Patient Safety, and People-Centred Services

The availability of health services is necessary but insufficient for effective chronic disease management. Patients must also receive care that is safe, clinically appropriate, timely, respectful, and responsive to their circumstances. Poor-quality care can undermine the benefits of diagnosis and treatment, even where facilities and medicines are available.

Quality includes several related dimensions. Clinical effectiveness concerns whether care follows appropriate evidence and improves health outcomes. Safety involves reducing avoidable harm, including medication errors and failures to recognize

complications. Timeliness concerns whether patients receive assessment and treatment when needed. People-centredness requires respectful communication, understandable information, and attention to individual preferences and needs.

Chronic disease care is particularly dependent on the relationship between patients and health professionals. Patients may need to discuss medication adherence, diet, physical activity, tobacco use, alcohol consumption, or concerns about complications. These conversations are more likely to be useful when patients feel respected and can explain the practical challenges they face without fear of blame or humiliation.

For example, a patient who misses follow-up appointments may be experiencing transportation difficulties, financial pressure, work responsibilities, or misunderstanding about the importance of monitoring. Treating missed appointments solely as evidence of non-compliance may overlook these barriers. A people-centred approach seeks to understand the reasons for interrupted care and identify practical solutions.

Kruk et al. (2018) emphasize that high-quality health systems must deliver care that improves health outcomes and responds to people's needs. For Timor-Leste, this perspective supports the development of clear clinical protocols, appropriate supervision, patient feedback mechanisms, and routine review of service performance.

Quality improvement should use meaningful indicators rather than relying exclusively on activity counts. The number of consultations delivered, for instance, does not establish whether blood pressure is adequately controlled or whether people with diabetes receive appropriate follow-up. Where feasible, service monitoring should include clinical outcomes, continuity indicators, patient experience, and evidence of equitable access.

Patient safety also requires clear processes for medication review, recognition of warning signs, appropriate referral, and communication across levels of care. As chronic disease management becomes more integrated, attention should be paid to the risks associated with multiple medicines and multiple providers.

A high-quality system does not merely treat a condition. It helps people understand their illness, navigate services, maintain treatment, and receive support when their needs change. This is the standard toward which chronic disease services should develop.

4.1.9 Integrating Chronic Disease Services with Existing Health Priorities

Timor-Leste must address NCDs within a broader public health environment that includes infectious diseases, maternal and child health, nutrition, and other essential services. The growth of chronic disease care should not create unnecessary competition between programmes or divert attention from conditions that continue to impose important health burdens.

Integration offers one way to address these overlapping needs. Certain health system functions, such as health education, surveillance, medicine procurement, workforce training, information systems, and community engagement, can support multiple programmes. Strengthening these shared functions may improve efficiency and reduce duplication.

However, integration must be planned carefully. Adding responsibilities to already stretched health workers without additional training, time, or support may reduce the quality of existing services. Similarly, combining programmes administratively does not guarantee effective coordination at the point of care. Clear roles, realistic workloads, appropriate supervision, and adequate resources remain necessary.

For example, routine primary care contacts may provide opportunities to assess relevant NCD risk factors while continuing to deliver other essential services. Yet the scope of assessment should be appropriate to the clinical setting, available equipment, and national guidance. Screening should be linked to a clear pathway for diagnosis, treatment, and follow-up rather than implemented as an isolated activity.

Integrated care also requires attention to patients with multiple conditions. A person may require management for diabetes while also receiving treatment for an infectious disease or another chronic condition. Coordinated clinical decisions can help reduce fragmented care, conflicting instructions, and unnecessary duplication of appointments.

For Timor-Leste, a phased approach is likely to be more sustainable than attempting to establish every possible NCD service simultaneously across all facilities. National planning can identify essential interventions, define minimum service packages, strengthen referral pathways, and progressively expand capacity as resources and workforce skills develop.

The purpose of integration is not to make every facility provide every service. It is to ensure that the health system as a whole responds coherently to the needs of the

population. Such an approach supports resilience by strengthening common foundations while allowing services to address different conditions appropriately.

4.1.10 Strategic Directions for Strengthening Health System Capacity

The preceding analysis identifies several strategic directions for strengthening chronic disease care in Timor-Leste. These directions should be understood as interconnected priorities rather than independent interventions.

First, strengthen primary health care capacity to support prevention, early identification, routine management, and follow-up of common chronic conditions. This requires appropriately trained personnel, clinical protocols, basic diagnostic capacity, essential medicines, and access to referral support.

Second, improve medicine and diagnostic reliability through stronger procurement, distribution, stock monitoring, equipment maintenance, and referral arrangements. Service plans should reflect the resources required to sustain treatment, not only the initial establishment of a programme.

Third, develop the health workforce through continuing education, supportive supervision, appropriate task-sharing, and fair distribution of trained personnel. Training should be linked to the responsibilities health workers are expected to perform and the resources available to them.

Fourth, strengthen continuity and coordination by improving referral procedures, discharge communication, patient records, and follow-up mechanisms. These arrangements are particularly important for patients who move between primary care, hospitals, rehabilitation, and community services.

Fifth, advance financial protection and equitable access by identifying the direct and indirect costs that prevent people from obtaining continuing care. Planning should pay particular attention to the needs of remote communities and households with limited resources.

Sixth, improve health information and accountability through reliable data collection, appropriate confidentiality safeguards, meaningful indicators, and regular review of service performance. Information should be used to improve clinical care and guide resource allocation.

Seventh, embed quality and people-centredness in all stages of service delivery. Clinical effectiveness, patient safety, respectful communication, and responsiveness to local circumstances should be considered core elements of the health system rather than optional additions.

These priorities are consistent with broader international approaches emphasizing integrated services, strong primary health care, and universal health coverage (WHO, 2016, 2019, 2020a). Their implementation in Timor-Leste should be guided by national policy, available evidence, realistic financing, and consultation with health professionals and communities.

Progress should be measured through improvements in access, treatment continuity, quality, health outcomes, and equity. The objective is not simply to expand the number of services but to ensure that people can receive the right care, at the appropriate time, in a form that is effective and sustainable.

4.1.11 Conclusion: Building a System Capable of Sustained Care

Chronic diseases require health systems capable of delivering continuous, coordinated, and people-centred services. Workforce competence, essential medicines, diagnostic capacity, primary health care, referral systems, information infrastructure, quality assurance, and financial protection all contribute to effective management. Weaknesses in these areas can disrupt treatment and increase the risk of avoidable complications.

For Timor-Leste, strengthening NCD services should be part of broader health system development. Primary health care can provide a foundation for prevention and long-term management, while referral hospitals and specialized services remain essential for complex conditions. Reliable coordination between these levels is necessary to ensure that patients receive appropriate care throughout the course of illness.

The pursuit of universal health coverage also requires attention to the practical costs of obtaining treatment. Service availability alone is insufficient if transportation, medicine shortages, or other barriers prevent people from maintaining care. Equity must therefore inform decisions about service distribution, workforce planning, and resource allocation.

Ultimately, the effectiveness of a health system should be judged not only by the number of facilities or consultations it provides, but by whether people can obtain safe,

appropriate, and continuous care without unnecessary hardship. Building such a system requires sustained investment, reliable evidence, transparent accountability, and collaboration between health professionals, communities, and policymakers.

The next section examines how prevention, early detection, and integrated community-based care can reduce avoidable disease burden and bring essential NCD services closer to the people who need them.

4.2 Prevention, Early Detection, and Integrated Community-Based Care

4.2.1 Introduction: Prevention Across the Life Course

The prevention and control of chronic diseases and noncommunicable diseases (NCDs) require a comprehensive public health approach that extends beyond clinical treatment and hospital-based services. Although effective diagnosis and medical management are essential, many chronic diseases develop over years through the interaction of biological susceptibility, individual behaviours, environmental exposures, socioeconomic conditions, and the availability of preventive health services. Consequently, an effective response must address the factors that contribute to disease development while strengthening the capacity of health systems to identify risks early, provide timely interventions, and maintain long-term support for people living with chronic conditions (World Health Organization [WHO], 2013, 2023a).

Prevention is particularly important because many major NCDs share common risk factors. Tobacco use, harmful alcohol consumption, unhealthy diets, physical inactivity, air pollution, and elevated blood pressure, blood glucose, and blood lipids contribute to the development and progression of cardiovascular diseases, diabetes, cancers, and chronic respiratory diseases. Interventions addressing these risks can therefore generate benefits across several disease categories rather than targeting one condition in isolation. Such an approach is especially valuable in health systems where financial, human, and infrastructural resources must be allocated carefully across competing public health priorities (WHO, 2017).

For Timor-Leste, prevention should be understood as a long-term investment in population health, human development, and the sustainability of national health services. As the country continues to strengthen its health system, the prevention of chronic diseases offers an opportunity to reduce avoidable illness, preserve the

productive capacity of adults, protect household resources, and improve the quality of life of older people and individuals living with long-term conditions. The national response must therefore connect health promotion, early detection, clinical care, rehabilitation, and community participation within a coherent framework of primary health care.

A life-course perspective is essential to this strategy. The foundations of chronic disease risk may emerge during childhood and adolescence, become more pronounced during adulthood, and eventually manifest as clinical disease or disability in later life. Prevention cannot therefore be limited to screening adults who already present with symptoms or elevated clinical measurements. It must also support healthy development, improve nutrition, encourage physical activity, reduce exposure to harmful substances, and create environments in which healthier choices are practical and accessible. This perspective aligns with the broader principle that health inequalities are shaped by social and economic conditions throughout life, rather than solely by personal decisions made at the individual level (Marmot et al., 2008).

An effective prevention strategy must consequently operate at several interconnected levels: public policy, community environments, primary health care facilities, households, and individual clinical encounters. Each level has a distinct function, but none can achieve sustainable results independently. Public policy can establish healthier environments; community programmes can strengthen awareness and participation; primary health care can identify risks and initiate treatment; and families can support the daily practices required for long-term disease prevention and management.

4.2.2 Population-Level Prevention and Healthy Environments

Population-level prevention seeks to reduce exposure to major health risks across an entire population rather than relying exclusively on individuals to change their behaviour. Its importance lies in the fact that the distribution of disease is strongly influenced by the environments in which people are born, grow, work, travel, eat, and live. Even when individuals understand the benefits of healthier behaviour, their ability to act on that knowledge may be constrained by the affordability of nutritious food, the availability of safe spaces for exercise, the accessibility of health information, or the commercial availability of tobacco and alcohol.

Public health policies should therefore create conditions that make healthier choices more achievable. Relevant measures include tobacco control, effective regulation of alcohol, improvements in food environments, promotion of physical activity, and action to reduce exposure to harmful air pollution. WHO's recommended NCD interventions emphasise that a combination of population-wide policies and individual health services can reduce preventable disease and improve the effectiveness of national responses (WHO, 2017).

In Timor-Leste, these principles should be adapted to local cultural practices, settlement patterns, economic conditions, and the realities of urban and rural life. Health promotion cannot assume that every household has the same access to nutritious food, safe exercise opportunities, clean household energy, or nearby health facilities. An intervention that is practical in an urban neighbourhood may require a different delivery model in a remote community where transport, communication, and access to services are more limited.

Healthy environments also involve the places where children and young people spend their time. Schools can support healthy eating, regular physical activity, tobacco prevention, and the development of health literacy. Workplaces and public institutions can encourage healthier routines and make preventive information available to adults who may not frequently visit health facilities. Community organisations can reinforce these messages through activities that are culturally appropriate and responsive to local needs.

The effectiveness of these measures depends on coordination between health authorities and other sectors. Education, agriculture, transport, environmental management, local administration, and economic policy can all influence NCD risk. For example, nutritional health is affected not only by individual dietary knowledge but also by food availability, household purchasing power, agricultural production, and food marketing. Similarly, opportunities for physical activity depend partly on the safety and accessibility of public spaces, roads, and community facilities.

A multisectoral approach is therefore necessary to ensure that health considerations are incorporated into decisions made beyond the health sector. WHO's global NCD strategy and implementation roadmap emphasise coordinated action across government and society, recognising that the determinants of chronic disease cannot be addressed by clinical services alone (WHO, 2013, 2023a).

For Timor-Leste, population-level prevention should be supported by clear institutional responsibilities, realistic financing arrangements, community consultation, and mechanisms for monitoring progress. National policies must be translated into activities that can be implemented at municipal and community levels. This requires not only formal policy commitments but also practical guidance, workforce preparation, appropriate resources, and accountability for implementation.

4.2.3 Tobacco, Alcohol, Unhealthy Diet, and Physical Inactivity

Tobacco use remains a major preventable contributor to cardiovascular diseases, chronic respiratory diseases, and several forms of cancer. Tobacco control requires more than advising individuals to stop smoking. It involves reducing exposure to tobacco products, preventing initiation among young people, supporting cessation among current users, protecting people from second-hand smoke, and implementing appropriate regulatory and fiscal measures. WHO's tobacco-control recommendations highlight the importance of comprehensive action that combines population-level protection with accessible support for people seeking to quit (WHO, 2020b).

Within Timor-Leste, tobacco prevention should be integrated into school health programmes, community education, primary health care consultations, and routine assessment of patients' health risks. Health professionals should be equipped to ask about tobacco use in a non-judgmental manner, explain its relationship with chronic disease, and provide practical cessation support. Where specialised cessation services are limited, primary health care personnel can still play an important role through brief advice, follow-up consultations, and referral when additional assistance is available.

Prevention messages must also recognise the social contexts in which tobacco is used. Young people may encounter peer pressure, commercial promotion, or social practices that normalise tobacco consumption. Adults may use tobacco within established social routines or continue using it despite awareness of its health consequences. Effective prevention therefore requires consistent public communication, supportive community environments, and policies that reduce both the availability and social acceptability of tobacco use.

Harmful alcohol consumption is another important risk factor. Its health consequences extend beyond individual disease, potentially affecting family relationships, road safety, occupational functioning, and household financial security. Alcohol-related harm may

also complicate the prevention and management of existing chronic conditions. Accordingly, alcohol prevention should combine public education, appropriate regulation, early identification of harmful consumption, and access to counselling or treatment where needed (WHO, 2014a).

Health professionals should approach alcohol-related discussions with confidentiality, respect, and sensitivity to cultural context. Stigmatising people who experience alcohol-related problems may discourage disclosure and delay access to care. A constructive approach focuses on health risks, personal circumstances, family wellbeing, and available support rather than moral condemnation. At the population level, policy measures should be informed by public health evidence and implemented in ways that protect vulnerable groups.

Dietary risk requires equally comprehensive attention. Diets that are high in salt, unhealthy fats, free sugars, and highly processed foods can contribute to elevated blood pressure, excess weight, metabolic disorders, and cardiovascular disease. Conversely, diets that include appropriate quantities of vegetables, fruits, legumes, whole grains, and other nutritious foods can support long-term health. However, dietary recommendations must be grounded in the economic and cultural realities of the population rather than presented as abstract instructions that households cannot reasonably follow.

For Timor-Leste, nutrition-related prevention should recognise the importance of local food systems, traditional dietary practices, household resources, food availability, and changing consumption patterns. Public education can explain healthier preparation methods, the importance of dietary variety, and practical ways to reduce excessive salt and sugar consumption. Nevertheless, education alone will not be sufficient where nutritious foods are difficult to obtain or unaffordable. Collaboration with relevant sectors is needed to improve access to safe and nutritious food while supporting sustainable local food production.

Physical inactivity is another modifiable risk factor associated with cardiovascular disease, diabetes, and other chronic conditions. Regular physical activity can improve cardiovascular fitness, metabolic health, muscular strength, and psychological wellbeing. WHO recommends that physical activity be promoted throughout the life course, with advice adapted to age, health status, functional ability, and individual circumstances (WHO, 2010a).

In Timor-Leste, physical activity promotion should not be restricted to formal exercise programmes or gym-based activities. Walking, active transport where safe, recreational movement, school sports, household activities, and community-based exercise can all contribute to healthier routines. Public messages should emphasise feasible and sustainable activities rather than prescribing practices that require expensive facilities or specialised equipment.

At the same time, health professionals must recognise that physical activity advice requires individual adaptation. People with established cardiovascular disease, diabetes complications, mobility limitations, or other significant health concerns may need clinical assessment before beginning more intensive activity. The purpose of prevention is to improve health safely and progressively, not to expose individuals to unnecessary risks.

These four behavioural risk areas are closely interconnected. A comprehensive prevention programme should therefore avoid fragmented campaigns that address tobacco, alcohol, diet, and physical activity as entirely separate issues. Instead, health promotion can communicate how these factors interact with blood pressure, blood glucose, body weight, and overall cardiovascular risk. This integrated approach can improve the consistency of prevention messages and make better use of limited health education resources (WHO, 2017, 2023a).

4.2.4 Early Detection and Risk Assessment

Prevention also depends on identifying elevated health risks before complications become severe. Many chronic diseases progress gradually, and some individuals remain unaware of their condition until they experience a serious event or develop symptoms that interfere with daily life. Elevated blood pressure and blood glucose, for example, may remain undetected for extended periods when access to routine assessment is limited. Early detection can create opportunities for timely counselling, clinical evaluation, treatment, and follow-up.

Primary health care facilities are particularly important in this process because they can provide accessible entry points for preventive assessment. Depending on national protocols, local resources, and individual clinical circumstances, assessments may include blood pressure measurement, evaluation of diabetes risk, blood glucose testing, assessment of tobacco and alcohol use, nutritional evaluation, and review of

family and medical history. The selection and frequency of tests should be guided by evidence-based recommendations and the capacity to provide appropriate follow-up.

Screening, however, should not be understood as an isolated technical activity. A screening programme has limited value if individuals with abnormal results cannot access confirmatory assessment, treatment, counselling, or continuing care. The purpose of early detection is not simply to identify more cases; it is to connect people with appropriate services that improve health outcomes and reduce the likelihood of preventable complications.

WHO's Package of Essential Noncommunicable Disease Interventions for primary health care provides a framework for integrating evidence-based interventions into settings where specialised resources may be constrained. The package emphasises practical approaches to the prevention, diagnosis, treatment, and follow-up of major NCDs within primary health care systems (WHO, 2020a).

For Timor-Leste, implementing early detection requires attention to service availability, equipment, staff competencies, diagnostic supplies, and reliable referral pathways. A health facility may have personnel prepared to measure blood pressure, for example, but the usefulness of that service depends on functioning equipment, consistent measurement practices, appropriate clinical interpretation, access to medicines where indicated, and a mechanism for reassessment. Similar considerations apply to diabetes screening and other preventive assessments.

The organisation of screening should also reflect local population needs. Outreach activities may be useful for people who rarely attend facilities, while routine assessment during existing health consultations may improve efficiency for patients who already use primary health care services. Where feasible, community-based activities should be linked to established health facilities so that abnormal findings lead to appropriate clinical evaluation rather than ending with the screening event.

Risk assessment should also avoid unnecessary duplication. When health services collect relevant information during one encounter, that information can inform subsequent consultations if records are accurate, secure, and accessible to authorised health professionals. Standardised protocols can help staff identify which patients require repeat assessment, more intensive counselling, referral, or ongoing clinical monitoring.

At the same time, screening policies must be evidence-based and proportionate. Testing should be selected according to the likely benefits, potential harms, local epidemiological evidence, and the availability of follow-up services. Unnecessary testing can place additional burdens on patients and health facilities, while poorly organised screening may create anxiety without ensuring that individuals receive appropriate care.

Equity must remain central to early detection. Programmes should examine whether rural populations, people with disabilities, older adults, women, low-income households, and other underserved groups can access preventive services on fair terms. A system that identifies risk primarily among people who live near health facilities may produce incomplete information about the population and reinforce existing inequalities. Outreach, accessible communication, and appropriate referral arrangements are therefore essential components of a fair screening strategy (Marmot et al., 2008; WHO, 2019).

4.2.5 Integrated Primary Care and Community-Based Care

Chronic diseases often require simultaneous attention to several related health problems. A person living with diabetes may also have elevated blood pressure, cardiovascular risk, visual impairment, kidney complications, or other conditions requiring continuing assessment. Treating each problem through disconnected programmes can create unnecessary duplication, increase the burden on patients, and make it difficult for health professionals to coordinate care. Integrated services offer a more coherent alternative by connecting prevention, diagnosis, treatment, counselling, follow-up, and referral within a coordinated system.

Integrated primary care should provide a foundation for this approach. Rather than functioning only as the first point of contact before referral to a hospital, primary health care should have the capacity to manage appropriate chronic conditions over time. This includes identifying risks, establishing treatment plans, monitoring progress, providing health education, supporting adherence, detecting complications, and arranging specialist assessment when clinically necessary.

The design of integrated services should be guided by the principle that patients experience health care as a continuing relationship rather than a series of unrelated encounters. Consistent clinical records, standard treatment protocols, clear referral criteria, and regular follow-up can help ensure that important information is not lost

when patients move between facilities or consult different health professionals. WHO's framework for integrated, people-centred health services emphasises coordination around people's needs throughout the continuum of care (WHO, 2016).

In Timor-Leste, integrated care must be responsive to the distribution of health facilities, the availability of trained personnel, transportation challenges, and differences in access between urban centres and more remote communities. A model that depends excessively on frequent hospital visits may be difficult for patients who face long travel times or transport costs. Where clinically appropriate, routine monitoring and health education should therefore be supported closer to the communities in which people live.

Community-based care can help connect formal health services with households and local organisations. Community health workers and other appropriately trained personnel may support health education, encourage attendance at scheduled appointments, identify practical barriers to care, and communicate relevant information to health professionals. Their roles must be clearly defined, supported by training and supervision, and connected to functioning clinical services. Community participation should complement professional care rather than substitute for necessary clinical assessment and treatment.

The involvement of families may also strengthen long-term disease management. Relatives can assist with appointment planning, medicine routines, dietary changes, mobility, and recognition of warning signs. However, family involvement should respect the patient's preferences, privacy, independence, and right to make decisions about personal health. Health professionals should not assume that every patient has a reliable caregiver or that family relationships always provide adequate support.

Integrated care also requires coordination across professional roles. Nurses, physicians, pharmacists, laboratory personnel, nutrition professionals, and community health workers can contribute different forms of expertise. Their collaboration becomes especially important when patients have multiple chronic conditions or need support that extends beyond the immediate clinical encounter. Clear communication and shared responsibilities can reduce fragmented care and improve the use of available resources.

The quality of integrated care should be evaluated not only through the number of consultations or patients registered but also through meaningful outcomes. Relevant

indicators may include continuity of follow-up, availability of essential medicines, timely referral, control of clinical risk factors, patient understanding of treatment, and avoidance of preventable complications. These indicators help distinguish the existence of services from their actual capacity to deliver effective and respectful care (Kruk et al., 2018).

4.2.6 Health Education, Health Literacy, and Community Participation

Health education is most effective when it enables people to understand health risks, evaluate information, make informed decisions, and navigate available services. The ability to understand a message does not automatically mean that an individual can act on it. People may recognise the importance of healthier eating but lack the financial resources to purchase appropriate foods. They may understand the benefits of regular monitoring but face transportation difficulties or competing family responsibilities. Prevention programmes must therefore connect health knowledge with realistic opportunities for action.

Health literacy includes the ability to obtain, understand, assess, and use health information. In chronic disease prevention, this may involve understanding blood pressure measurements, recognising the significance of elevated blood glucose, following treatment instructions, identifying warning signs, and knowing when to seek professional assistance. It also involves the capacity to distinguish reliable health information from misleading claims and to ask questions when medical advice is unclear.

In Timor-Leste, health communication should be culturally and linguistically appropriate. Materials and educational activities should consider the languages used within communities, differences in literacy, local beliefs about illness, and the communication needs of people with disabilities. Oral explanations, visual materials, demonstrations, group discussions, and individual counselling may be more appropriate than written information alone in some settings. The choice of method should be informed by community needs rather than by assumptions about how people learn.

Health professionals should also practise clear and respectful communication during clinical encounters. Technical terminology can create barriers when patients are unfamiliar with medical concepts. Explaining a diagnosis in accessible language, checking the patient's understanding, and inviting questions can improve

communication and support more informed participation in care. Such practices are particularly important when treatment involves long-term lifestyle adjustments or medicines that must be taken consistently.

Community participation is equally important because prevention programmes are more likely to respond to local needs when communities have meaningful opportunities to shape them. Community leaders, schools, civil society organisations, religious institutions, youth groups, and other local organisations may help identify barriers to prevention, develop appropriate educational activities, and communicate concerns to health authorities. Their involvement can strengthen trust and increase the relevance of public health initiatives.

Nevertheless, community participation should not be treated as a substitute for government responsibility. Communities cannot independently resolve structural barriers such as inadequate health infrastructure, shortages of essential medicines, limited diagnostic capacity, or unaffordable transport. Their role is to contribute knowledge, strengthen collective action, and participate in designing solutions, while public institutions remain responsible for ensuring the availability and quality of essential services.

Prevention programmes should also avoid blaming individuals for conditions shaped partly by circumstances beyond their control. Chronic disease risk reflects an interaction between personal behaviour, biological factors, social conditions, and the environments in which people live. Messages that portray illness as evidence of personal failure may increase stigma and discourage individuals from seeking assistance. A more constructive approach recognises personal agency while acknowledging the social and economic barriers that affect health (Marmot et al., 2008).

For this reason, health literacy should be accompanied by service literacy: practical knowledge about where services are available, how referrals work, when follow-up is required, and what assistance may be accessible. A person who understands the importance of blood pressure control but does not know where to obtain monitoring or treatment has not yet been fully supported by the health system. Education and service organisation must therefore reinforce each other.

4.2.7 Outreach, Digital Health, and Access in Remote Communities

Geographical access is an important consideration when designing prevention and early-detection programmes for Timor-Leste. Differences in travel time, transport availability, workforce distribution, and access to communication infrastructure can influence whether people receive regular preventive care. For individuals living far from health facilities, the cost and practical difficulty of attending appointments may become barriers to routine monitoring, even when services are nominally available.

Outreach programmes can help reduce these barriers by bringing selected preventive services closer to communities. Depending on available resources and clinical requirements, outreach may include health education, blood pressure assessment, diabetes risk assessment, follow-up support, and referral coordination. Such activities should be planned in collaboration with local health facilities and community representatives to ensure that services reach the intended population and respond to identified needs.

Outreach must, however, be integrated into a continuing system of care. A single community screening event may identify individuals with elevated clinical measurements, but its long-term value depends on whether those individuals receive confirmation, counselling, treatment when indicated, and follow-up. Outreach activities should therefore include clear documentation, referral procedures, and mechanisms for tracking whether patients successfully reach the next stage of care.

Digital health may provide additional opportunities to strengthen communication and continuity. Where infrastructure and staffing permit, electronic records, telephone follow-up, appointment reminders, and remote professional consultation may help health workers coordinate care and maintain contact with patients. Digital tools can also support the collection and analysis of routine information, potentially helping managers identify missed appointments, medicine interruptions, or gaps in follow-up.

However, digital interventions should be introduced according to demonstrated needs and local feasibility. Reliable electricity, connectivity, equipment maintenance, staff training, language accessibility, data protection, and technical support all influence whether digital services can operate effectively. An intervention that depends on stable connectivity may not be appropriate for every location, and an electronic system that is poorly maintained may create additional administrative burdens without improving care.

Digital health must also avoid creating new inequalities. Patients may have limited access to mobile devices, affordable data, digital literacy, or private spaces for communication. Older adults and people with disabilities may require additional assistance, while confidentiality concerns may affect willingness to use certain technologies. Consequently, digital services should complement rather than eliminate accessible in-person care and non-digital communication options.

The most appropriate model for Timor-Leste is therefore likely to be a context-sensitive combination of facility-based care, community outreach, telephone communication where feasible, and digital systems where they can be maintained reliably. Decisions should be based on local evaluation rather than the assumption that technological innovation automatically improves health outcomes.

WHO's emphasis on primary health care and integrated service delivery supports the broader principle that technologies should strengthen accessible, coordinated care rather than become separate programmes disconnected from everyday clinical practice (WHO, 2019, 2023a). In this context, digital health is valuable when it helps patients receive appropriate care at the right time and supports health professionals in fulfilling their responsibilities.

4.2.8 Preventing Complications and Maintaining Continuity of Care

Prevention does not end when a chronic disease has been diagnosed. For people already living with diabetes, cardiovascular disease, chronic respiratory disease, or other long-term conditions, preventing complications is a central component of effective care. Disease management should aim to reduce avoidable deterioration, preserve functional ability, support psychological wellbeing, and enable individuals to participate as fully as possible in family and community life.

For diabetes, complication prevention may involve appropriate monitoring of blood glucose, blood pressure, and other cardiovascular risk factors, together with clinical assessment for relevant complications. Depending on individual circumstances and national protocols, care may include foot assessment, evaluation of kidney function, eye assessment, medication review, nutritional counselling, and support for recognising symptoms that require urgent attention. These services should be coordinated according to clinical need rather than delivered as unrelated procedures.

For cardiovascular disease, effective long-term management may involve monitoring blood pressure and lipid levels, appropriate medicines, support for smoking cessation,

dietary counselling, and rehabilitation following serious cardiovascular events. Patients who have experienced a stroke or heart attack may require continuing assessment of physical function, emotional wellbeing, and capacity to resume daily activities. The objective is not merely to prevent another clinical event but also to reduce disability and support recovery.

People with chronic respiratory disease may require assessment of symptoms, appropriate pharmacological management, advice about inhaler use when relevant, vaccination according to applicable recommendations, and measures to reduce exposure to tobacco smoke and other respiratory irritants. Prevention of complications also requires timely recognition of worsening symptoms and access to appropriate clinical assessment.

These examples illustrate why chronic disease services should be organised around continuing clinical needs rather than short-term encounters. Treatment plans need to be reviewed, medicines monitored, complications assessed, and patients supported as their circumstances change. WHO's essential NCD interventions provide a practical basis for organising key services within primary health care, particularly where specialised resources are limited (WHO, 2020a).

Continuity is also influenced by the patient's capacity to follow treatment. Medicine availability, cost and transport barriers, confusing instructions, adverse effects, family responsibilities, and limited understanding of the disease can interfere with adherence. Health professionals should explore these barriers respectfully before concluding that a patient is unwilling to follow advice. In many cases, the problem lies partly in how services are organised rather than in an individual's motivation.

Medication continuity is particularly important. Interruptions caused by stock shortages, difficulties obtaining prescriptions, or travel barriers can undermine long-term management. Health facilities should therefore establish appropriate systems for medicine procurement, stock monitoring, prescription review, and communication with patients. Where clinically appropriate, appointment arrangements and medicine collection procedures may be adapted to reduce avoidable burdens without compromising safety.

Patients and caregivers should receive clear guidance about warning signs that require urgent evaluation. They should also understand the purpose of scheduled follow-up and the importance of reporting changes in symptoms or difficulties with treatment.

Such communication can improve the likelihood that complications are recognised early, although education must always be supported by accessible services capable of responding to identified problems.

Preventing complications also requires attention to dignity and quality of life. Chronic disease may affect independence, employment, relationships, emotional wellbeing, and participation in community activities. Care should therefore consider the person's goals and preferences, not only laboratory values or clinical measurements. Integrated and people-centred services seek to coordinate interventions around the individual's needs and life circumstances rather than treating each diagnosis as an isolated technical problem (WHO, 2016; Kruk et al., 2018).

4.2.9 Monitoring, Evaluation, and Health Equity

A prevention strategy requires systematic monitoring to determine whether planned interventions are being implemented and whether they are producing meaningful improvements. Without reliable information, policymakers may know that activities have taken place but remain uncertain about who benefited, which populations were missed, and whether the burden of chronic disease is changing. Monitoring and evaluation should therefore be incorporated into programme design from the beginning rather than added after implementation.

A practical monitoring framework should distinguish between resources, activities, service outputs, and health outcomes. Resources may include trained personnel, equipment, essential medicines, and programme funding. Activities may include community education, preventive consultations, and outreach visits. Outputs may include the number of people assessed or referred, while outcomes may include improved follow-up, better management of risk factors, fewer avoidable complications, or more equitable access to services.

These indicators serve different purposes and should not be treated as interchangeable. An increase in the number of people screened may indicate expanded access, but it does not necessarily demonstrate improved health. Similarly, a high number of educational sessions does not establish that participants understood the messages or could adopt the recommended practices. Programme evaluation must therefore examine the quality, continuity, reach, and consequences of interventions.

For Timor-Leste, national and local data should be used together wherever possible. National surveys, routine health information systems, facility records, and relevant

international estimates provide different perspectives on population health. Each source has strengths and limitations, and discrepancies should be investigated rather than automatically interpreted as evidence that one source is incorrect. Differences in sampling, diagnostic criteria, reporting completeness, age coverage, and estimation methods can affect comparability.

The Ministry of Health's STEPS 2023 survey provides an important national source for understanding NCD-related risk factors among the population covered by the survey. Its findings should be interpreted according to the survey's methods, definitions, age range, and reported limitations. The results can inform prevention priorities, but they should not be used to make claims about populations or outcomes that the survey did not directly measure (Ministry of Health, Timor-Leste, 2023).

Equity-focused monitoring is particularly important because national averages can conceal substantial differences between population groups. Data should be examined, where reliable and appropriate, by age, sex, geographical location, socioeconomic circumstances, disability status, and other relevant characteristics. Such analysis can help identify whether particular groups face lower screening coverage, longer delays in diagnosis, less consistent access to treatment, or greater barriers to continuing care.

However, collecting disaggregated information requires appropriate safeguards. Personal health information should be protected, data collection should have a clear public health purpose, and reporting should avoid exposing individuals or small communities to identification or stigma. Data quality and confidentiality are both essential to maintaining public trust.

Evaluation should also include feedback from patients, caregivers, health workers, and communities. Quantitative indicators may show whether appointments were attended, but qualitative information can help explain why people missed follow-up, whether communication was understandable, and whether services were experienced as respectful. Combining these perspectives can support more responsive programme design and help identify practical improvements that routine statistics alone may not reveal.

The ultimate objective of monitoring is to strengthen accountability and learning. Evidence should inform decisions about resource allocation, workforce development, outreach arrangements, medicine availability, and preventive policy. When programmes are not achieving their intended results, the response should include

analysis of implementation barriers and opportunities for improvement rather than simply increasing activity counts.

This approach is consistent with the wider principle that health systems should be assessed according to their capacity to deliver effective, safe, equitable, and people-centred care, not merely the number of services provided (Kruk et al., 2018; WHO, 2019).

4.2.10 Strategic Priorities for Prevention in Timor-Leste

The preceding analysis suggests that prevention and early detection in Timor-Leste should be organised around a set of mutually reinforcing priorities. First, national policy should strengthen population-level measures addressing tobacco, harmful alcohol consumption, unhealthy diets, physical inactivity, and relevant environmental exposures. These interventions can reduce shared risk factors across several major NCD categories and should be supported by coordination between health authorities and other sectors.

Second, primary health care should be equipped to provide accessible risk assessment, early diagnosis, counselling, treatment initiation, and continuing monitoring. Preventive services are most effective when they are linked to functioning clinical pathways, essential medicines, diagnostic capacity, and clear referral arrangements. Expanding screening without strengthening follow-up risks creating unmet expectations and increasing the burden on facilities without securing corresponding health benefits.

Third, community-based prevention should be developed in collaboration with local institutions and adapted to differences in geography, language, literacy, and available resources. Community participation can strengthen trust, improve the relevance of educational messages, and help health services identify barriers experienced by households. Such participation should be supported by appropriate supervision, resources, and links to professional health care.

Fourth, prevention should be integrated into existing services across the life course. Schools, maternal and child health services, adult health consultations, and services for older people can all provide appropriate opportunities for health promotion and risk reduction. The precise interventions should reflect the needs of each group and should avoid treating all populations as though they face identical risks.

Fifth, health authorities should strengthen monitoring and evaluation so that prevention programmes can be assessed using reliable and meaningful indicators. National survey findings, routine data, and community feedback should inform priorities while their methodological limitations are acknowledged. Particular attention should be paid to groups whose access to prevention and treatment may be constrained by geographical or socioeconomic barriers.

Finally, preventive policy should be guided by long-term sustainability. Programmes should be designed around available institutional capacity and realistic financing, while progressively strengthening the resources required for wider implementation. Short-term campaigns may raise awareness, but lasting reductions in chronic disease risk require sustained public policy, consistent clinical practice, and continuing engagement with communities.

These priorities are aligned with WHO's recommendations on NCD prevention, essential interventions, and the strengthening of primary health care (WHO, 2017, 2020a, 2023a). Their implementation in Timor-Leste should be guided by national evidence, local consultation, and periodic review of outcomes.

4.2.11 Conclusion

Prevention, early detection, and integrated community-based care are essential components of a sustainable response to chronic diseases and NCDs in Timor-Leste. Their importance extends beyond reducing the incidence of disease. They also help prevent avoidable complications, preserve functional ability, protect household resources, and improve the capacity of individuals to participate in family, economic, and community life.

The analysis demonstrates that effective prevention cannot rely exclusively on individual behaviour change or isolated screening activities. It requires supportive public policies, healthier environments, accessible primary health care, competent health workers, reliable essential medicines, appropriate diagnostic services, and continuing relationships between health professionals and patients. Community participation and culturally appropriate health communication can strengthen these efforts, particularly when interventions are designed in response to local realities.

At the same time, prevention must be equitable. People should not be excluded from essential services because they live far from a health facility, have limited financial resources, encounter language barriers, or lack access to digital technologies.

Outreach and community-based services can help address these challenges, provided they are connected to a functioning system capable of delivering appropriate follow-up and treatment.

For Timor-Leste, the strategic objective should be to establish a prevention-oriented health system in which early identification of risk leads to timely action, community engagement strengthens rather than replaces public responsibility, and long-term disease management remains accessible and person-centred. Achieving this objective will require sustained coordination, evidence-informed decisions, and accountability for both health outcomes and equity.

The next section examines how these preventive and clinical priorities can be translated into policy commitments, institutional responsibilities, and a research agenda suited to the evolving health needs of Timor-Leste.

4.3 Policy Priorities, Strategic Recommendations, and a Research Agenda for Timor-Leste

4.3.1 Introduction: From Public Health Commitments to Sustainable Action

The prevention and control of chronic diseases and noncommunicable diseases (NCDs) require more than technical knowledge, clinical interventions, or the publication of national policy documents. Sustainable progress depends on the ability of a health system to translate policy commitments into coordinated action, allocate resources according to population needs, strengthen institutional capacity, and maintain accountability for results. For Timor-Leste, this challenge is particularly important because the development of effective NCD services must take place alongside continuing responsibilities in maternal and child health, infectious disease control, nutrition, emergency care, and the broader strengthening of primary health care.

Chronic diseases require a long-term policy perspective. Their risk factors may develop over many years, their clinical consequences may emerge gradually, and their management often continues throughout a person's life. Consequently, short-term interventions that focus exclusively on individual campaigns or isolated clinical activities cannot provide a sufficient response. National strategies must establish a coherent framework connecting prevention, early detection, treatment, rehabilitation,

palliative care where appropriate, and continuing support for people living with chronic conditions.

The World Health Organization (WHO) has repeatedly emphasised the importance of comprehensive, multisectoral action against NCDs, including the integration of preventive and clinical interventions into national health systems. Its global action plan and subsequent implementation roadmap provide a framework for countries to establish priorities, strengthen governance, and monitor progress according to national circumstances (WHO, 2013, 2023a). These principles are relevant to Timor-Leste, but their implementation must be grounded in the country's institutional capacity, epidemiological evidence, geographical realities, and available resources.

The National Health Sector Strategic Plan 2011–2030 provides an important historical and policy reference for understanding the country's long-term health system development priorities (Ministry of Health, Timor-Leste, 2011). However, the effectiveness of any national strategy depends on its implementation, financing, institutional coordination, and ability to respond to changing population needs. Chronic disease policy should therefore be assessed not only according to the existence of strategic documents but also according to whether essential services are accessible, medicines are available, health workers are adequately supported, and patients receive appropriate continuing care.

A central argument of this section is that Timor-Leste needs a coordinated, evidence-informed, and equity-oriented approach to chronic disease policy. This approach should not treat NCDs as an isolated programme competing with other health priorities. Instead, it should strengthen the health system's ability to prevent disease, manage multiple health conditions, and protect people from avoidable illness and financial hardship. The following analysis identifies key policy priorities, proposes strategic directions, and outlines a research agenda that can support the development of a more resilient and responsive national health system.

4.3.2 Strengthening National Governance and Policy Coordination

Effective NCD prevention begins with clear governance. National authorities need to establish policy objectives, define institutional responsibilities, coordinate relevant sectors, and create mechanisms through which implementation can be monitored. Without clear responsibilities, health initiatives may become fragmented across departments, programmes, municipalities, and development partners. Such

fragmentation can lead to duplicated activities in some areas and persistent service gaps in others.

Governance should begin with a shared understanding of national priorities. Cardiovascular diseases, diabetes, cancers, and chronic respiratory diseases require different clinical interventions, but they also share important risk factors and depend on several common health system capacities. National policy should therefore combine disease-specific measures with integrated approaches to prevention, risk assessment, essential medicines, clinical follow-up, and health information management.

The Ministry of Health should play a central coordinating role in establishing standards, setting priorities, supporting implementation, and monitoring progress. However, the determinants of chronic disease extend beyond the formal health sector. Education influences health literacy; agriculture and food policy affect nutritional environments; transport and urban planning influence opportunities for physical activity; environmental management affects exposure to air pollution; and fiscal policy can influence the affordability and availability of products associated with health risks.

A multisectoral approach requires practical coordination rather than occasional consultation. Relevant institutions need to understand their respective responsibilities, identify shared objectives, and agree on how progress will be assessed. For example, a national commitment to improve healthy eating should be linked to feasible actions involving nutrition education, food availability, public communication, and appropriate regulatory measures. Similarly, policies intended to increase physical activity should consider the safety and accessibility of public spaces and transport environments.

Governance also requires meaningful engagement with local administrations, health professionals, communities, and civil society. National policy may establish the overall direction, but implementation often depends on decisions made closer to the population. Local institutions can help identify service barriers, communicate community priorities, and adapt implementation to geographical and social circumstances. Their participation should be supported by clear guidance, appropriate resources, and accountability arrangements.

International partners may contribute technical assistance, funding, training, research capacity, and support for programme evaluation. Such contributions can be valuable, especially when national institutions are strengthening their capacity to address complex chronic diseases. Nevertheless, external support should reinforce national

ownership rather than create parallel structures that are difficult to maintain. Coordination should promote alignment with national priorities, strengthen local expertise, and support the gradual development of sustainable domestic capacity.

Policy coordination must also include periodic review. Epidemiological patterns, health service needs, technologies, and available evidence change over time. National priorities should therefore be revisited using updated population data, implementation findings, financial assessments, and feedback from patients and health workers. A policy framework that cannot learn from experience risks becoming disconnected from the realities it was designed to address.

Ultimately, effective governance is demonstrated through the consistent delivery of services and measurable improvements in population health. For Timor-Leste, the objective should be to establish a policy environment in which institutions understand their responsibilities, communities can participate meaningfully, resources are allocated transparently, and progress is assessed according to both health outcomes and equity (WHO, 2013, 2023a).

4.3.3 Establishing Sustainable Financing and Resource Allocation

The implementation of an effective chronic disease strategy requires predictable financing. Prevention programmes, diagnostic services, essential medicines, professional training, health information systems, and continuing clinical care all depend on resources that must be planned and sustained over time. When financing is inadequate or unpredictable, programmes may begin without sufficient capacity to maintain their activities, leaving patients with interrupted care and health workers with limited means to respond to identified needs.

A central policy challenge is to balance immediate health service demands with investments that generate benefits over the longer term. Chronic disease prevention may not always produce immediate, visible reductions in hospital admissions or mortality. Nevertheless, reducing exposure to major risk factors, identifying disease earlier, and improving long-term management can help prevent avoidable complications and reduce future pressure on health services. Resource allocation should therefore consider both current service needs and the longer-term consequences of delayed intervention.

The economic rationale for prevention should not be interpreted as a claim that every preventive programme automatically saves money. The costs and benefits of

interventions depend on their design, target population, implementation quality, and time horizon. Some interventions may improve health while requiring additional expenditure; others may reduce certain future costs without generating immediate budget savings. Policy decisions should therefore consider health benefits, affordability, feasibility, equity, and opportunity costs together.

For Timor-Leste, priority setting should be based on a transparent assessment of disease burden, intervention effectiveness, service gaps, and implementation capacity. Programmes addressing major shared risk factors may offer benefits across several disease categories. At the same time, patients with established chronic diseases need reliable access to clinical services and essential medicines. A balanced financing strategy must therefore support both population-level prevention and the continuity of care for people who already require treatment.

Resource allocation should also reflect geographical differences in service access. Facilities serving remote communities may face additional costs related to transport, outreach, supply distribution, equipment maintenance, and workforce retention. A financing model based solely on equal allocations to every facility may fail to account for these differences. Equitable resource distribution should consider population needs, geographical barriers, service responsibilities, and the cost of providing reliable care.

Essential medicines and diagnostic supplies deserve particular attention. A facility may have trained personnel and an appropriate clinical protocol, but these capacities have limited value when medicines, test supplies, or functioning equipment are unavailable. Financing arrangements should therefore support reliable procurement, stock monitoring, distribution, and maintenance, while reducing avoidable interruptions in treatment.

Financial protection is equally important. People living with chronic diseases may require repeated consultations, long-term medicines, laboratory assessments, transportation, and occasional specialist care. Even where direct medical services are subsidised, indirect expenses can impose significant burdens on households. Policies should assess whether patients can obtain necessary services without sacrificing food, education, housing, or other essential needs.

The WHO and World Bank monitoring framework for universal health coverage highlights the importance of assessing both service coverage and financial hardship

(WHO & World Bank, 2023). For Timor-Leste, this perspective suggests that progress should not be measured solely by whether a service exists. It should also consider whether people can reach it, obtain the necessary treatment, and continue care without experiencing avoidable financial distress.

Financial planning should include a clear distinction between short-term programme funding and recurrent expenditure. Pilot initiatives may depend on external funding, but their long-term value is limited if the country cannot sustain essential components after the pilot ends. New interventions should therefore be assessed for workforce requirements, medicine costs, supervision, data management, equipment maintenance, and other recurrent commitments before being expanded.

Sustainable financing is ultimately a question of public priorities and institutional responsibility. By linking budgets to evidence, service needs, and measurable outcomes, Timor-Leste can strengthen the foundation for a more equitable and resilient response to chronic diseases.

4.3.4 Developing the Health Workforce and Strengthening Professional Capacity

A national chronic disease strategy cannot succeed without a capable, adequately supported, and appropriately distributed health workforce. Chronic conditions require repeated clinical assessment, interpretation of diagnostic findings, treatment planning, counselling, medication management, recognition of complications, and coordination across different levels of care. These responsibilities place sustained demands on health professionals and require competencies that extend beyond the management of acute illness.

Workforce development should therefore be treated as a continuing investment rather than a one-time training activity. Initial education provides an essential foundation, but clinical knowledge and practice must be reinforced through continuing professional development, supervision, mentoring, and opportunities to apply evidence-based guidance. Health professionals also need access to clear clinical protocols that support consistent decision-making across facilities.

Nurses are particularly important in chronic disease prevention and management. Their responsibilities may include health education, screening support, monitoring, medication administration, patient counselling, follow-up coordination, and recognition of warning signs. In settings where physicians and specialists are limited, nurses can

contribute significantly to the continuity of primary care when their responsibilities are clearly defined and supported by appropriate training and referral arrangements.

Physicians remain essential for diagnosis, clinical decision-making, treatment adjustment, management of complications, and referral to specialist services when needed. Pharmacists and pharmacy personnel contribute to medication safety, patient education, appropriate dispensing, and monitoring of medicine availability. Laboratory personnel support diagnostic accuracy, while community health workers can help connect households with formal health services. The effectiveness of this multidisciplinary approach depends on communication and coordination rather than the isolated performance of individual professions.

For Timor-Leste, workforce planning should examine not only the total number of health professionals but also their geographical distribution, competencies, working conditions, and access to professional support. A concentration of expertise in urban facilities may leave remote communities with limited access to appropriate assessment and continuing management. Addressing this imbalance requires context-sensitive deployment strategies, support for rural practice, and mechanisms that enable health workers to maintain professional competence outside major centres.

Clinical protocols and structured training can help standardise essential care, but they should not replace professional judgement. Patients differ in age, comorbidities, functional capacity, treatment preferences, and social circumstances. Health workers need the capacity to adapt interventions safely while following established standards and seeking specialist support when a case exceeds their competence.

Workforce wellbeing also deserves attention. High workloads, limited supplies, inadequate supervision, and insufficient opportunities for professional development can affect staff retention and the quality of care. Chronic disease management requires sustained relationships with patients, making workforce continuity particularly valuable. Policies should therefore consider supportive supervision, fair working conditions, continuing education, and professional recognition as components of health system strengthening.

The health workforce must also be prepared to communicate respectfully with patients and families. Technical competence is indispensable, but it does not guarantee that patients understand their diagnosis or feel able to discuss barriers to treatment.

Training should therefore include communication skills, cultural responsiveness, ethical decision-making, confidentiality, and shared care planning.

Finally, professional development should be linked to service outcomes. Training programmes should be evaluated according to whether they improve clinical practice, patient understanding, appropriate referral, continuity of care, and the management of risk factors. Counting training sessions or participants is insufficient if the new knowledge cannot be applied because medicines, equipment, supervision, or referral services are unavailable.

A stronger chronic disease workforce will require coordination between educational institutions, professional bodies, health facilities, government agencies, and relevant development partners. Such collaboration can support locally relevant curricula, continuing professional development, applied research, and the gradual strengthening of national expertise (Atun et al., 2013; Kruk et al., 2018).

4.3.5 Strengthening Surveillance, Data Governance, and Evidence-Informed Policy

Reliable data are essential for determining which diseases and risk factors require priority attention, identifying inequalities, allocating resources, and evaluating the effects of public health interventions. Without adequate surveillance, policymakers may be unable to distinguish between genuine changes in disease patterns and changes in reporting, diagnosis, or data collection. A strong national response therefore requires a coherent information system that connects population surveys, routine health service records, disease-specific reporting, and research.

Timor-Leste's STEPS 2023 survey provides an important foundation for understanding NCD-related risk factors among the population covered by the survey (Ministry of Health, Timor-Leste, 2023). Its value lies not only in documenting risk but also in providing a basis for policy discussion, identifying evidence gaps, and informing future surveillance. However, survey findings represent particular measurements collected under defined methodological conditions. They should be interpreted alongside information about sampling, age coverage, measurement procedures, and limitations.

National surveys alone cannot provide every form of information required for chronic disease planning. They may describe selected risk factors in a population but cannot necessarily establish the quality of ongoing clinical care, the frequency of medicine interruptions, or the reasons patients fail to attend follow-up appointments. Routine

health information systems and focused operational research are needed to address these questions.

A strengthened surveillance system should support the consistent collection of relevant information on diagnoses, risk factors, treatment, referral, complications, and health outcomes. Data definitions should be sufficiently standardised to permit comparison across facilities and over time. Where possible, records should support the monitoring of individual care pathways while protecting personal information and ensuring that access is restricted to authorised users.

Data completeness and accuracy are particularly important. If diagnoses are not recorded consistently, if facilities report information using incompatible definitions, or if certain communities are systematically underrepresented, the resulting analysis may provide a misleading picture of population health. Strengthening surveillance therefore requires staff training, clear reporting procedures, appropriate technology, supportive supervision, and feedback that demonstrates how collected information contributes to better decisions.

International estimates, including those produced by the Global Burden of Disease study, can provide useful comparative perspectives on mortality, disability, and broader disease burden. However, these estimates may use statistical modelling to address incomplete or unavailable data. They should not automatically be treated as equivalent to directly observed national measurements. Their assumptions, uncertainty, and methodological differences need to be considered when comparing estimates with national survey findings or routine records (Institute for Health Metrics and Evaluation [IHME], 2024).

For policy purposes, evidence should be triangulated. If survey data, routine clinical records, and international estimates appear to tell different stories, analysts should investigate whether the differences arise from population coverage, diagnostic access, data quality, time periods, or estimation methods. Such discrepancies can reveal weaknesses in the evidence base and help identify where additional research or surveillance is needed.

Data governance is equally important. Health information can be sensitive, and individuals must be protected from unnecessary disclosure, discrimination, or misuse. Clear rules are needed regarding data collection, storage, access, sharing, retention, and publication. Researchers and health authorities should use appropriate ethical

review and privacy safeguards, especially when information can identify individuals or small communities.

The long-term objective should be to develop a system in which information is used routinely for learning and improvement. Facility managers should be able to identify service gaps, municipal and national authorities should be able to compare performance fairly, and communities should receive appropriate information about health priorities and progress. Data collection becomes meaningful when it supports decisions, improves accountability, and contributes to better health outcomes.

4.3.6 Integrating Prevention and Chronic Disease Services into Primary Health Care

Primary health care should serve as the operational foundation of the national response to chronic diseases. It provides an opportunity to combine prevention, early detection, treatment, health education, continuing monitoring, and referral within services that are closer to where people live. This is particularly important for conditions that require repeated contact with health professionals and long-term management rather than a single episode of treatment.

Integration requires more than placing several services in the same building. It involves coordinated clinical pathways, shared information, clear responsibilities, appropriate supplies, and continuity between community-level services and higher levels of care. Patients should not have to navigate unnecessarily complicated systems to obtain routine monitoring, receive prescriptions, or arrange a referral.

For Timor-Leste, the design of integrated primary care should reflect local capacity and avoid introducing complex service models that cannot be maintained. A practical approach may begin with standardised risk assessment, clear protocols for common chronic conditions, reliable medicine management, scheduled follow-up, and referral arrangements for complications. Additional services can be developed progressively as workforce capacity, infrastructure, and financing improve.

The management of multiple chronic conditions deserves particular attention. A patient may have diabetes, hypertension, and another long-term condition requiring different assessments and treatments. If each condition is managed through an entirely separate programme, the patient may experience repeated appointments, conflicting advice, and unnecessary travel. Coordinated care can reduce fragmentation while maintaining appropriate disease-specific clinical standards.

Clinical pathways should define which cases can be managed within primary care, which require consultation with a more experienced clinician, and which need urgent or specialist referral. These pathways should be supported by communication between facilities so that the transfer of a patient does not result in loss of essential clinical information. When patients return to their local facility after specialist assessment, the follow-up plan should be clearly documented.

Continuity also depends on the reliable availability of essential medicines and diagnostic services. Clinical protocols cannot deliver their intended benefits when treatment is interrupted by stock shortages or when necessary tests cannot be completed. Procurement, distribution, maintenance, and supply monitoring should therefore be considered integral parts of chronic disease policy rather than separate administrative concerns.

WHO's Package of Essential Noncommunicable Disease Interventions offers a framework for organising appropriate interventions in primary health care, including in resource-constrained settings (WHO, 2020a). Its application in Timor-Leste should be guided by national clinical priorities, the capabilities of local facilities, and the availability of referral services.

Integration should also include rehabilitation and support for people who experience chronic disease complications. Depending on individual needs, patients may require assistance with mobility, daily activities, psychological adjustment, or return to work and community participation. Where specialised rehabilitation resources are limited, service planning should identify feasible referral arrangements and opportunities to strengthen support within existing health and community structures.

The success of integrated primary care should ultimately be assessed through the experience and outcomes of patients. Relevant measures include whether individuals receive timely assessment, understand their treatment plans, can obtain prescribed medicines, attend appropriate follow-up, and receive coordinated care when complications arise. This patient-centred perspective is consistent with international evidence that effective health systems must provide not merely contact with services but high-quality care that improves health and responds to people's needs (Kruk et al., 2018; WHO, 2019).

4.3.7 Advancing Health Equity and Protecting Vulnerable Populations

Health equity should be treated as a central objective of chronic disease policy rather than an additional consideration applied after programmes have been designed. Chronic diseases affect people across society, but the ability to prevent illness, obtain diagnosis, continue treatment, and manage complications is influenced by socioeconomic conditions, geographical access, education, disability, and the availability of family or community support.

An equitable policy framework recognises that equal treatment does not always require identical service arrangements. People facing greater barriers may need additional outreach, transport support, accessible communication, or more flexible service delivery to obtain a comparable opportunity for good health. The purpose is not to provide unnecessary differences in care but to address avoidable disadvantages that prevent certain groups from benefiting from available services.

For Timor-Leste, geographical barriers deserve particular attention. Distance from facilities, transport costs, difficult travel conditions, and the availability of health professionals can influence whether people receive preventive assessments and continuing treatment. Policy responses should be based on local evidence identifying where these barriers are most significant and which service arrangements are feasible.

Socioeconomic conditions also influence chronic disease risk and management. Households with limited resources may have difficulty obtaining nutritious foods, travelling for repeated consultations, or paying indirect costs associated with long-term treatment. Prevention programmes that focus exclusively on individual responsibility may therefore have limited effectiveness unless they also consider the conditions that shape people's choices.

Health equity should be incorporated into programme design, budgeting, and evaluation. Authorities should examine whether preventive services are reaching underserved groups, whether essential medicines are consistently available across facilities, and whether referral pathways are accessible to people living far from specialised services. Where data permit, service indicators should be disaggregated to identify systematic differences between population groups.

Equity also requires respectful and non-discriminatory care. People living with chronic diseases may experience stigma related to body weight, tobacco use, alcohol consumption, disability, or other aspects of their health and social circumstances. Such stigma can discourage individuals from seeking assistance and weaken the

relationship between patients and health professionals. Staff training and institutional standards should promote confidentiality, respect, and shared decision-making.

Older adults and people living with disabilities may require additional support to access and continue care. Health facilities should consider physical accessibility, communication needs, medication complexity, mobility limitations, and the involvement of caregivers where appropriate. However, support should preserve the individual's autonomy and should not assume that every older person or person with a disability has the same needs.

Gender-sensitive service planning may also be important where responsibilities, social expectations, safety, or access to resources influence health-seeking behaviour. The appropriate response should be guided by local evidence and consultation rather than unsupported assumptions about the experiences of women or men.

The social determinants of health framework emphasises that reducing health inequalities requires action on the conditions in which people are born, grow, live, work, and age (Marmot et al., 2008). Applied to chronic disease policy, this means that clinical services must be accompanied by broader efforts to improve access, strengthen health literacy, reduce preventable exposure to risk, and ensure that disadvantaged populations are not systematically excluded from the benefits of prevention and treatment.

4.3.8 Building a National Research Agenda for Chronic Diseases

A sustainable chronic disease strategy requires a continuing programme of research. Research helps clarify the distribution of disease, identify determinants of risk, evaluate interventions, understand patient experiences, and determine how health services can be organised more effectively. For Timor-Leste, a national research agenda should prioritise questions that can inform decisions and strengthen local capacity rather than simply reproduce studies conducted in other settings.

The first priority is to strengthen epidemiological evidence. Nationally representative surveys and routine health information can help describe risk factors and disease patterns, but important gaps may remain regarding diagnosis, treatment, complications, disability, mortality, and differences between geographical or socioeconomic groups. Future studies should identify these gaps and use appropriate methods to address them.

The second priority is to investigate the implementation of chronic disease services. Evidence is needed on whether clinical protocols are used consistently, whether essential medicines and diagnostic services are available, how patients move through referral systems, and what factors contribute to missed appointments or interrupted treatment. Such research can identify practical improvements that may be more immediately useful than introducing complex new programmes without understanding existing barriers.

The third priority is to evaluate prevention interventions. Health education, community outreach, tobacco-control measures, dietary interventions, physical activity programmes, and early-detection initiatives should be assessed according to their reach, acceptability, feasibility, costs, and outcomes. Evaluations should consider whether programmes are effective in different settings and whether their benefits extend to populations that face the greatest barriers to care.

The fourth priority is to investigate the economic consequences of chronic disease. Research should examine direct health expenditure, transport costs, lost income, unpaid caregiving, and other effects on households. These findings can help policymakers understand the relationship between chronic disease, poverty, workforce participation, and financial protection. Economic evaluations can also support transparent decisions about which interventions should be prioritised within available resources.

The fifth priority is to strengthen research on patient and community experiences. Quantitative measures can indicate whether services are being used, but qualitative studies can explain why people seek care late, discontinue treatment, experience difficulties with health communication, or prefer particular service arrangements. Interviews, focus groups, participatory research, and carefully designed mixed-methods studies can contribute valuable insights when conducted ethically and interpreted in context.

The sixth priority is to develop research on health system resilience and workforce capacity. Studies may examine training needs, staff distribution, medicine supply chains, referral coordination, the use of health information systems, and the feasibility of integrating chronic disease services into existing primary health care. These questions are particularly important where resources are limited and policy choices must balance several competing demands.

A seventh priority is to assess the appropriate role of digital health. Research should determine whether digital tools improve continuity, communication, data quality, and access in specific Timorese settings. Evaluations should include not only potential benefits but also costs, maintenance requirements, workforce demands, privacy risks, and the possibility of excluding people who lack reliable access to technology.

Finally, research should be developed through partnerships that strengthen national expertise. Collaboration among the Ministry of Health, universities, health facilities, professional organisations, communities, and international research institutions can support training, mentorship, data analysis, and publication. Partnerships should promote local leadership and ensure that findings are communicated to decision-makers and communities in forms that can inform practice.

Proposed research priorities

Research area	Priority research questions	Potential policy contribution
Epidemiology and surveillance	What are the major gaps in national NCD data, and how can they be addressed?	Improved monitoring and priority setting
Primary health care	What barriers affect diagnosis, follow-up, referral, and treatment continuity?	More effective service organisation
Prevention	Which interventions are feasible, acceptable, and effective in different local settings?	Better targeting of prevention resources
Health equity	Which populations face the greatest barriers to preventive and clinical services?	Fairer distribution of resources
Household economics	How do chronic diseases affect household expenditure, income, and caregiving?	Stronger financial protection policies

Health workforce	What competencies, supervision, and distribution strategies are needed?	Better workforce planning
Digital health	Which technologies improve care without creating new inequalities?	Evidence-based digital investment

This agenda should be treated as a flexible framework rather than a fixed list of projects. Priorities should be refined through consultation with national institutions, health professionals, communities, and researchers. Research questions should also be assessed according to feasibility, ethical requirements, resource availability, and their likely contribution to policy and practice.

4.3.9 Ethical Principles, Accountability, and Responsible Implementation

Chronic disease policy has an important ethical dimension because decisions about resource allocation, screening, treatment, and prevention affect people's opportunities to live healthy and dignified lives. Public health authorities must balance population-level benefits with individual rights, including privacy, informed decision-making, non-discrimination, and access to appropriate care.

The ethical principle of justice requires that essential services be distributed according to defensible criteria, taking account of health needs and barriers to access. Where resources are limited, priorities should be established transparently, with attention to effectiveness, urgency, feasibility, and equity. Decisions should not disadvantage individuals simply because they live in remote communities or have limited ability to advocate for themselves.

Respect for autonomy is equally important. Patients should receive information that enables them to understand their condition and participate meaningfully in treatment decisions. Health professionals should explain the expected benefits and risks of interventions in accessible language and respect the individual's right to ask questions and express preferences. Family participation can be valuable, but it should not override the patient's wishes or confidentiality.

Prevention programmes must also protect against stigma. Public communication that portrays chronic disease as the result of personal weakness may undermine dignity and discourage people from seeking assistance. An ethical approach recognises the complex interaction between individual behaviour, social conditions, biological factors,

and access to resources. It supports personal responsibility without ignoring structural barriers.

Research and surveillance require similar safeguards. Data should be collected for legitimate purposes, protected against unauthorised disclosure, and used in ways that do not expose individuals or communities to avoidable harm. Research involving human participants should follow applicable ethical review requirements and use appropriate consent procedures, privacy protections, and transparent communication about the intended use of information.

Accountability requires that institutions explain what they intend to achieve, how resources will be used, and how progress will be evaluated. Indicators should be selected carefully so that institutions are not rewarded merely for increasing the number of consultations or screening activities without improving the quality of care. Evaluation should consider service accessibility, safety, continuity, health outcomes, patient experience, and financial protection.

Public accountability can be strengthened through clear reporting, periodic review, and opportunities for community feedback. Such mechanisms should not be reduced to administrative compliance. Their purpose is to identify problems, correct ineffective practices, and ensure that policy decisions remain responsive to population needs.

Responsible implementation also requires realistic expectations. A national strategy cannot resolve every chronic disease challenge immediately. Progress may depend on improvements in workforce capacity, medicine procurement, infrastructure, data systems, and intersectoral coordination that take time to establish. Phased implementation can help institutions test approaches, learn from experience, and expand programmes when adequate capacity is available.

The ethical objective is to ensure that the development of chronic disease services improves not only technical performance but also the fairness and humanity of the health system. A programme should be judged by whether it helps people obtain appropriate care, protects their dignity, and reduces avoidable disadvantage—not merely by whether it meets an administrative target.

4.3.10 A Phased Strategic Framework for Timor-Leste

The recommendations developed in this section can be organised into a phased strategic framework. A phased approach does not imply that prevention, clinical care,

and equity should be addressed separately. Rather, it recognises that different activities may require different levels of preparation, investment, and institutional capacity.

The first phase should focus on establishing a clear baseline and strengthening essential functions. Priorities may include reviewing existing policies, identifying service gaps, assessing workforce and medicine availability, clarifying clinical protocols, and improving the use of existing surveillance information. This phase should also establish responsibilities for coordination and define indicators that can be monitored consistently.

The second phase should focus on strengthening the delivery of essential services. Activities may include improving routine risk assessment, strengthening follow-up systems, expanding appropriate community outreach, supporting professional development, and improving referral communication. Implementation should be adapted to the needs and capacities of different facilities rather than assuming that every location can immediately provide an identical range of services.

The third phase should consolidate quality improvement and equity-focused implementation. Authorities should examine whether essential services are reaching underserved populations, whether clinical pathways are functioning reliably, and whether medicine interruptions or referral delays remain common. Programme adjustments should be informed by routine data, patient feedback, and targeted research.

A further phase should focus on sustained institutional learning and progressive expansion. Interventions that demonstrate value and can be maintained should be integrated into regular health system operations. New technologies or specialised services should be introduced where their expected benefits, costs, workforce requirements, and maintenance needs have been adequately assessed.

The timing of these phases should be determined by national priorities, available resources, and institutional readiness rather than by arbitrary deadlines. Some measures, such as clear clinical protocols and improved communication, may be feasible relatively quickly, while broader changes in infrastructure, workforce distribution, and financing may require longer-term planning.

The following framework summarises the principal areas for action.

Strategic domain	Priority actions	Illustrative indicators
Governance	Clarify responsibilities and strengthen multisectoral coordination	Documented plans, assigned responsibilities, periodic reviews
Financing	Link resource allocation to needs and service continuity	Budget execution, essential medicine availability, financial protection measures
Workforce	Strengthen competencies, supervision, and workforce distribution	Training coverage, supervision records, service availability
Primary health care	Integrate prevention, diagnosis, treatment, and follow-up	Continuity of care, appropriate follow-up, referral completion
Prevention	Address shared behavioural and environmental risk factors	Reach of interventions, risk assessment coverage, programme outcomes
Surveillance	Improve data quality, interpretation, and use	Completeness, timeliness, consistency, and use of reports
Equity	Identify and address barriers affecting underserved groups	Disaggregated service coverage and access indicators
Research	Develop locally relevant studies and evaluation capacity	Completed evaluations, research partnerships, documented policy use

These indicators are illustrative and should be refined according to national information systems, available data, and agreed policy objectives. They should not be interpreted as evidence that current performance is either adequate or inadequate. Baseline measurements are needed before realistic targets can be established.

A phased framework also requires periodic reassessment. If new evidence reveals important service gaps or if implementation produces unexpected consequences,

priorities should be revised accordingly. The ability to adapt is a sign of responsible governance, not policy failure.

4.3.11 Conclusion: Building a Sustainable and People-Centred Response

The future of chronic disease prevention and management in Timor-Leste will depend on the country's capacity to connect national policy with effective implementation. Chronic diseases require sustained action across prevention, early detection, clinical management, rehabilitation, research, and financial protection. These responsibilities cannot be fulfilled through isolated projects or short-term campaigns alone; they require coordinated institutions, reliable resources, a capable workforce, and a health system that learns from evidence and experience.

The policy priorities identified in this section emphasise the importance of governance, sustainable financing, workforce development, surveillance, integrated primary health care, health equity, and locally relevant research. These priorities are mutually reinforcing. Reliable information supports better resource allocation; adequate financing strengthens service continuity; workforce development improves the quality of care; and equity-focused implementation helps ensure that progress reaches people who face the greatest barriers.

For Timor-Leste, a central objective should be to strengthen existing health system foundations while progressively expanding the capacity to prevent and manage chronic conditions. National policy should be ambitious enough to respond to the long-term burden of NCDs but sufficiently realistic to account for workforce, infrastructure, financial, and geographical constraints. Evidence-informed sequencing can help translate broad commitments into practical and sustainable action.

The success of this strategy should ultimately be measured in human terms. It should be reflected in whether people can obtain timely assessment, receive appropriate treatment, maintain continuity of care, avoid preventable complications, and live with dignity despite chronic illness. It should also be reflected in whether households are protected from avoidable financial hardship and whether geographical or socioeconomic circumstances cease to determine who benefits from essential health services.

The prevention and control of chronic diseases are therefore not solely clinical or administrative responsibilities. They are part of a wider commitment to human development, social justice, and the strengthening of national wellbeing. By combining

effective public policy with accessible services, community participation, responsible research, and accountable implementation, Timor-Leste can build a more resilient response to the evolving health needs of its population.

Conclusion

Chronic diseases and noncommunicable diseases have become central concerns for public health systems worldwide. Their significance lies not only in the mortality they cause but also in their contribution to disability, reduced quality of life, household financial pressure, and wider socioeconomic disadvantage. Addressing these conditions requires an understanding of epidemiological patterns, risk factors, disease burden, and the capacity of health systems to deliver sustained and equitable care.

This book has examined the global and local landscapes of chronic diseases, the epidemiological characteristics of major NCDs, the use of disease burden indicators such as disability-adjusted life years (DALYs), and the strategic challenges facing health services in Timor-Leste. Across these areas, a consistent conclusion emerges: chronic disease prevention and management cannot be separated from the broader social, economic, institutional, and environmental conditions that shape population health.

The epidemiological analysis demonstrates the importance of distinguishing between disease occurrence, risk exposure, premature mortality, and non-fatal health loss. Indicators such as years of life lost (YLLs), years lived with disability (YLDs), and DALYs can help policymakers compare health problems and establish priorities. However, these indicators must be interpreted carefully, with attention to data quality, methodological assumptions, uncertainty, and the differences between directly observed information and modelled estimates.

For Timor-Leste, the development of a strong evidence base remains essential. National surveys, routine health information, clinical records, and international estimates each provide different forms of evidence. Used together and interpreted appropriately, they can help identify priorities, reveal information gaps, and guide the progressive strengthening of surveillance and service delivery. The STEPS 2023 survey provides an important contribution to understanding NCD-related risk factors within its defined population and methodological scope (Ministry of Health, Timor-Leste, 2023).

The analysis also highlights that the major chronic disease categories share important risk factors and health system requirements. Tobacco use, harmful alcohol consumption, unhealthy diets, physical inactivity, and environmental exposures contribute to preventable disease, while elevated blood pressure, blood glucose, and other clinical risk factors can increase the likelihood of complications. Effective prevention therefore requires coordinated population-level policies, appropriate early detection, and continuing support for people already living with chronic conditions (WHO, 2017, 2023a).

Nevertheless, the capacity to benefit from prevention and treatment is not equally distributed. Geographical barriers, socioeconomic disadvantage, limited health literacy, shortages of health professionals, and interruptions in essential medicines can affect access to care. Consequently, a national response must pursue health equity as an explicit objective. The existence of a service is not sufficient if people cannot reach it, afford the indirect costs of attending, or continue treatment over time.

Primary health care should remain the foundation of an integrated response. It provides opportunities to connect risk assessment, health education, diagnosis, treatment, monitoring, and referral within a continuing relationship between health professionals and patients. Community-based services can strengthen this approach by supporting communication and reaching populations who face access barriers. Digital health may also contribute where infrastructure, technical support, privacy protections, and local feasibility permit. These approaches should complement one another within a coordinated system rather than operate as disconnected initiatives (WHO, 2016, 2019, 2020a).

The socioeconomic consequences of chronic disease further demonstrate the importance of prevention and financial protection. Illness can reduce income, increase household expenditure, create caregiving responsibilities, and affect educational and employment opportunities. Such consequences may reinforce existing inequalities, particularly when families lack the resources needed to maintain long-term treatment. Health policy must therefore consider the effects of chronic disease beyond clinical outcomes and recognise the relationship between health, poverty, social participation, and human development (Marmot et al., 2008; WHO & World Bank, 2023).

Moving forward, Timor-Leste will need a coordinated strategy that strengthens national governance, secures sustainable financing, develops workforce capacity, improves surveillance, and supports locally relevant research. Progress should be based on

transparent priorities and realistic implementation plans, with regular evaluation of service quality, health outcomes, and equity. National institutions, universities, health professionals, communities, and development partners can contribute to this process when collaboration reinforces national ownership and builds long-term local capacity.

Ultimately, the response to chronic diseases should be judged not only by the number of programmes established or services delivered but also by the lives improved through their implementation. A resilient health system is one that can prevent avoidable illness, recognise disease early, provide appropriate treatment, protect patients from unnecessary hardship, and maintain care with dignity and respect.

For Timor-Leste, this commitment is inseparable from the country's broader development aspirations. Investing in chronic disease prevention and care means investing in the health of families, the capacity of communities, the productivity of the population, and the wellbeing of future generations. The challenge is substantial, but a sustained, evidence-informed, equitable, and people-centred approach can provide a practical foundation for strengthening population health and advancing human development.

APPENDIX A

Technical Guide to NCD Epidemiological Indicators and Disability-Adjusted Life Years (DALYs)

A.1. Introduction

The assessment of chronic diseases and noncommunicable diseases (NCDs) requires reliable epidemiological indicators that describe how frequently diseases occur, which populations are most affected, how many people die prematurely, and how much disability is experienced over time. These indicators provide an essential foundation for understanding population health, comparing disease patterns, evaluating interventions, and identifying priorities for health policy.

In Timor-Leste, epidemiological measurement is particularly important because health planning must respond to changing population needs while accounting for geographical differences, resource constraints, and the capacity of health services to collect and use reliable information. A clear understanding of epidemiological indicators can help researchers, health professionals, policymakers, and students interpret available evidence without confusing observed national data with estimates produced by international modelling exercises.

This appendix presents key indicators relevant to NCD surveillance and planning. It also explains the concepts of years of life lost (YLLs), years lived with disability (YLDs), and disability-adjusted life years (DALYs), which are widely used to estimate the burden of disease (Bonita et al., 2006; GBD 2019 Diseases and Injuries Collaborators, 2020).

A.2. Core Epidemiological Indicators

A.2.1. Prevalence

Prevalence measures the proportion of a population that has a particular disease or health condition at a specified time or during a defined period. It is useful for estimating the number of people who may require ongoing treatment, monitoring, rehabilitation, and long-term support.

$$\text{Prevalence} = \frac{\text{Existing cases}}{\text{Total population}} \times 100$$

Illustrative example: If 500 people in a population of 10,000 have diabetes at the time of assessment, the prevalence is 5%.

This example is hypothetical and does not represent an estimate for Timor-Leste. In practice, prevalence estimates should specify the population studied, the diagnostic criteria applied, the reference period, and whether cases were identified through self-report, clinical assessment, laboratory testing, or medical records.

A.2.2. Incidence

Incidence measures the occurrence of new cases in a population over a defined period. It helps researchers understand the rate at which a disease develops and may be particularly useful for evaluating prevention strategies.

Cumulative incidence can be calculated as:

\[\text{Cumulative incidence}= \frac{\text{New cases during a period}} {\text{Population at risk at the beginning}} \]

Illustrative example: If 40 new cases of diabetes are identified among 2,000 people initially free of diabetes during one year, the cumulative incidence is 2% over that year, assuming the population and case ascertainment are appropriately defined.

Incidence is not interchangeable with prevalence. Prevalence reflects the existing burden of disease, while incidence measures new disease occurrence. A chronic condition may have high prevalence because people live with it for many years, even when the number of new cases remains relatively stable.

A.2.3. Mortality Rate

The mortality rate measures the frequency of deaths in a population during a specified period. Disease-specific mortality refers to deaths attributed to a particular cause, while all-cause mortality includes deaths from all causes.

\[\text{Cause-specific mortality rate}= \frac{\text{Deaths from a specified cause}} {\text{Population during the period}} \times 100{,}000 \]

Rates are often expressed per 100,000 population to facilitate comparisons between populations of different sizes. Interpretation requires attention to the quality of death registration, cause-of-death certification, population estimates, and the period under analysis.

A.2.4. Case-Fatality Proportion

The case-fatality proportion estimates the proportion of identified cases of a disease that result in death within a specified period.

\[\text{Case-fatality proportion}= \frac{\text{Deaths among identified cases}} {\text{Total identified cases}} \times 100 \]

It can help assess the severity of an acute event, such as a stroke, or outcomes among people diagnosed with a specific condition. However, it should not be interpreted as a direct measure of population mortality because it depends on which cases are detected, how cases are defined, and the length of follow-up.

A.2.5. Age-Standardised Rates

Age-standardisation adjusts rates to a common reference age distribution. It is useful when comparing populations with different age structures because the risk of many NCDs increases with age.

An age-standardised rate should not be confused with a crude rate. A crude rate describes the observed population experience, whereas an age-standardised rate supports comparisons under a standardised age structure. Reports should identify the standard population used and should apply consistent methods when comparing countries, regions, or periods.

A.3. Measuring Premature Mortality: Years of Life Lost (YLLs)

Years of life lost quantify the loss of expected years of life resulting from premature death. The concept recognises that deaths occurring at younger ages generally represent a greater loss of potential life-years than deaths occurring at older ages.

A simplified expression is:

\[YLL=\sum_x D_x \times L_x \]

Where:

- \(D_x\) represents the number of deaths at age \(x\).
- \(L_x\) represents the standard remaining life expectancy at that age.

The calculation uses a specified reference life table or standard life expectancy schedule. Consequently, YLL estimates depend on the standard used and on the quality and completeness of mortality data.

For NCD planning, YLLs can help identify conditions that cause substantial premature mortality. For example, cardiovascular disease that causes deaths in working-age adults may contribute considerably to YLLs, even if another condition produces a larger number of deaths at older ages. This makes YLLs useful alongside death counts and mortality rates rather than as a replacement for them.

A.4. Measuring Non-Fatal Health Loss: Years Lived with Disability (YLDs)

Years lived with disability estimate the time spent in a health state associated with a degree of health loss. In contemporary Global Burden of Disease (GBD) methodology, YLD calculations generally combine the prevalence of a health state with a disability weight reflecting its severity.

A simplified formula is:

$$ YLD=\sum_s P_s \times DW_s $$

Where:

- P_s represents the time lived in health state s, commonly operationalised using prevalence in the relevant calculation.
- DW_s represents the disability weight assigned to that health state.

Disability weights range from 0, representing a state equivalent to full health, to 1, representing a state equivalent to death in the DALY framework. They are not individual clinical scores and should not be interpreted as a complete assessment of a person's dignity, social participation, or lived experience.

YLDs are important for chronic conditions that may cause long-term disability without immediately causing death. Examples include complications of diabetes, limitations following stroke, chronic respiratory disease, and some cancers. The magnitude of YLDs depends on the prevalence of the relevant health states, their severity, and the methodology used to estimate disability.

A.5. Disability-Adjusted Life Years (DALYs)

Disability-adjusted life years combine premature mortality and non-fatal health loss into a single summary measure.

$$ \boxed{DALYs=YLLs+YLDs} $$

One DALY represents one year of healthy life lost under the DALY framework. DALYs can be used to compare the relative contribution of different diseases and risk factors to population health loss, although their interpretation depends on consistent definitions and methods (GBD 2019 Diseases and Injuries Collaborators, 2020).

Hypothetical example of DALY calculation

Component	Illustrative value
Years of life lost (YLLs)	800
Years lived with disability (YLDs)	350
Total DALYs	1,150

In this hypothetical example, the disease accounts for 1,150 DALYs. These figures are provided solely to demonstrate the calculation; they are not observed data or estimates for Timor-Leste.

DALYs should be interpreted alongside other indicators. A condition may cause substantial premature mortality, a high level of long-term disability, or both. Separating YLLs and YLDs helps policymakers understand these different pathways of health loss and select responses appropriate to the disease burden.

A.6. Comparing Indicators and Interpreting Evidence

No single epidemiological indicator provides a complete account of a disease. Prevalence estimates the existing number or proportion of affected people; incidence describes new disease occurrence; mortality measures deaths in a population; YLLs quantify premature mortality; YLDs estimate non-fatal health loss; and DALYs combine fatal and non-fatal health loss.

For Timor-Leste, responsible comparisons require careful attention to the source and nature of the evidence. National surveys, routine health information systems, civil registration, hospital records, and international GBD estimates may use different populations, time periods, diagnostic criteria, and modelling assumptions. Their results should not be treated as directly interchangeable.

When reporting an estimate, researchers should document, wherever possible:

1. The source and year of the data.

2. The population and geographical coverage.
3. The case definition and diagnostic method.
4. The numerator and denominator used.
5. Whether the result is crude or age-standardised.
6. Whether the figure is directly observed, survey-derived, modelled, or otherwise estimated.
7. The principal limitations affecting interpretation.

These practices improve transparency and reduce the risk that uncertainty will be mistaken for a precise measurement of the national situation.

A.7. Using Epidemiological Indicators for Health Planning

Epidemiological indicators become useful for policy when they are linked to clearly defined decisions. Prevalence may inform estimates of the demand for continuing medication and follow-up. Incidence may help evaluate prevention programmes. Mortality and YLLs can identify conditions associated with premature death, while YLDs can draw attention to long-term disability and rehabilitation needs. DALYs can support comparative priority-setting when interpreted together with cost, feasibility, equity, and community preferences.

In Timor-Leste, these indicators should be used to strengthen—not substitute for—clinical judgement, community knowledge, and local consultation. The absence of reliable data should be recognised as an information gap rather than interpreted as evidence that a health problem does not exist.

A.8. Conclusion

A technically sound understanding of epidemiological indicators is essential for assessing chronic diseases and NCDs. Prevalence, incidence, mortality, YLLs, YLDs, and DALYs describe different dimensions of population health and should be interpreted together. For Timor-Leste, the value of these measures will depend not only on their mathematical definitions but also on data quality, transparent reporting, consistent methodology, and their meaningful use in health planning. Properly applied, they can support more equitable allocation of resources and a more evidence-informed response to chronic disease.

APPENDIX B

Proposed NCD Monitoring and Research Framework for Timor-Leste

An Illustrative Framework for Strengthening Health Information, Evidence-Informed Policy, and Population Health Equity

B.1. Introduction

Effective prevention and control of chronic diseases and noncommunicable diseases (NCDs) require more than clinical services and public health campaigns. They also depend on the capacity of a health system to collect reliable information, identify population needs, monitor changes over time, evaluate interventions, and translate evidence into practical decisions. Without a consistent monitoring framework, policymakers may struggle to determine which populations experience the greatest burden, whether prevention programmes are reaching those who need them, and whether investments in health services are producing meaningful improvements.

For Timor-Leste, strengthening NCD monitoring is particularly important in the context of geographical differences in access to healthcare, resource limitations, changing population health needs, and the need to integrate long-term disease management into primary healthcare. National surveys, routine health information systems, hospital records, civil registration, and research studies can each contribute valuable information, but their usefulness depends on data quality, consistency, accessibility, and appropriate interpretation (Bonita et al., 2006; Kruk et al., 2018; World Health Organization [WHO], 2023).

This appendix proposes an illustrative framework for monitoring NCDs and developing a national research agenda. It is intended as an academic and policy-planning resource, not as a description of an officially adopted Timor-Leste monitoring system. The indicators, proposed reporting intervals, and potential data sources should be reviewed with the Ministry of Health, relevant institutions, healthcare professionals, researchers, and communities before implementation.

B.2. Objectives of the Proposed Framework

The proposed framework has six principal objectives:

1. Strengthen epidemiological surveillance by supporting the systematic collection and interpretation of information on NCD prevalence, incidence where measurable, mortality, disability, and risk factors.
2. Improve health service monitoring by examining access to screening, diagnosis, treatment, referral, and continuing care.
3. Identify health inequalities by analysing differences according to age, sex, geographical location, socioeconomic circumstances, and other ethically appropriate population characteristics.
4. Support evidence-informed policy by linking epidemiological findings to planning, resource allocation, prevention, and service improvement.
5. Develop national research capacity by identifying priority questions and encouraging collaboration among health institutions, universities, researchers, and communities.
6. Promote accountability and transparency through clearly defined indicators, responsible institutions, regular reporting, and appropriate protection of personal information.

These objectives reflect the principle that measurement should serve a practical purpose. Data collection is most valuable when it helps health professionals and decision-makers identify problems, assess possible solutions, and improve the quality and equity of healthcare.

B.3. Proposed Core NCD Monitoring Indicators

The following matrix identifies potential indicators that could contribute to a strengthened NCD monitoring framework in Timor-Leste. The list is illustrative and should be adapted to the availability of reliable data, national priorities, and the operational capacity of the health system.

Table B.1. Proposed Core Indicators for NCD Monitoring

Monitoring domain	Proposed indicator	Potential data source	Suggested reporting interval

Population risk	Prevalence of tobacco use among adults	STEPS or comparable population surveys	Every 3–5 years, subject to resources
Population risk	Prevalence of insufficient physical activity	Population health surveys	Every 3–5 years
Population risk	Prevalence of raised blood pressure	Population surveys and clinical measurements	Every 3–5 years
Population risk	Prevalence of raised blood glucose or diagnosed diabetes	Population surveys, laboratory data, and clinical records	Every 3–5 years for population surveys; routine service data more frequently
Cardiovascular health	Number of patients receiving care for hypertension	Primary healthcare and facility records	Quarterly or annually
Diabetes care	Number of registered patients receiving continuing diabetes care	Facility records and patient registers	Quarterly or annually
Cancer	Number of recorded cancer diagnoses by type and relevant demographic characteristics	Hospital records and any established cancer registration arrangements	Annually, subject to data completeness
Chronic respiratory disease	Number of patients receiving care for chronic respiratory conditions	Facility records	Quarterly or annually

Premature mortality	Cause-specific mortality rates, where reliable data are available	Civil registration, mortality records, and validated estimates	Annually
Disease burden	YLLs, YLDs, and DALYs for selected NCDs	GBD estimates and suitable national studies	Periodic review, with source and methodology stated
Access to care	Proportion of patients with documented follow-up after diagnosis	Primary healthcare and referral records	Quarterly or annually
Continuity of care	Proportion of registered patients attending scheduled follow-up	Facility registers or suitable electronic records	Quarterly
Medicine availability	Availability of selected essential NCD medicines at designated facilities	Facility stock records and supervision reports	Monthly or quarterly
Health workforce	Availability of personnel trained in essential NCD prevention and management	Workforce records and training databases	Annually
Health equity	Differences in selected NCD indicators by geography and relevant population groups	Surveys and disaggregated routine data	According to the source

Financial protection	Reported out-of-pocket expenditure associated with chronic disease care	Household surveys and health expenditure studies	Periodically, according to feasibility

Interpretation note: The reporting intervals above are proposals, not confirmed national reporting requirements. Some indicators may be unavailable or may not yet be measured consistently. Where a reliable baseline does not exist, the first priority should be to establish a clear definition, assess data availability, and determine whether the indicator can be measured validly.

The framework should also distinguish population-level prevalence from facility-based counts. For example, the number of patients recorded in a clinic does not represent the total number of people living with diabetes in the country. It may reflect access to diagnosis, service availability, referral patterns, and the completeness of patient registration as well as the underlying burden of disease.

B.4. Operational Definitions and Measurement Principles

Indicators require standard definitions if their results are to be compared across districts, facilities, and years. A monitoring system should therefore establish an operational specification for every selected indicator.

Table B.2. Illustrative Indicator Specifications

Indicator	Proposed operational definition	Important measurement consideration
Adult tobacco-use prevalence	Proportion of the defined adult survey population meeting the survey's specified tobacco-use definition	State the age range, tobacco products covered, survey period, and sampling method
Raised blood pressure	Proportion of the assessed population meeting a prespecified blood-pressure threshold under a documented measurement protocol	Specify the guideline and protocol used; distinguish a survey measurement from a confirmed clinical diagnosis

Diabetes prevalence	Proportion of the defined population meeting the prespecified diagnostic criteria or reporting a previous diagnosis, as appropriate	Identify laboratory criteria, medication use, self-report, and the denominator
Hypertension follow-up	Proportion of registered patients due for follow-up who have a documented follow-up encounter within the defined period	Define eligibility, the follow-up window, and how transfers or deaths are handled
Essential medicine availability	Proportion of assessed facilities with the specified medicine available on the day of assessment	Identify medicines, facility types, stock-out rules, and assessment dates
NCD-related premature mortality	Cause-specific deaths occurring within the prespecified age range and period	State the age range, cause-of-death classification, and completeness of mortality data
DALYs	Sum of YLLs and YLDs for a specified disease, population, and period	Identify the estimation source, year, uncertainty intervals, and whether results are modelled or nationally calculated

Operational definitions should be established before routine reporting begins. Changes to diagnostic thresholds, survey instruments, coding systems, or data collection procedures should be documented because they may affect comparisons over time.

For population surveys, sampling design and non-response should be considered when interpreting estimates. For routine health records, the completeness of reporting and the possibility of duplicate registrations should be assessed. When national estimates are drawn from international modelling studies, their methods and uncertainty should be described rather than treating the estimates as direct counts from health facilities.

B.5. Proposed Data Sources and Their Complementary Roles

No single data source can adequately describe every dimension of the NCD burden. A sustainable monitoring framework should combine complementary sources while recognising the limitations of each.

B.5.1. Population-Based Surveys

Population surveys can provide information on risk factors, measured blood pressure, anthropometric indicators, and selected biochemical measures. They are particularly useful for identifying people who may not have been diagnosed or reached by health services.

The Timor-Leste STEPS survey can serve as an important reference for adult NCD risk-factor surveillance. Future survey planning should consider comparability with previous rounds, adequate geographical coverage, consistent measurement procedures, and the financial and logistical requirements of repeated data collection.

B.5.2. Routine Primary Healthcare Records

Primary healthcare records can help monitor patient registration, follow-up attendance, treatment continuity, and service utilisation. Their principal advantage is the potential to generate information more frequently than population surveys.

However, routine records generally describe people who reach services rather than the entire population. They may therefore underrepresent people who face geographical, financial, cultural, or other barriers to accessing care.

B.5.3. Hospital and Referral Records

Hospital records can contribute information about severe complications, emergency presentations, specialist treatment, and outcomes among admitted patients. They can help identify service gaps and inform referral planning.

Hospital-based information should not be interpreted as a complete representation of national disease prevalence. People who cannot reach hospitals, receive care elsewhere, or are never diagnosed may be missing from these records.

B.5.4. Civil Registration and Mortality Information

Reliable death registration and cause-of-death information are essential for monitoring premature mortality. Improvements in completeness, medical certification, and

consistent cause-of-death coding can strengthen the capacity to identify the contribution of NCDs to population health loss.

Where mortality information is incomplete, limitations should be explicitly reported. Modelled estimates may help describe possible patterns, but they should remain distinguishable from directly registered deaths.

B.5.5. Research and International Estimates

University research, clinical studies, health systems research, and international datasets can provide information that routine systems do not capture. GBD estimates, for example, can support comparisons of mortality, disability, and DALYs across diseases and geographical areas.

These sources should be triangulated rather than combined uncritically. Differences in years, case definitions, age structures, and modelling methods may explain apparent discrepancies between estimates.

B.6. Health Equity and Population Disaggregation

National averages can conceal important inequalities. Two districts may report similar overall disease prevalence while differing considerably in diagnosis, treatment access, medicine availability, and the ability of patients to attend follow-up appointments. An effective monitoring framework should therefore examine not only the total burden but also how health risks and services are distributed.

Where sample size, data quality, and privacy protections permit, indicators may be disaggregated by:

- Age group and sex.
- Municipality or other appropriate geographical unit.
- Urban, peri-urban, and rural residence, where these categories are clearly defined.
- Socioeconomic characteristics measured through suitable surveys.
- Disability status, where collection is ethical and methodologically appropriate.
- Relevant barriers to access, including distance to facilities or transport difficulties.

Disaggregation should not be used to stigmatise communities or imply that disease is the result of individual failure. Differences in health outcomes may reflect structural factors, including income, education, food environments, living conditions, transport, service availability, and the distribution of healthcare personnel.

Small population counts may create risks of identifying individuals or producing unstable estimates. Reporting rules should therefore include minimum-cell protections, appropriate aggregation, and secure handling of identifiable records.

B.7. Proposed National Research Agenda

Monitoring describes what is happening; research helps explain why it is happening and which responses are likely to be effective. Timor-Leste's NCD research agenda should therefore address epidemiological patterns, health system performance, social determinants, patient experiences, and the feasibility of interventions in different settings.

Table B.3. Proposed Priority Research Questions

Research priority	Illustrative research question	Potential research approach	Potential policy contribution
Epidemiological surveillance	What are the current patterns of major NCD risk factors across relevant population groups?	Population-based surveys and secondary data analysis	Strengthen prevention priorities and establish baselines
Hypertension and cardiovascular disease	What factors influence early diagnosis and sustained hypertension control?	Facility-based studies and mixed-methods research	Improve screening, follow-up, and treatment continuity
Diabetes care	What barriers affect diabetes diagnosis, medicine access, and prevention of complications?	Clinical audits, patient interviews, and service studies	Strengthen diabetes pathways and patient support

Cancer services	What are the principal barriers to timely recognition, diagnosis, referral, and treatment of selected cancers?	Referral-pathway analysis and qualitative research	Improve service coordination and identify feasible priorities
Chronic respiratory disease	Which environmental and behavioural exposures are associated with respiratory symptoms and service utilisation?	Epidemiological studies and environmental assessment	Inform prevention and clinical management
Rural and remote access	How do travel time, transport costs, and service availability influence continuing NCD care?	Geographic analysis and community-based studies	Improve outreach and service distribution
Health workforce	What training and supervision arrangements best support essential NCD care in primary healthcare?	Workforce assessment and implementation research	Inform professional development and staffing plans
Essential medicines	How frequently do selected facilities experience stock-outs of priority NCD medicines, and what factors contribute?	Supply-chain assessment and facility surveys	Strengthen procurement and distribution
Household economic consequences	How does long-term NCD care affect household spending, employment, and caregiving?	Household surveys and qualitative studies	Inform financial protection and social support

Community prevention	Which community-based approaches are acceptable, feasible, and sustainable in different settings?	Participatory research and implementation evaluation	Support locally appropriate prevention
Digital health	Can feasible digital tools improve follow-up, referral coordination, or patient education in areas with limited access?	Pilot studies and implementation evaluation	Guide responsible investment in digital services
Health equity	Which populations are least likely to receive timely diagnosis and continuing care, and why?	Disaggregated data analysis and qualitative research	Reduce avoidable inequalities

The table identifies research priorities for consideration; it does not imply that the proposed studies have already been conducted or that their findings are known. The final research agenda should be developed collaboratively and aligned with national priorities, available expertise, ethical requirements, and funding capacity.

B.8. Research Ethics, Data Governance, and Community Participation

NCD monitoring and research involve responsibilities beyond technical accuracy. Health information can be sensitive, and careless collection or disclosure may expose individuals to stigma, discrimination, or breaches of privacy. Data governance should therefore be integrated into the design of surveys, patient registers, clinical studies, and digital health initiatives.

Research involving human participants should undergo appropriate ethical review and obtain informed consent where required. Researchers should collect only the information necessary to answer the research question, protect identifiable records, limit access to authorised personnel, and report findings in ways that reduce the risk of identifying individuals or small communities.

Data-sharing arrangements should clarify ownership, access permissions, permitted uses, retention periods, and responsibilities for protecting information. When digital

tools or artificial intelligence are considered, the assessment should include data security, reliability, human oversight, accessibility, and the risk that poorly designed systems could reproduce existing inequalities.

Community participation is equally important. Patients, caregivers, community representatives, health workers, and local organisations should have opportunities to contribute to the definition of research priorities and the interpretation of findings. Their participation can help ensure that monitoring addresses practical concerns rather than focusing exclusively on what is easiest to measure.

B.9. Proposed Phased Implementation Framework

A phased approach may help avoid creating a monitoring system that is too complex or expensive to sustain. The following sequence is illustrative and should be adapted after assessing existing national arrangements and institutional capacity.

Table B.4. Illustrative Implementation Phases

Phase	Main activities	Illustrative outputs
Phase 1: Assessment and prioritisation	Review existing surveys, reporting systems, indicators, and data gaps; consult stakeholders	Agreed priority indicators and documented baseline assessment
Phase 2: Standardisation and pilot testing	Define indicators, test data collection tools, clarify responsibilities, and assess data quality	Operational indicator guide and pilot evaluation
Phase 3: Initial implementation	Introduce feasible indicators in selected settings and establish reporting procedures	Initial monitoring reports and documented implementation lessons
Phase 4: Review and improvement	Evaluate completeness, usability, costs, and equity implications; revise procedures where needed	Revised framework and prioritised improvement plan

Phase 5: Institutionalisation and scale-up	Integrate feasible procedures into established systems, training, supervision, and planning	More consistent reporting and routine use of findings in policy decisions

These phases are not an official government timetable. Their duration should depend on the resources available, the maturity of existing information systems, and the feasibility of implementing the proposed activities without undermining essential healthcare services.

Implementation should be accompanied by a clear assignment of responsibilities. Relevant institutions should determine who is accountable for data collection, quality assurance, analysis, reporting, and the use of findings. A monitoring framework will be more sustainable when responsibilities are explicit and integrated into existing structures rather than dependent on temporary projects alone.

B.10. Monitoring, Evaluation, and Accountability

The success of a monitoring framework should not be judged solely by the volume of information collected. It should also be assessed according to data quality, usefulness for decision-making, equity, cost, and the extent to which findings lead to appropriate action.

A periodic review could examine five dimensions:

1. Data completeness: whether participating facilities and reporting units submit the required information.
2. Data quality: whether indicators are defined consistently and records are sufficiently accurate for their intended use.
3. Timeliness: whether reports are available when planning and resource-allocation decisions are made.
4. Policy use: whether findings inform programme adjustments, training, procurement, referral planning, or prevention activities.
5. Equity and accountability: whether monitoring identifies underserved populations and whether responsible institutions respond to the problems documented.

Where possible, evaluations should distinguish between outputs and outcomes. Training sessions delivered, facilities reporting, or surveys completed are outputs. Improved diagnosis, more consistent treatment, fewer avoidable complications, and reduced health inequalities are potential outcomes that require appropriate measurement over time.

Changes in health outcomes should not automatically be attributed to a single programme. Population health is influenced by multiple interacting factors, and evaluations should consider alternative explanations, implementation conditions, and the period required for effects to become observable.

B.11. Strategic Recommendations for Timor-Leste

Based on the proposed framework, the following recommendations merit consideration by relevant national institutions and research partners.

First, prioritise a small set of feasible, high-value indicators. A limited set of clearly defined measures is preferable to a comprehensive list that cannot be collected consistently. The initial selection should reflect the burden of disease, the needs of primary healthcare, and the practical capacity of the information system.

Second, strengthen the connection between population surveys and routine health information. Surveys can identify risk factors and people who remain undiagnosed, while clinical records can help assess service utilisation and continuing care. Used together, they provide a more complete picture than either source alone.

Third, improve the quality and interpretation of mortality information. More reliable cause-of-death information would support assessment of premature mortality, comparisons across conditions, and the planning of prevention and treatment services.

Fourth, incorporate equity into routine analysis. Where data permit, findings should be examined across geographical areas and relevant population groups so that national averages do not obscure unequal access to prevention and care.

Fifth, establish a collaborative research agenda. National health authorities, universities, healthcare professionals, and communities should identify questions that can realistically inform policy and improve patient outcomes. Research priorities should be reviewed periodically as new evidence and service needs emerge.

Sixth, protect privacy and ensure responsible data use. Data governance, ethical review, secure information handling, and transparent reporting should be built into the monitoring framework from the beginning.

Finally, connect monitoring to action. Every selected indicator should have a clear purpose, an identified responsible institution, and a defined process for reviewing results and deciding whether corrective action is required.

B.12. Conclusion

A sustainable response to chronic diseases and NCDs requires a reliable foundation of epidemiological evidence, health service information, and locally relevant research. For Timor-Leste, a monitoring framework should combine population-based surveys, routine clinical records, mortality information, and carefully interpreted research estimates. It should also recognise the limits of available data and avoid presenting proposed indicators as though they were already part of an established national system.

The proposed framework offers a starting point for discussion and adaptation. Its value will depend on institutional ownership, technical feasibility, adequate resources, ethical governance, and meaningful participation by communities and healthcare professionals. Above all, monitoring should not become an administrative exercise disconnected from people's needs. Its ultimate purpose is to help ensure that evidence informs better prevention, more continuous care, fairer resource allocation, and improved health and dignity for people living with chronic diseases in Timor-Leste.

REFERENCES

Atun, R., Jaffar, S., Nishtar, S., Knaul, F. M., Barreto, M. L., Nyirenda, M., Banatvala, N., & Piot, P. (2013). Improving responsiveness of health systems to non-communicable diseases. The Lancet, 381(9867), 690–697. https://doi.org/10.1016/S0140-6736(13)60063-X

Bonita, R., Beaglehole, R., & Kjellström, T. (2006). Basic epidemiology (2nd ed.). World Health Organization.

Ezzati, M., Lopez, A. D., Rodgers, A., Vander Hoorn, S., & Murray, C. J. L. (2002). Selected major risk factors and global and regional burden of disease. The Lancet, 360(9343), 1347–1360.

GBD 2019 Diseases and Injuries Collaborators. (2020). Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: A systematic analysis for the Global Burden of Disease Study 2019. The Lancet, 396(10258), 1204–1222. https://doi.org/10.1016/S0140-6736(20)30925-9

GBD 2019 Risk Factors Collaborators. (2020). Global burden of 87 risk factors in 204 countries and territories, 1990–2019: A systematic analysis for the Global Burden of Disease Study 2019. The Lancet, 396(10258), 1223–1249. https://doi.org/10.1016/S0140-6736(20)30752-2

GBD 2021 Causes of Death Collaborators. (2024). Global burden of 288 causes of death and life expectancy decomposition in 204 countries and territories and 811 subnational locations, 1990–2021: A systematic analysis for the Global Burden of Disease Study 2021. The Lancet, 403(10440), 2100–2132. https://doi.org/10.1016/S0140-6736(24)00367-2

GBD 2021 Diseases and Injuries Collaborators. (2024). Global incidence, prevalence, years lived with disability (YLDs), disability-adjusted life-years (DALYs), and healthy life expectancy (HALE) for 371 diseases and injuries in 204 countries and territories and 811 subnational locations, 1990–2021: A systematic analysis for the Global Burden of Disease Study 2021. The Lancet. https://doi.org/10.1016/S0140-6736(24)00757-8

GBD 2021 Demographics Collaborators. (2024). Global age-sex-specific mortality, life expectancy, and population estimates in 204 countries and territories and 811 subnational locations, 1950–2021, and the impact of the COVID-19

pandemic: A comprehensive demographic analysis for the Global Burden of Disease Study 2021. The Lancet. https://doi.org/10.1016/S0140-6736(24)00476-8

Institute for Health Metrics and Evaluation. (2024). Global Burden of Disease Study 2021 results. University of Washington. https://vizhub.healthdata.org/gbd-results/

Kruk, M. E., Gage, A. D., Arsenault, C., Jordan, K., Leslie, H. H., Roder-DeWan, S., Adeyi, O., Barker, P., Daelmans, B., Doubova, S. V., English, M., García-Elorrio, E., Guanais, F., Gureje, O., Hirschhorn, L. R., Jiang, L., Kelley, E., Lemango, E. T., Liljestrand, J., ... Pate, M. (2018). High-quality health systems in the Sustainable Development Goals era: Time for a revolution. The Lancet Global Health, 6(11), e1196–e1252. https://doi.org/10.1016/S2214-109X(18)30386-3

Marmot, M., Friel, S., Bell, R., Houweling, T. A. J., & Taylor, S. (2008). Closing the gap in a generation: Health equity through action on the social determinants of health. The Lancet, 372(9650), 1661–1669. https://doi.org/10.1016/S0140-6736(08)61690-6

Ministry of Health, Timor-Leste. (2011). National health sector strategic plan 2011–2030. Government of Timor-Leste.

Ministry of Health, Timor-Leste. (2014). National strategy for noncommunicable diseases, injuries, disabilities and care of the elderly 2014–2018. Government of Timor-Leste.

Ministry of Health, Timor-Leste. (2023). STEPS 2023 [Noncommunicable disease risk factor survey dataset]. World Health Organization. https://extranet.who.int/ncdsmicrodata/index.php/catalog/998

Murray, C. J. L., & Lopez, A. D. (Eds.). (1996). The global burden of disease: A comprehensive assessment of mortality and disability from diseases, injuries, and risk factors in 1990 and projected to 2020. Harvard School of Public Health.

Roth, G. A., Mensah, G. A., Johnson, C. O., Addolorato, G., Ammirati, E., Baddour, L. M., Barengo, N. C., Beaton, A. Z., Benjamin, E. J., Benziger, C. P., Bonny, A., Brauer, M., Brodmann, M., Cahill, T. J., Carapetis, J., Catapano, A. L., Chugh, S. S., Cooper, L. T., Coresh, J., ... Fuster, V. (2020). Global burden of cardiovascular diseases and risk factors, 1990–2019: Update from the GBD 2019

Study. Journal of the American College of Cardiology, 76(25), 2982–3021. https://doi.org/10.1016/j.jacc.2020.11.010

United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development. https://sdgs.un.org/2030agenda

United Nations General Assembly. (2011). Political declaration of the High-level Meeting of the General Assembly on the Prevention and Control of Non-communicable Diseases (A/RES/66/2).

United Nations General Assembly. (2018). Political declaration of the third high-level meeting of the General Assembly on the prevention and control of non-communicable diseases (A/RES/73/2).

World Bank. (2019). Timor-Leste health sector review: Health system strengthening and service delivery. World Bank.

World Bank. (2023). World development indicators. https://data.worldbank.org/country/timor-leste

World Health Organization. (2004). Global burden of disease 2002: Data sources, methods and results. World Health Organization.

World Health Organization. (2008a). Closing the gap in a generation: Health equity through action on the social determinants of health. World Health Organization.

World Health Organization. (2008b). The world health report 2008: Primary health care—Now more than ever. World Health Organization.

World Health Organization. (2010a). Global recommendations on physical activity for health. World Health Organization.

World Health Organization. (2010b). Monitoring the building blocks of health systems: A handbook of indicators and their measurement strategies. World Health Organization.

World Health Organization. (2013). Global action plan for the prevention and control of noncommunicable diseases 2013–2020. World Health Organization.

World Health Organization. (2014a). Global status report on alcohol and health 2014. World Health Organization.

World Health Organization. (2014b). Global status report on noncommunicable diseases 2014. World Health Organization.

World Health Organization. (2016). Framework on integrated, people-centred health services. World Health Organization.

World Health Organization. (2017). Tackling NCDs: “Best buys” and other recommended interventions for the prevention and control of noncommunicable diseases. World Health Organization.

World Health Organization. (2018). Noncommunicable diseases country profiles 2018. World Health Organization.

World Health Organization. (2019). Primary health care on the road to universal health coverage: 2019 monitoring report. World Health Organization.

World Health Organization. (2020a). WHO package of essential noncommunicable (PEN) disease interventions for primary health care. World Health Organization.

World Health Organization. (2020b). WHO report on the global tobacco epidemic 2020: Addressing new and emerging products. World Health Organization.

World Health Organization. (2021). WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. World Health Organization.

World Health Organization. (2022). Noncommunicable diseases progress monitor 2022. World Health Organization.

World Health Organization. (2023a). WHO implementation roadmap 2023–2030 for the global action plan for the prevention and control of NCDs 2013–2030. World Health Organization.

World Health Organization. (2023b). World health statistics 2023: Monitoring health for the SDGs, sustainable development goals. World Health Organization.

World Health Organization. (2024a). Health and health-related SDG country profile: Timor-Leste. World Health Organization, Regional Office for South-East Asia.

World Health Organization. (2024b). Noncommunicable diseases. https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases

World Health Organization. (2025). Noncommunicable diseases progress monitor 2025. World Health Organization.

World Health Organization, & World Bank. (2023). Tracking universal health coverage: 2023 global monitoring report. World Health Organization and World Bank.